ESSAYS ON HETEROGENEOUS RETURNS, TRUST, AND HOUSEHOLD WEALTH

by
Decory Edwards

A dissertation submitted to The Johns Hopkins University in conformity with the requirements for the degree of Doctor of Philosophy

Baltimore, Maryland
April 2026

Abstract

This dissertation studies heterogeneity in returns, the structural modeling of household wealth accumulation, and theoretical measures of wealth inequality through three essays. The first essay extends the standard consumption-saving framework in heterogeneous-agent modeling to allow for differences in returns to assets across households and shows that the resulting model distribution of wealth closely matches the empirical distribution of wealth measured in the Survey of Consumer Finances (SCF). The distributional and welfare implications of wealth taxation and capital income taxation when households earn different returns are also compared here. The second essay uses the RAND Longitudinal product version of the Household Retirement Survey (HRS) to measure returns to net wealth and documents a hump-shaped relationship between the measure of returns and generalized trust for respondents, in line with evidence of a similar hump-shaped relationship between trust and economic performance measures found in the literature. Across cross-sectional and panel specifications, moderate trust is associated with the highest returns even after accounting for demographic characteristics and portfolio composition. Moreover, I can use the HRS data to describe the persistent component of returns between respondents, which is an important empirical object in the literature on returns on assets. The third essay develops a decision-theoretic framework for ranking wealth distributions under ambiguity and applies the social welfare approach to derive a measure of wealth inequality aversion.

Keywords: wealth inequality, heterogeneous returns, household finance, trust, macroe-

conomics

Primary Reader and Dissertation Advisor

Christopher Carroll
Department of Economics
Johns Hopkins University

Secondary Readers

Nick Papageorge
Department of Economics
Johns Hopkins University

Stelios Fourakis
Department of Economics
Johns Hopkins University

Acknowledgments

I am indebted to Christopher Carroll, Nick Papageorge, and John Huston, Kelly Lyons, Harry Wallace, Teresa Morrison, and the rest of the McNair Scholar Program and Economics Department at Trinity University for funding support conducting research. For helpful comments and suggestions, I thank my doctoral advisors Christopher Carroll, Nick Papageorge, and Stelios Fourakis; Mohammed Ali Khan and Isaiah Andrews for providing useful feedback. I would like to thank Matthew White, Alan Lujan, Mateo Velasquez-Giraldo, and Will Du, as well as my fellow cohort members Adrian Monninger, Kyung Woong Koh, and David Osten for comments and feedback as well. This work makes use of the HA toolkit made available by the Econ-Ark project.

Table of Contents

List of Tables

List of Figures

Chapter 1

Matching Wealth Moments with Heterogeneous Returns

Abstract

Empirical evidence shows that individuals face widely varying returns on their assets. This heterogeneity has important implications for modeling wealth inequality. I incorporate heterogeneity in returns into a standard heterogeneous-agent model, and find the distribution required to match the empirical wealth distribution. Introducing return heterogeneity yields a much better match to the wealth moments measured using Survey of Consumer Finances (SCF) data than assuming homogeneous returns. In particular, the bottom 60% of the wealth distribution is matched well enough that the aggregate marginal propensity to consume will be aligned with modest estimates. In addition, it has been noted that the equivalence between a wealth tax and capital income tax fails when individuals no longer earn the same return. I compare a wealth tax and a capital income tax set to raise equal revenue. A capital income tax reduces wealth inequality more by targeting high-return households, but each policy entails different welfare trade-offs across the return distribution. There are a number of final exercises, such as estimating a lognormal distribution of returns across households and interpreting the returns heterogeneity required by the model as arising from banking-sector frictions (e.g., incomplete deposit rate pass-through).

1.1 Introduction

The unequal distribution of wealth is a well documented phenomenon across many countries. Disparities have not only persisted over time but intensified in recent years. In 2018, the aggregate wealth of the nation's three richest individuals surpassed that of the entire bottom 50% of the U.S. wealth distribution. This point is stressed in a recent article based on Forbes magazine rankings (Institute for Policy Studies, November 2022)[1].

The unequal distribution of wealth has also long been a focus of study across disciplines. The statistics literature, for instance, has linked the distribution of income to the observable skewness in wealth distribution, and work in economics established microfoundations for wealth accumulation over the life cycle. Macroeconomic studies have examined the distribution of wealth among households to understand how the economy as a whole responds to aggregate fiscal shocks, as well as the decomposition of those effects by group (like wealth level).

The macroeconomics literature has undergone significant changes in recent years, with the widespread adoption of models that abandon the traditional representative agent assumption in their analysis. As this setting will require that in equilibrium all agents hold the same level of wealth, it is not a desirable laboratory in terms of producing model objects, like the distribution of wealth, that can be compared to real world counterparts.

[1]See Inequality.org articles data November 21, 2022: "Wealth Inequality in the United States" and "Updates: Billionaire Wealth, U.S. Job Losses and Pandemic Profiteers" (date accessed: March 27, 2023)

Researchers began incorporating an exogenously determined income process that generates a distribution of income among households. The baseline form of (ex-post) heterogeneity in the model is the description of a permanent and transitory process for income. To account for business cycle dynamics, one can further assume that individuals face some level of potential unemployment in each period, creating a precautionary savings motive for consumers. Given that such uncertainty cannot be fully insured against, the availability of a riskless asset that partially insures against income risk results in households choosing to hold different levels of market resources optimally.

Krusell and Smith 1998's seminal work suggests that models assuming heterogeneity in individual income perform well in matching the aggregate capital stock but poorly in matching the distribution of wealth. The next step is to assume there is some ex-ante heterogeneity among households, leading more households to optimally hold lower levels of wealth. [2] Carroll et al. 2017 adopt this approach and assume that agents differ in their time preferences, which reflects implicit characteristics of households relevant to their lifetime wealth accumulation. The authors find that this assumption of modest heterogeneity in time preferences is sufficient to match both the shape and skewness of the empirical distribution of wealth.

The time preference factor (β) is one of the key parameters that influences an individual's equilibrium target level of market resources, but it is not directly observable. Estimating would require one to collect data through surveys or other methods that

[2] Kaplan and Violante 2022's recent work provides a comprehensive survey of incomplete markets models with heterogeneous agents featuring (i) uninsurable idiosyncratic income risk, (ii) a precautionary savings motive, and (iii) an endogenous wealth distribution.

provide direct information from households. In contrast, differences in the rate of return on financial assets across households is possible, as this variable *is* observable.

In this paper, I provide further evidence of the heterogeneous agent framework's ability to match wealth moments by adding a single source of heterogeneity across households beyond the realization of ex-post shocks to their income. I allow households to differ in the return earned on assets first in the infinite horizon setting. I further extend the model to allow for rich life cycle dynamics. From there, I compare the effects of revenue-equivalent wealth and capital income tax policies on wealth inequality and welfare. Additionally, I interpret the heterogeneity in the returns on safe assets earned by households in the context of the transmission channel of monetary policy and its variation across the banking sector. As a final step, I rerun the entire exercise under the assumption that the returns across households are lognormally distributed.

1.2 Literature Review

1.2.1 Explaining Inequality in the Distribution of Wealth

Although wealth inequality gets a considerable amount of media coverage in modern times, it has been the object of deep reflection by academic and political thinkers for centuries. For this reason, the literature on wealth inequality is rich and has been studied by several disciplines.

Benhabib and Bisin 2018 review the literature on the documented skewness in the distribution of wealth with historical accounts of the origins of the shape of the wealth distribution. The authors then provide the traditional theoretical explanations of this unequal distribution: (i) skewness in the (exogenous) distribution of earnings, (ii) stochastic returns to wealth and savings, and, importantly, (iii) microfoundations for the evolution of wealth resulting from the consumption and saving behavior of households[3].

De Nardi and Fella 2017 survey the literature on the microfoundations of wealth accumulation. A number of possible models of household consumption and saving behavior marked by observable differences between households (beyond demographic differences) lead to greater inequality in wealth accumulation over time. Earnings and rate of return risk, ex-ante heterogeneity in preferences, medical expenses, bequest motives, and entrepreneurship are all potential mechanisms which influence the shape of the distribution of wealth.

Gabaix et al. 2016 introduces the concept of speed of convergence to explain the observed evolution of income inequality over time, particularly within the upper tail of the distribution over the past 40 years. Notably, they show that, replicating the empirical dynamics of inequality requires incorporating two additional forms of heterogeneity in the income process for households than are included in the standard consumption and saving models.[4] The first form is *type dependence* in the income growth rate distribution, which accounts for some households having a higher average income growth rate. Second, *scale dependence* captures the fact that higher income

[3]As explored in the next section, the emergence of heterogeneous agent models has been a significant development in investigating this issue. Bewley 1983, Aiyagari 1994, and Huggett 1993 are among the earliest examples.

[4]Note that, although this analysis is about the distribution of income, this literature notably asserts that the distribution of wealth inherits some of its skewness from the distribution of income

levels are more susceptible to shocks to their income growth. The authors find that the former does a good job of explaining the rapid rise in income inequality, and the latter can generate infinitely fast transitions in inequality.

1.2.2 Measurements of heterogeneous rates of return

The rationale behind incorporating heterogeneity in rates of return to asset holdings lies in the use of novel datasets in recent empirical research to quantify the differences in returns among individuals. Using 12 years of Norwegian tax records, Fagereng et al. 2020 document the heterogeneity in realized returns. These differences occur both across individuals (*type dependence*), within and across asset classes with varying levels of risk, and within wealth deciles, where returns are positively correlated with wealth (*scale dependence*). Moreover, they find that heterogeneous returns exhibits significant persist over time and are positively correlated across generations. These findings support the assumption of ex-ante heterogeneous rates of return in the buffer-stock savings model of households, and provide a benchmark for comparing the distribution of rates of return to the empirical distribution of wealth, as in Carroll et al. 2017.

Bach et al. 2018 use administrative panel data on the balance sheets of Swedish residents to gauge historical and expected returns, as well as risks associated with asset holdings. Their analysis of portfolio performance supports the finding that heterogeneous returns substantially shape in the levels and growth of top wealth shares over time.

Campbell et al. 2019 examine equity holdings in India between 2002 and 2011 and find that heterogeneity in investment returns arising from the inherent randomness associated with risky assets and differences in investment strategies, is a key driver of rising inequality in portfolio holdings during the time period. The authors attribute the scale dependence of equity returns to the tendency of smaller accounts to be less diversified than larger ones.

Deuflhard et al. 2018 analyze household savings account investments within a heterogeneous agent, incomplete markets model with a precautionary saving motive and a *single asset* to partially insure against risk. They document substantial type dependence in the rate of return to these safe assets and attribute the heterogeneity in returns to differences in financial sophistication. Notably, explaining differences in investment returns for households is a vital step toward endogenizing this form of ex-ante heterogeneity among households in future models.

Altmejd et al. 2024 provide causal evidence that financial education leads to significant differences in portfolio returns. Using university application records from the Swedish National Archives and data from the Swedish Income and Wealth registry, they show that individuals marginally admitted to business or economics programs hold more money in stocks and earn higher raw returns on these holdings than those not admitted.

1.2.3 Recent heterogeneous agent models with varying rates of return

Several studies extend the heterogeneous agent framework to allow for households to earn different returns on their assets. Of the models with a single, riskless asset and a partial equilibrium analysis of a distribution of returns across agents, my paper is unique in that it generates a realistic mass of agents below the 60-th percentile of the wealth distribution. This is important because, as we will see, this region of the wealth distribution is important for generating a reasonable estimate of the aggregate marginal propensity to consume. That said, I will discuss models that go beyond the modeling choices made in my paper.

Daminato and Pistaferri 2024 incorporate heterogeneous returns into the solution of a model of consumption-saving for households. They use data from the PSID to document heterogeneity in returns, which they find is comparable to the returns distribution measured using the Norwegian registry by Fagereng et al. 2020.

Benhabib et al. 2019 propose an overlapping-generations model that incorporates intergenerational wealth transfers, with agents facing uncertainty regarding both labor and capital income. In an earlier work (Benhabib et al. 2017) the authors more explicitly define household preferences for bequests to the next generation. Both studies conclude that the distribution of earnings and differences in rates of savings and bequests are crucial for accurately replicating the tails of the observed wealth distribution.

Guler et al. 2022 develop a life-cycle model with endogenous heterogeneity in the rate of return by incorporating households' optimal housing and mortgage decisions. Using this framework, the authors investigate the effects of aggregate fiscal shocks, including one-time stimulus payments and mortgage debt relief programs.

Menzio and Spinella 2025 introduce search frictions in financial markets within a standard infinite-horizon macroeconomic model. A distribution of returns across households arises endogenously in their model, and they use the empirical findings from Fagereng et al. 2020 regarding the distribution of returns to net worth as notable targeted moments.

1.2.4 Estimates of the aggregate MPC

A substantial empirical literature measures households' marginal propensities to consume (MPCs), consistently finding that the aggregate MPC is far larger—and far more heterogeneous—than implied by representative-agent benchmarks. Across quasi-experimental settings, estimated MPCs typically fall in the range of 0.2 to 0.4. For example, Johnson et al. 2006 show that households spent roughly 20–40 percent of the 2001 tax rebates in the quarter they were received, while Parker et al. 2013 find similar magnitudes for the 2008 stimulus payments. Structural and covariance-based approaches deliver comparable estimates: Blundell et al. 2008 imply quarterly MPCs of about 0.2–0.3, and meta-analytic evidence from Havránek and Sokolova 2020 concludes that most studies cluster in this same range. At the same time, research using administrative or survey data highlights that MPCs can be substantially higher among households with low wealth or low liquidity, often exceeding 0.5 (see Jappelli and Pistaferri 2014a; Fagereng et al. 2021). Taken together, these findings indicate that empirically plausible MPCs span a wide but well-documented interval.

The model developed in this paper produces MPCs that fall squarely within these empirically established ranges, even though the MPC distribution is not directly targeted in the calibration. This serves as an additional, independent validation of the framework, demonstrating that a model calibrated primarily to match wealth inequality can nevertheless replicate realistic consumption responses across the wealth distribution.

1.2.5 My contributions

This paper contributes to the literature in at least two ways. First, I model the labor income uncertainty as a random walk, rather than an AR(1) process. The AR(1) specification implies less uncertainty in household earnings over the life cycle, leading to less accumulation of wealth over the life cycle. By using a the permanent income framework, I account for *as much* of the dispersion of wealth across households as possible with labor income uncertainty. The remaining dispersion in wealth across households that cannot be explained by differences in earnings is attributed to heterogeneity in returns, ultimately leading to modest estimates of differences in returns across households.

Second, the life-cycle version of my model is much richer in its calibration of earnings and mortality rates than other studies. Specifically, I use the earnings profile of Cagetti 2003 that distinguishes mean earnings not only by age but by education cohort. I use age-education-dependent mortality rates from Brown et al. 2007. This approach allows households to be distinguished by both age and education, providing an additional mechanism for explaining the dispersion in wealth holdings. However, the impact of this mechanism is limited, and other sources of ex-ante heterogeneity among households, such as time preferences or the rate of return, are still needed to match wealth moments precisely. Said differently, life-cycle dynamics and labor income uncertainty are not enough to generate a reasonably skewed wealth distribution.

1.3 Model

The heterogeneous agent macro literature provides a class of models whose output is a distribution of wealth which can be easily compared to data on the wealth holdings across individuals in terms of moments. This section will introduce the model I solve and simulate computationally in order to do this particular comparison of wealth moments.

1.3.1 Defining the stochastic income process

Each household's income (y_t) during a given period depends on three main factors: aggregate wage rate (W_t) that all households in the economy face, the permanent income component (p_t), which represents an agent's present discounted value of human wealth, and the transitory shock component (ξ_t) which reflects the potential risks that households may face in receiving their income payment during that period. Thus, household income can be expressed as:

$$y_t = p_t \xi_t W_t.$$

The level of permanent income for each household is subject to a stochastic process. In line with the labor income process described by Friedman 1957, I assume that this process follows a geometric random walk, which can be written as:

$$p_t = p_{t-1} \psi_t.$$

The white noise permanent shock to income with a mean of one is represented by ψ_t, which is a significant component of household income. The probability of receiving income during a given period is determined by the transitory component, which is modeled to reflect the potential risks associated with becoming unemployed. Specifically, if the probability of becoming unemployed is $\mho$, the agent will receive unemployment insurance payments of $\mu > 0$. On the other hand, if the agent is employed, which occurs with a probability of $1 - \mho$, the model allows for tax payments τ_t to be collected as insurance for periods of unemployment. The transitory component is then written as:

$$\xi_t = \begin{cases} \mu & \text{with probability } \mho, \\ (1 - \tau_t) l \theta_t & \text{with probability } 1 - \mho, \end{cases}$$

where l is the time worked per agent and the parameter θ captures the white noise component of the transitory shock.

1.3.2 Decision problem for households

Next, I present the baseline version of the household's optimization problem for consumption-savings decisions, assuming no ex-ante heterogeneity. In this case, each household aims to maximize its expected discounted utility of consumption $u(c) = \frac{c^{1-\rho}}{1-\rho}$ by solving the following:

$$\max \mathbb{E}_t \sum_{n=0}^{\infty} (\cancel{D}\beta)^n u(c_{t+n}).$$

Note that this setting follows a perpetual youth model of buffer stock savings, similar to the seminal work of Krusell and Smith 1998. To solve this problem, I use the Bellman equation, which means that the sequence of consumption functions $\{c_{t+n}\}_{n=0}^{\infty}$ associated with a household's optimal choice over a lifetime must satisfy[5]

$$\begin{aligned}
v(m_t) &= \max_{c_t} u(c_t(m_t)) + \beta \cancel{D} \mathbb{E}_t[\psi_{t+1}^{1-\rho} v(m_{t+1})] \\
&\text{s.t.} \\
a_t &= m_t - c_t(m_t), \\
k_{t+1} &= \frac{a_t}{\cancel{D}\psi_{t+1}}, \\
m_{t+1} &= (1 + r_t)k_{t+1} + \xi_{t+1}, \\
a_t &\geq 0.
\end{aligned}$$

1.4 Results

The estimation of the model proceeds in several steps. I begin by analyzing an infinite-horizon version of the model, in which households face uninsurable labor income risk and a borrowing constraint, and solve the consumption-saving problem using the endogenous grid method as implemented in `HARK`. When there is no heterogeneity in returns, I find the capital-to-output ratio necessary to match the capital-to-output target of $\frac{K}{Y} = 3$. Quantitative macroeconomic models calibrated to U.S. data commonly

[5] Here, each of the relevant variables is normalized by the level of permanent income ($c_t = \frac{C_t}{p_t}$, and so on), following the standard state-space reduction of the problem for numerical tractability.

target an annual capital-to-output ratio in the range of roughly 2.5 to 3.0. For example, Lee et al. 2021 calibrate their heterogeneous-agent model with financial frictions to match a capital-output ratio of 2.5, while Suen 2012 targets an annual capital-output ratio of 3.0 in a heterogeneous-agent model with preference heterogeneity.

When heterogeneity is present, there is a candidate distribution of returns and I simulate a large population of households forward to obtain the stationary wealth distribution. The parameters governing the return distribution are then chosen via Simulated Method of Moments (SMM) to minimize the distance between simulated and empirical wealth shares from the Survey of Consumer Finances. Following existing literature, the 20th, 40th, 60th, and 80th percentiles of the wealth distribution from the 2004 wave of the survey are used as empirical targets.

After establishing the results in the infinite-horizon setting, I extend the model to a life-cycle framework with age- and education-specific income profiles, mortality risk, and initial conditions. The model is then re-solved and re-estimated when heterogeneity is and is not present.

1.4.1 The infinite horizon setting

1.4.1.1 The model with no returns heterogeneity

To solve and simulate the model, I follow the calibration scheme described in table 1.1.

The solution of the model with no heterogeneity in returns (the R-point model) is to find the value for the rate of return R which minimizes the distance between the

Description	Parameter	Value	Source
Time discount factor	β	0.99^4	Den Haan et al. 2010
CRRA	ρ	1	Den Haan et al. 2010
Capital share	α	0.36	Den Haan et al. 2010
Depreciation rate	δ	0.025	Den Haan et al. 2010
Time worked per employee	ℓ	1/.09	Den Haan et al. 2010
Wage rate	W	2.37	Den Haan et al. 2010
Unempl. insurance payment	μ	0.15	Den Haan et al. 2010
Probability of survival	$\not D$	$(1 - 0.00625)^4$	Yields 40-year working life
Std. dev of $\log \theta_{t,i}$	σ_θ^2	0.010 x 4 x $\sqrt{4}$	Carroll 1992, Carroll et al. 2015
Std. dev of $\log \psi_{t,i}$	σ_ψ^2	0.010 x 4/11 x $\sqrt{4}$	Carroll 1992, Debacker et al. 2013, Carroll et al. 2015
Unemployment rate	$\mho$	0.07	Mean in Den Haan et al. 2010

Table 1.1: Parameter values (annual frequency) for the perpetual youth model.

models capital-to-ouput ratio $\frac{K}{Y}$ and the calibrated $\frac{K}{Y}$ ratio value of 3. The estimation procedure finds this optimal value to be $R = 1.0602$.

1.4.1.2 Incorporating heterogeneous returns

To estimate the model with heterogeneity in returns, I follow a procedure similar to the one outlined by Carroll et al. 2017. Specifically, I assume that different types of households have a time preference factor drawn from a uniform distribution over the interval $(\grave{R} - \nabla, \grave{R} + \nabla)$, where ∇ represents the level of dispersion. I then simulate the model to estimate $\grave{R}$ and ∇ so that the resulting wealth distribution aligns with the observed distribution in terms of inequality. Practically, I solve the following minimization problem:

$$\{\grave{R}, \nabla\} = \arg\min_{R,\nabla} \left(\sum_{i=20,40,60,80} (w_i(R,\nabla) - \omega_i)^2 \right)^{\frac{1}{2}}$$

subject to the constraint that the aggregate capital-to-output ratio in this model matches the calibrated value 3. Note that w_i and ω_i represent the porportion of total aggregate net worth held by the top i percent in the model and in the data, respectively. The estimation procedure yields the optimal values of $R = 1.0204$ and $\nabla = 0.06833$ which pin down the estimated uniform distribution. The Lorenz curve associated with the wealth distribution from the model and the SCF data are compared in figure 1.1.

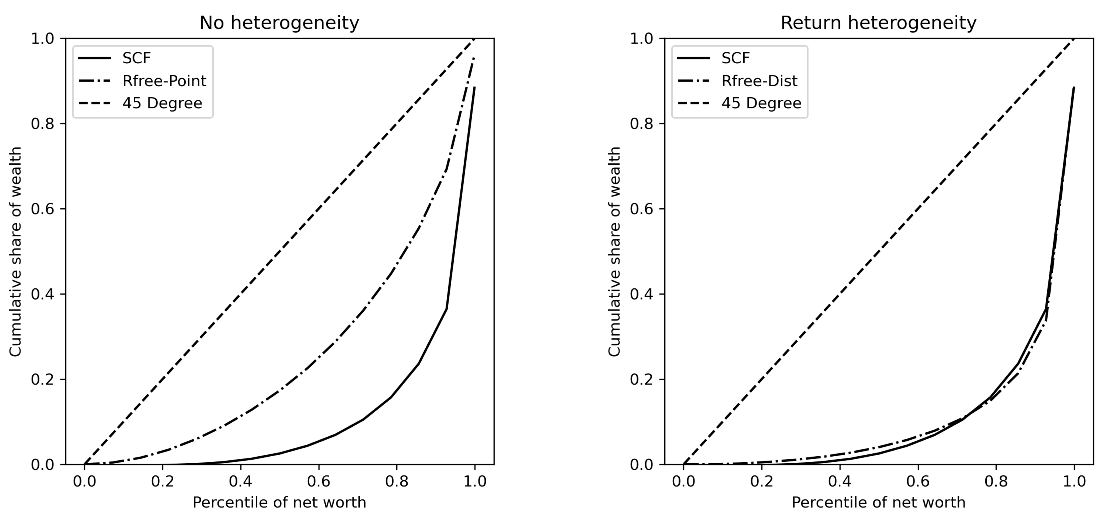


Figure 1.1: Comparison of R-Point and R-Dist Models.

The 45 degree line represents perfect equality. Lorenz curves further away from this line represent more unequal distributions of wealth. As we can see from the figure, the Lorenz curve generate by allowing for heterogeneous returns does a much better job at matching the shape of the SCF wealth distribution than when everyone earns the same return. Additionally, the following table compares the relevant moments of the empirical and simulated wealth distributions.

Table 1.2: Wealth Distribution (PY)

	0–20	**20–40**	**40–60**	**60–80**	**80–95**	**Top 5%**
Wealth share (data)	-0.002	0.011	0.044	0.118	0.255	0.574
Wealth share (no het.)	0.030	0.085	0.142	0.224	0.285	0.233
Wealth share (het.)	0.005	0.019	0.042	0.096	0.251	0.588

1.4.2 Incorporating life cycle dynamics into the model

Assumptions about the age and education level of households can have important implications for the income and mortality process of households. Next I extend the model to incorporate these life-cycle dynamics.

Households enter the economy at time t aged 24 years old with an education level $e \in \{D, HS, C\}$, an initial permanent income level $\mathbf{p}_0$, and a capital stock k_0. The life cycle version of household income is given by:

$$y_t = \xi_t \mathbf{p}_t = (1-\tau)\theta_t \mathbf{p}_t,$$

where $\mathbf{p}_t = \psi_t \bar{\psi}_{es} \mathbf{p}_{t-1}$ and $\bar{\psi}_{es}$ captures the age-education-specific average growth factor. Households that have lived for s periods have permanent shocks drawn from a lognormal distribution with a mean of 1 and a variance of $\sigma^2_{\psi s}$ and transitory shocks drawn from a lognormal distribution with a mean of $\frac{1}{\not\mho}$ and a variance of $\sigma^2_{\theta s}$ with probability $\not\mho = (1-\mho)$ and μ with probability $\mho$.

The normalized version of the age-education-specific consumption-saving problem for households is given by

$$
\begin{aligned}
v_{es}(m_t) &= \max_{c_t} u(c_t(m_t)) + \beta \not{D}_{es} \mathbb{E}_t[\psi_{t+1}^{1-\rho} v_{es+1}(m_{t+1})] \\
&\text{s.t.} \\
a_t &= m_t - c_t, \\
k_{t+1} &= \frac{a_t}{\psi_{t+1}}, \\
m_{t+1} &= (1 + r_t)k_{t+1} + \xi_{t+1}, \\
a_t &\geq 0.
\end{aligned}
$$

The additional parameters necessary to calibrate the life-cycle version of the model are given in table 1.3. The age-education dependent mean income levels come from Cagetti 2003. The permanent and transitory shock variances come from Sabelhaus and Song 2010. The age-education dependent mortality rates come from Brown et al. 2007.

The estimation yields an optimal value of $R = 1.0431$ for the R-point model in this setting. For the R-dist model in the life-cycle setting, $R = 1.002$ and $\nabla = 0.0953$. Note the improved fit to the data, as illustrated in figure 1.2.

Table 4 compares the relevant moments of the empirical and simulated wealth distributions again, this time for the setting with life-cycle dynamics..

Description	Parameter	Value
Population growth rate	N	0.0025
Technological growth rate	Γ	0.0037
Rate of high school dropouts	θ_D	0.11
Rate of high school graduates	θ_{HS}	0.55
Rate of college graduates	θ_C	0.34
Labor income tax rate	τ	0.0942

Table 1.3: Parameter values (annual frequency) for the life-cycle model.

Table 1.4: Wealth Distribution (LC)

	0–20	**20–40**	**40–60**	**60–80**	**80–95**	**Top 5%**
Wealth share (data)	-0.002	0.011	0.044	0.118	0.255	0.574
Wealth share (no het.)	0.018	0.056	0.110	0.207	0.321	0.287
Wealth share (het.)	0.003	0.014	0.037	0.115	0.358	0.473

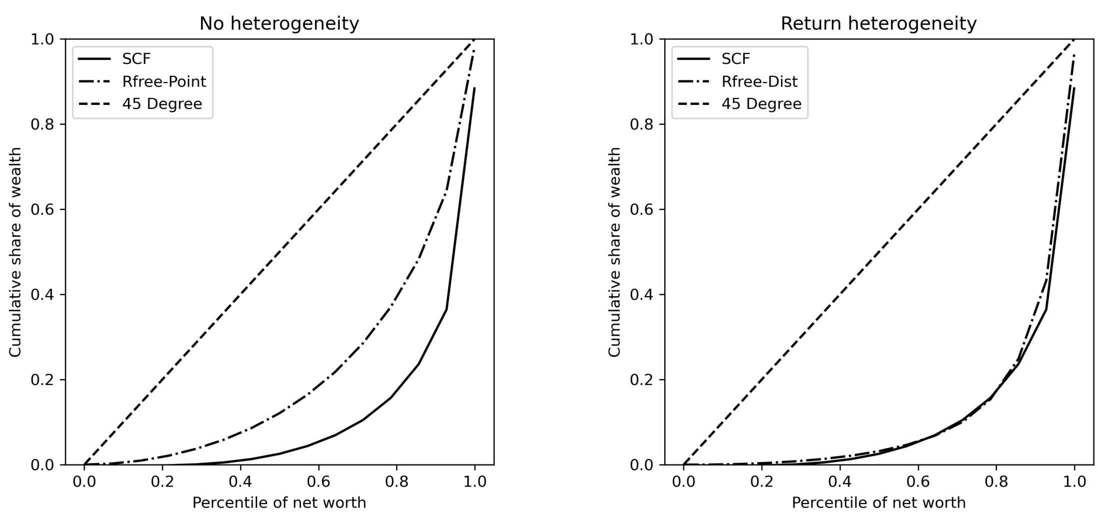


Figure 1.2: Comparison of R-Point and R-Dist Models in the Life-Cycle Setting.

1.4.3 Untargeted moments

Introducing an additional source of (ex-ante) heterogeneity beyond labor income uncertainty into the representative agent framework allows the simulated distribution of wealth to better match moments of the empirical wealth distribution. But this is an expected result when allowing for an additional dimension of complexity in a model. So, although the model with heterogeneous returns does a good job of matching the given lorenz targets, we need another way to assess the model's performance.

1.4.3.1 The marginal propensity to consume

The literature on heterogeneous agents consistently shows that models with a precautionary savings motive will lead to a dispersion in the marginal propensity to consume (MPC) across households. This feature is key to why the HA framework is able to produce estimates of the aggregate MPC that are close to empirical measurements, such as those measured in Jappelli and Pistaferri 2014b. [6]

In the infinite horizon setting, introducing heterogeneity changes the model's implications for both the aggregate MPC and its dispersion across households. When all households earn the same return on their savings, the aggregate MPC is 11.3%, rising to 27.5% when heterogeneous returns are introduced. We decompose this by looking at average MPCs by wealth decile in figure 1.3.

As Figure 3 illustrates, MPCs are much higher when households earn different returns. When heterogeneous returns are present, the model replicates the extreme

[6] Jappelli and Pistaferri 2014b estimate an MPC of about 48% and document significantly larger MPCs for households with low cash-on-hand.

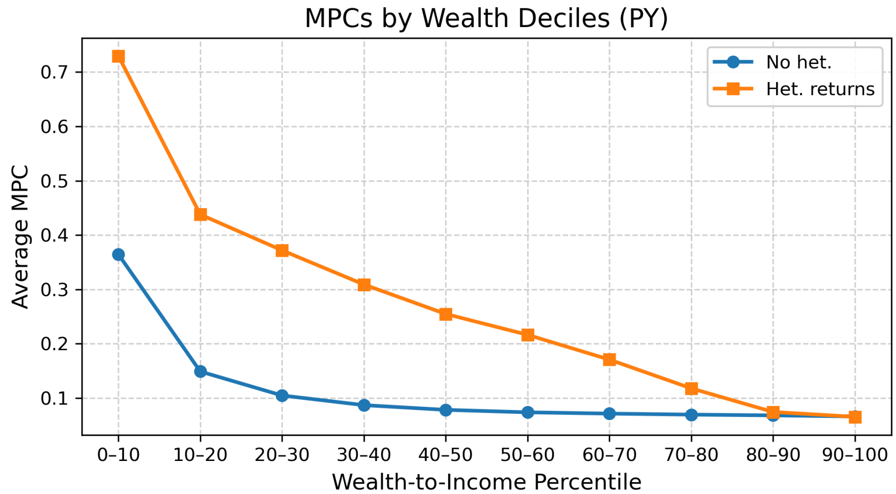


Figure 1.3: Infinite-horizon marginal propensities to consume.

wealth concentration observed in the data. Poor households are not just somewhat poorer—they are dramatically poorer, pushing them into the hand-to-mouth region where consumption tracks income almost one-for-one.

Because incorporating life-cycle dynamics generates a more realistic distribution of wealth (even without heterogeneous returns), MPCs are slightly higher here: an aggregate MPC of 12.6% without heterogeneity and 27.8% with heterogeneous returns (see Figure 1.4). However, the stark difference between the model specifications with (orange line) and without heterogeneity (blue line in Figure 4) does persist.

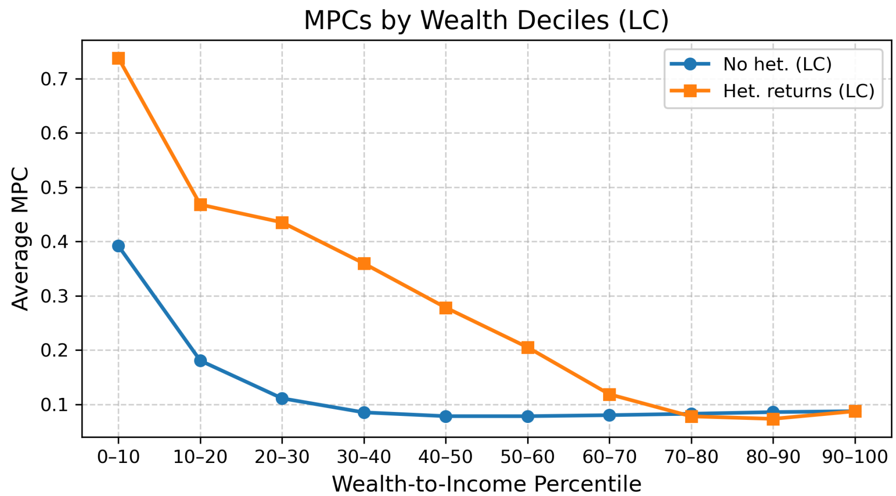


Figure 1.4: Life-cycle marginal propensities to consume.

1.4.3.2 Wealth shares by age cohort

Next, I include age-dependent wealth moments from the same wave of the SCF to serve as another set of untargeted moments in figure 1.5.

Empirical Lorenz Shares by Age (2004)

age	20th	40th	60th	80th
25-30	-0.0723	-0.0657	-0.0266	0.1099
30-40	-0.008	0.0054	0.057	0.1813
40-50	-0.0001	0.0187	0.0776	0.2178
50-60	0.0018	0.0215	0.0766	0.2126
60-70	0.0011	0.0188	0.0726	0.2081

Figure 1.5: Empirical Lorenz Curve Targets from the 2004 SCF.

Figures 6 and 7 present the simulated version of the untargeted moments for the model without heterogeneity 1.6, and with heterogeneity 1.7. The age-dependent Lorenz targets that arise from the model again fit the data much better when returns heterogeneity is present versus when it is absent.

Simulated Lorenz Shares by Age

age	20th	40th	60th	80th
25-30	0.0449	0.1342	0.2773	0.5253
30-40	0.0421	0.1426	0.2984	0.5325
40-50	0.0465	0.1416	0.2875	0.5116
50-60	0.0506	0.1452	0.2878	0.5069
60-70	0.0466	0.1352	0.2706	0.4837

Figure 1.6: Simulated Untargeted Moments without Heterogeneity (R-point).

Simulated Lorenz Shares by Age

age	20th	40th	60th	80th
25-30	0.0229	0.0951	0.2203	0.4458
30-40	0.01	0.052	0.1423	0.3432
40-50	0.0057	0.0293	0.088	0.266
50-60	0.0044	0.0216	0.0695	0.2397
60-70	0.0039	0.0195	0.0662	0.2292

Figure 1.7: Simulated Untargeted Moments with Heterogeneity (R-dist).

1.5 Wealth versus Capital Income Tax

We've discussed the role of heterogeneous returns in generating substantial wealth inequality in the simulated model. Heterogeneity in returns also has interesting policy implications. Guvenen et al. 2023 documents that, when individuals earn different returns on their assets, a capital income tax imposes a larger burden on those more efficient with their capital. In contrast, a wealth tax imposes a larger burden on those who are unproductive with their capital. This redistributive effect of the wealth tax leads to higher aggregate productivity, output, and ultimately larger welfare gains than the capital income tax. Following this reasoning, in this section I consider the quantitative effects of each tax scheme on the distribution of wealth when heterogeneous.

I start with an economy in which the distribution of returns matches wealth moments measured in the data. The question then becomes: how does the distribution of wealth change if a revenue equivalent tax rate is applied to every household? A capital income tax will clearly decrease wealth inequality, since only households that earn capital income will see a tax but those with zero or negative returns will see a capital tax of zero. Consequently, less wealthy households will hold a higher share of the aggregate wealth than when the capital tax is not present.

The effects of the wealth tax are less obvious, since all households hold some wealth and thus will be taxed accordingly. In this setting, the wealth tax enters the household budget constraint as follows:

$$m_{t+1} \;=\; \Big(1 + r_t - \tau_w\Big)\, k_{t+1} + \xi_{t+1}.$$

Similarly, for the capital income tax,

$$m_{t+1} \;=\; \Big(1 + (1 - \tau_{ci})\, r_t\Big)\, k_{t+1} + \xi_{t+1}.$$

I compute aggregate income as GDP in this setting, and then find the wealth tax rate and the capital tax rate that would raise tax revenues equal to 1% of GDP. I follow this procedure for the estimated distribution of returns both for the infinite horizon and the life-cycle setting.

1.5.1 Effects on the wealth distribution

Next, I apply the tax schemes to each households and compare the resulting wealth distributions for both cases before and after each policy is implemented. Table 5 presents the results of applying each of the tax schemes in the infinite horizon 1.5 version of the model.

	Lorenz points			
Tax scheme	20%	40%	60%	80%
None	.49%	2.4%	6.7%	16.3%
Wealth $\tau_w = .33\%$	.54%	2.8%	7.8%	18.7%
Capital income $\tau_{ci} = 4.96\%$	.59%	2.97%	7.8%	18.7%

Table 1.5: Tax policies in the infinite horizon setting.

Applying the tax schemes increases the share of wealth held by households at each chosen percentile, reflecting the redistributive effect of taxation on extreme wealth. Although the effect is mininal in both cases, the capital income tax more effectively reduces wealth inequality than the flat wealth tax. This result is consistent with the extant literature: when heterogeneous returns are present, a capital income tax disproportionately burdens households who use capital most productively, thereby reducing the concentration of wealth.

Next, I apply the tax schemes in the life-cycle 1.6 version of the model and obtain similar results: the capital income tax has a larger effect on reducing wealth inequality than the wealth tax that raises the same level of tax revenue. Table 6 presents the results, which that , the effects in the life-cycle model are less pronounced than in the infinite horizon model.

	Lorenz points			
Tax scheme	20%	40%	60%	80%
None	.34%	1.8%	5.4%	16.98%
Wealth $\tau_w = .36\%$	.35%	1.8%	5.6%	17.1%
Capital income $\tau_{ci} = 7.1\%$	.37%	1.9%	5.9%	17.9%

Table 1.6: Tax policies in the life cycle setting.

1.5.2 Welfare effects of the tax policies

As an additional comparison of the two tax policies when heterogeneous returns are present, I follow Guvenen et al. 2023 and examine newborns' lifetime utility, where a newborn's return type is drawn from the estimated distribution that best fits the SCF wealth data. I summarize welfare using a consumption-equivalent measure: the percentage change in consumption under one policy that would make a newborn, before learning their return type, indifferent between the wealth tax and the capital income tax. I then decompose this comparison by estimated return type.

For this analysis, I retain my calibrated value of $\rho = 1$ such that $u(c) = \log c$. The consumption-equivalent change Δ between a wealth tax (WT) and a capital income tax (CIT) is defined by equating the lifetime utilities under each policy. Thus,

$$1 + \Delta = \exp\left(\frac{V^{CIT} - V^{WT}}{S}\right),$$

where $S = \sum_{t=0}^{\infty}(\beta \cdot \not{D})^t$, and V^j is the lifetime value of a newborn under regime $j \in \{WT, CIT\}$. The parameter Δ therefore represents the *constant percentage increase in consumption each period under the wealth tax regime that would make a newborn as well off as under the capital income tax regime.* Equivalently, if $\Delta > 0$, then the newborn prefers the capital income tax to the wealth tax.

Table 7 presents the welfare effects of the tax policies for both the infinite horizon and life-cycle settings. As a baseline for comparison, the table also includes the expected lifetime utility of the original regime with no tax policy present.

Table 1.7: Expected Consumption-Equivalent Welfare Changes from Tax Reform

	Infinite horizon	**Life-cycle**
WT vs CIT	0.20%	0.15%
WT vs No Tax	0.85%	0.35%
CIT vs No Tax	0.65%	0.20%

Notes: Entries are consumption-equivalent (CE) welfare changes, Δ, expressed as percent changes. For example, WT vs CIT is the percentage increase in consumption under the wealth tax required to make a newborn indifferent to the capital income tax. Positive values therefore indicate that the right-hand policy yields higher newborn lifetime welfare than the left-hand policy.

Results from Table 7 show that the no-tax regime is preferred to both tax policies.

This is expected because the analysis is partial equilibrium and therefore abstracts from any benefit associated with government spending of the tax revenue. Relative to the no-tax benchmark, both tax instruments reduce expected lifetime utility, but the wealth tax is more costly than the capital income tax in consumption-equivalent terms. In the infinite-horizon case, for example, a newborn under the wealth tax would require roughly a 0.2% increase in consumption each period to be as well off as under the capital income tax. This distinction from Guvenen et al. 2023, where the wealth tax is preferred in welfare terms, reflects the mechanism omitted here: my framework does not allow taxes to affect aggregate productivity, output, or consumption. Accordingly, the welfare comparison in this section should be interpreted as a measure of the direct utility cost of each tax regime, not as a full general-equilibrium welfare ranking.

I next decompose these welfare effects by realized return type in the infinite-horizon setting.

The per-type results clarify why expected welfare for a newborn is higher under the capital income tax than under the wealth tax. For the lower six return types, the consumption-equivalent measure is positive, meaning that these households would require compensation under the wealth tax to be as well off as they would be under the capital income tax. The intuition is straightforward. For low-return households, and especially for those with returns below one, capital-income taxation is small or zero because there is little or no taxable capital income. A wealth tax, by contrast, lowers the gross return on assets regardless of whether the household's return is high, low, or even below one. Only the highest return type prefers the wealth tax in these results.

Table 1.8: Per-Type Consumption-Equivalent Welfare Change and Baseline Return (WT vs CIT)

	Type 1	**Type 2**	**Type 3**	**Type 4**	**Type 5**	**Type 6**	**Type 7**
Baseline R (gross)	0.9618	0.9813	1.0009	1.0204	1.0399	1.0594	1.0790
CE Δ (WT vs CIT, %)	0.23%	0.27%	0.34%	0.32%	0.30%	0.24%	-0.31%

Notes: CE entries are consumption-equivalent welfare changes (pmv-weighted within type), expressed as percent. Positive values indicate that capital income taxation is preferred to wealth taxation for that return type. Baseline R values are pre-tax gross returns by type (low $\rightarrow$ high).

The life-cycle setting produces the same qualitative pattern, shown in Table 9. Here I further decompose the effects by comparing return types within education groups.

Table 1.9: Per-Education Per-Type Consumption-Equivalent Welfare Change and Baseline Return (WT vs CIT)

		Type 1	**Type 2**	**Type 3**	**Type 4**	**Type 5**	**Type 6**	**Type 7**
NoHS	Baseline R (gross)	0.9200	0.9472	0.9744	1.0016	1.0289	1.0561	1.0833
	CE Δ (WT vs CIT, %)	0.118%	0.143%	0.182%	0.264%	0.340%	0.135%	-0.124%
HS	Baseline R (gross)	0.9200	0.9472	0.9744	1.0016	1.0289	1.0561	1.0833
	CE Δ (WT vs CIT, %)	0.117%	0.142%	0.184%	0.271%	0.332%	0.115%	-0.108%
College	Baseline R (gross)	0.9200	0.9472	0.9744	1.0016	1.0289	1.0561	1.0833
	CE Δ (WT vs CIT, %)	0.116%	0.142%	0.184%	0.267%	0.309%	0.098%	-0.106%

Notes: CE entries are consumption-equivalent welfare changes (pmv-weighted within type), expressed as percent. Positive values indicate that capital income taxation is preferred to wealth taxation for that return type. Baseline R values are pre-tax gross returns by type (low $\rightarrow$ high) at $t = 0$.

For each education group, the consumption-equivalent measure is positive for six of the seven return types. Thus, within every education category, the lower six return types would need compensation under the wealth tax to be as well off as they would be under the capital income tax. The same economic logic applies here as in the infinite-horizon model: the capital income tax is less costly for households that are less productive with a unit of capital, whereas the wealth tax reduces returns more uniformly across the distribution.

1.6 Extension: Mechanism for Returns Heterogeneity

We've discussed the main contributions of this paper. From here on, we will discuss extensions of the model which arise naturally in the discussion of the role of heterogeneous returns in explaining wealth inequality. The first extension is regarding a potential source of returns heterogeneity in the model.

Several potential explanations have been proposed for the persistent component of returns heterogeneity. One commonly cited factor is that some individuals are business owners, and variability in their entrepreneurial talent creates greater heterogeneity in labor income than standard HA models assume. Cagetti and De Nardi 2006 and Cagetti and De Nardi 2009 are notable examples of studies that explicitly model entrepreneurial talent as a component of the consumption-saving problem for households and assess the ability of such models to match empirical wealth moments.

Although modeling differences in entrepreneurial talent as variability in labor market productivity improves the model's ability to match wealth moments at the upper tail of the distribution, it does not adequately explain mechanism considered here. This modeling choice will result in households (firms) with high levels of wealth (capital) earning lower rates of return, and vice versa for households (firms) with low levels of wealth (capital). As shown by Fagereng et al. 2020, both average returns and the persistent, idiosyncratic components of returns increase with wealth, reflecting clear scale dependence.

Another explanation, closer to my mechanism but still distinct is financial literacy or sophistication. Lusardi and Mitchell 2014 survey models that explicitly allow households to make costly investments in financial literacy, granting them access to an investment technology offering a higher average return. Lusardi et al. 2017 show that incorporating such endogenous financial accumulation into the standard consumption-saving framework enables models to match wealth moments particularly well, this time through the returns channel rather than the labor income channel associated with entrepreneurial talent.

As I am particularly interested in a setting with a single, safe asset available, I examine another alternative source of heterogeneity in the rate of return by drawing on literature documenting substantial variation in the banking sector, specifically the rates offered to depositors.

1.6.1 Deposit rates and the sensitivity of deposit levels to changes in the market interest rate

Deuflhard et al. 2018 among others, document substantial heterogeneity in rates of return earned by depositors. To incorporate this finding in the standard consumption-saving framework, I must identify potential sources of heterogeneity in the banking sector.

I begin by considering the balance sheet of banks and the problem they face in optimally choosing the rate to offer on deposits. In a simple setting, they accept deposits at an offered deposit rate and then hold reserves within the central banking system, earning the market interest rate. This discrepancy between the offered deposit rate and the market interest rate leaves room for banks to earn a profit.

Empirical evidence from the U.S. shows that the level of deposits at a given bank will change due to exogenous changes in the federal funds rate. In fact, Drechsler et al. 2017 propose a clear transmission channel for monetary policy: changes in the federal funds rate may lead banks to widen the interest spread they charge on deposits, which causes deposits to flee the bank. They find a strong, negative relationship between changes in the federal funds rate and the growth rate of deposits, with a 100-basis-point increase in the federal funds rate leading to higher deposit outflows in bank branches in more concentrated markets relative to those in less concentrated markets.[7]

Thus, variation in the strength of this transmission channel across banks can be viewed as a potential source of heterogeneity in the banking sector, as certain market characteristics would make the level of deposits offered by some banks less sensitive to changes in the federal funds rate than other banks. For example, Sarkisyan and Viratyosin 2021 make a distinction between *local* and *globally integrated* banks, and show that "global banks lose much more deposits relative to local banks in response to unexpected changed in the federal funds rate". Similarly, Adrien d'Avernas et al. 2024 document heterogeneity in deposit rates between larger and smaller banks.

Given these empirical findings, I extend the standard HA model describing household

[7] Branches in "more concentrated markets" operate in local deposit markets where a few banks hold large market shares.

consumption-saving decisions by explicitly modeling a banking sector. In this setting, banks will each solve a similar profit-maximization problem regarding accepting deposits at an offered deposit rate and holding reserves that earn the market interest rate. The key distinction between banks in the model is how sensitive the level of deposits is to changes in the market interest rate, in line with prior empirical evidence. This analysis will help explain how heterogeneity in returns may arise for households that are otherwise the same.

1.6.2 Model of heterogeneous deposit rates

I consider a small, open economy in which banks and households are the optimizing agents. This analysis operates in partial equilibrium as the world interest rate is taken as given. I present a simple framework for determining banks' optimal deposit rates. Assuming a Cobb-Douglas aggregate production function, I derive the marginal product of capital (less depreciation) consistent with the capital-to-output ratio and its empirical counterpart. This effective interest rate serves as the world or "market" interest rate and, together with the estimated distribution of returns, can be used to estimate the elasticities of foreign deposits to the deposit rates for each of the banks in the model.

1.6.2.1 Assumptions regarding the banking sector

In this setting, the economy features a continuum of banks, identical in all respects except for the elasticity of the level of deposits to changes in the market interest rate.

The model is static: each bank sets a deposit rate at the beginning of the horizon based solely on the market interest rate and its given elasticity, without the ability to expand its depositor base. [8].

Households, in turn, do not endogenously choose which banks to do business with but instead are assigned a bank at birth and remain with it until death. In this simplified setting, the banking sector merely replaces the assignment of idiosyncratic rates of returns over the time horizon in the standard model.

1.6.2.2 Decision problem for banks accepting deposits

The sole distinction between banks in this model is the sensitivity of their level of deposits to changes in the market interest rate, which is indexed by ε_i. This heterogeneity in deposit elasticity generates variation in returns in the model. I present the decision problem for banks using a simplified version of the framework found in Paul and Ulate 2024.

Let R^m be the market rate of return, R^d be the rate of return offered on deposits by a bank, and $S(R^d, R^m)$ be the level of deposits held at a given bank.

Banks solve

$$\max(R^m - R^d) \cdot S(R^d, R^m)$$

subject to

$$S(R^d, R^m) = A \left(\frac{R^d}{R^m} \right)^{\varepsilon}.$$

[8]For example, compare a bank in a suburb area of Montana (local) versus a bank near downtown Houston, Texas (globally integrated).

Importantly, the first-order condition implies that the optimal deposit rate for the i-th bank is

$$R_i^d = \frac{\varepsilon_i}{1+\varepsilon_i} R^m.$$

This relationship plays a key role in the model: once I calibrate the model for a particular value of R^m, estimating a uniform distribution of returns using the simulated method of moments will imply a corresponding distribution of elasticities (i.e., the distribution that minimizes the distance between simulated and empirical Lorenz wealth moments). The seven discretized points therefore capture seven different deposit rates offered, resulting in varying elasticities among seven different bank types in the model. From the expression above, banks with higher values of ε_i must set R_i^d closer to R^m.

1.6.3 The implied distribution of bank heterogeneity

Having estimated the distribution of returns that best matches the observed wealth moments, I can use these results, along with the assumptions regarding the bank's decision problem to back out an implied distribution of ε. This parameter reflects how strongly each bank's deposit base responds to market interest rate changes. In both this simplified setting and the transmission channel empirically documented by Drechsler et al. 2017, differences in these sensitivities ultimately leads to differences in the deposit rates offered by banks.

I begin by assuming that a Cobb-Douglas aggregate production function, such that the marginal product of capital can be written as $\alpha\frac{Y}{K}$. With calibrated values of

$\delta = .025$ and $\alpha = .36$, and a capital-to-output ratio 3, this setting has an effective interest rate of $R^m = 1.095$, which can be used as the market interest rate.

Since the model with heterogeneity (i.e., the R-dist model) has seven estimated points for the uniform distribution, the implied estimated points for ε can be uniquely determined by the expression

$$\varepsilon_i = \frac{R_i^d}{R^m - R_i^d}.$$

Table 10 shows, for the infinite horizon and life-cycle settings, the estimated returns distribution that best matches 2004 SCF data on net worth and the corresponding implied elasticities.

Table 1.10: Estimated Returns and Implied Elasticities

Infinite Horizon		Life-cycle	
Estimated returns	Implied elasticities	Estimated returns	Implied elasticities
0.962	7.222	0.920	5.256
0.981	8.634	0.947	6.408
1.001	10.632	0.974	8.081
1.020	13.676	1.002	10.728
1.040	18.877	1.029	15.553
1.059	29.786	1.056	27.127
1.079	67.239	1.083	92.489

1.6.3.1 Interpreting the implied distribution of elasticities

Next, I assess the model's performance by comparing its implications to established empirical findings on bank deposit sensitivities to changes in the federal funds rate.

The implied distribution of elasticities derived from my estimation method can be directly compared to those empirical estimates to assess the validity of my model. In addition, I include wealth shares by age cohort as a set of untargeted moments to further assess the model's fit.

By defining the deposit function as $S(\cdot) = A\left(\frac{R^d}{R^m}\right)^{\varepsilon}$ the parameter ε can be clearly interpreted as the elasticity of deposits to changes in the market interest rate. Formally,

$$-\varepsilon = \frac{\partial \ln S(\cdot)}{\partial \ln R^m}.$$

The elasticity parameter therefore indicates how a one-percentage-point change in the market interest rate translates into a percentage change in deposits. This formulation enables me to directly compare the implied elasticities following the estimation procedure to the empirical evidence on the transmission channel described by Drechsler et al. 2017 regarding the relationship between the federal funds rate and the level of deposits at banks. For example, Genay and Halcomb 2004 find that a 1% change in the Fed funds rate leads to about a 3% to 4% change in the level of deposits, depending on the size of the bank.

The returns heterogeneity required to match observed wealth inequality using only safe assets (i.e., bank deposits) produces vastly overstated elasticities for the banking sector. This outcome illustrates a limitation of the model: banks must raise deposit rates to attract depositors when the federal funds rate increases, regardless of their size. Consequently, there will be less variation in the optimal deposit rates offered across the banking sector, and banks that do not offer competitive deposit rates will likely find that their depositors switch to other safe investment technologies like money market funds. Since my model does not account for the number of banks in the economy or returns heterogeneity arising from choices between safe assets, it is not too surprising that deposit rate elasticities are not well matched in this setting. That said, the ability of the model to back out a distribution of elasticities under the given assumptions is still useful.

1.7 Extension: Estimating a Lognormal Distribution of Returns

Using Norwegian population data, Fagereng et al. 2020 document a persistent component to individual returns, illustrated by a nonuniform distribution of estimated fixed effects in the return to net worth for individuals. Motivated by this finding, I rerun the estimation of my model under the assumption that returns are instead lognormally distributed across households. The main findings of the paper are robust to this modification.

1.7.1 Comparing the simulated wealth distributions

Since the model without heterogeneous returns is the same regardless of the distributional assumptions, in Figure 8 I show the results for the infinite horizon and life cycle models with heterogeneous returns, lognormally distributed across households.

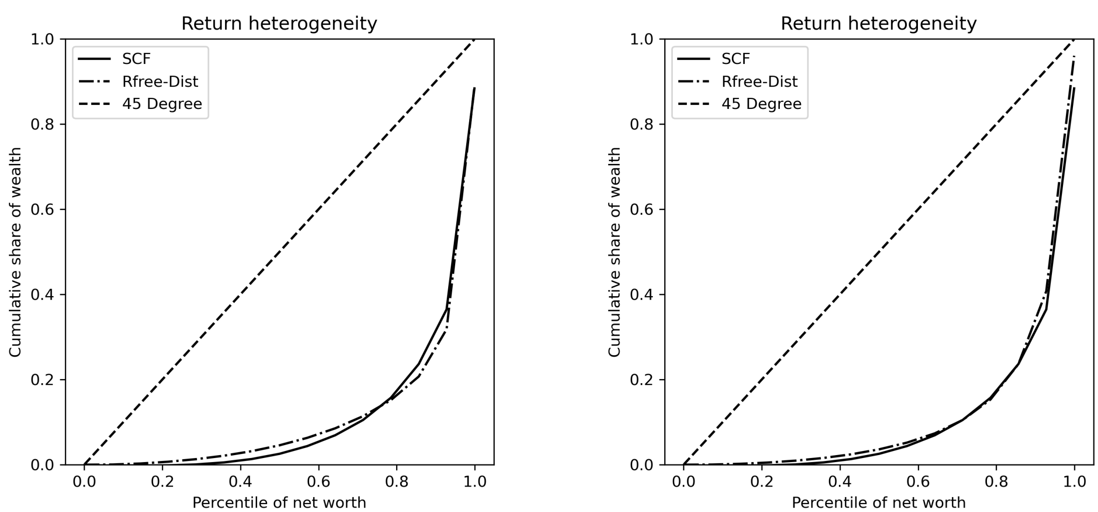


Figure 1.8: Comparison of PY (left) vs LC (right) R-Dist Models.

It is clear from the figure that the results from the infinite horizon setting persist here: the life cycle setting is better at matching the observed wealth distribution than the infinite horizon setting, but they both match wealth moments especially well. Again, I include the wealth moments for each of the models and the data in Table 11.

Table 1.11: Wealth Distribution (Lognormal Returns)

	0–20	**20–40**	**40–60**	**60–80**	**80–95**	**Top 5%**
Wealth share (data)	-0.002	0.011	0.044	0.118	0.255	0.574
Wealth share (PY)	0.006	0.021	0.044	0.088	0.225	0.616
Wealth share (LC)	0.004	0.016	0.039	0.107	0.335	0.498

1.7.2 Untargeted moments

The aggregate MPC is 25.7% and is nearly identical in the the life-cycle setting at 26%. Figure 9 presents a breakdown of average MPCs by wealth deciles for both settings. Again, the MPC's are generally higher in the life-cycle setting due to its superior ability to match lower moments of the wealth distribution.

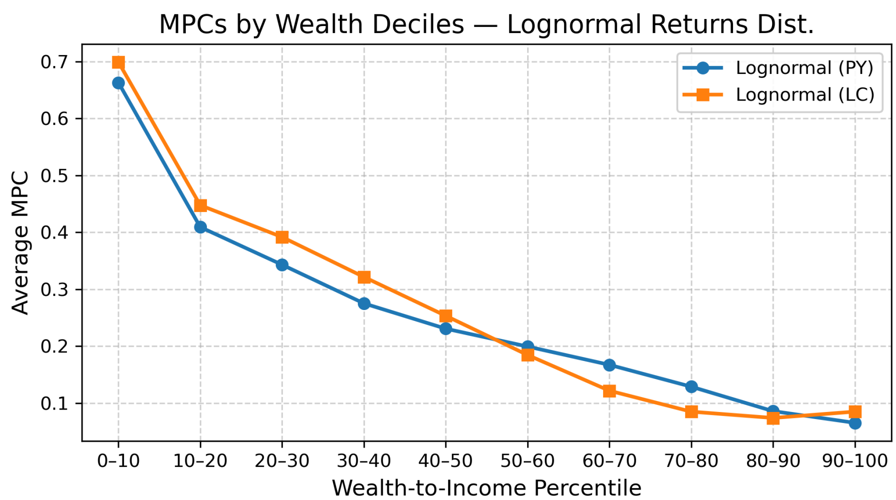


Figure 1.9: Infinite-horizon and Life-cycle marginal propensities to consume.

For the second set of untargeted moments, I include the cumulative wealth shares by age for this case of an estimated lognormal returns distribution. Again, these moments align more closely with the empirical data than the case with no heterogeneity, particularly for the wealth shares of older cohorts.

Empirical Lorenz Shares by Age (2004)

age	20th	40th	60th	80th
25-30	-0.0723	-0.0657	-0.0266	0.1099
30-40	-0.008	0.0054	0.057	0.1813
40-50	-0.0001	0.0187	0.0776	0.2178
50-60	0.0018	0.0215	0.0766	0.2126
60-70	0.0011	0.0188	0.0726	0.2081

Figure 1.10: Empirical Lorenz Curve Targets from the 2004 SCF.

Simulated Lorenz Shares by Age

age	20th	40th	60th	80th
25-30	0.0261	0.1017	0.2285	0.4545
30-40	0.0126	0.0595	0.1531	0.3432
40-50	0.0074	0.0349	0.097	0.2547
50-60	0.0059	0.0272	0.0797	0.2296
60-70	0.0054	0.0254	0.076	0.2199

Figure 1.11: Simulated Untargeted Moments with Heterogeneity (R-dist).

1.7.3 Tax implications

I also consider the tax implications of the model in this setting. I begin by comparing the effects of each of the tax policies on wealth inequality for both the infinite horizon and the life-cycle cases. The same relationship holds here: each effect of the tax is small (since the policies are revenue-equivalent and only raise 1% of aggregate labor income), but the capital income tax is more effective at reducing wealth inequality more.

	Lorenz points			
Tax scheme	20%	40%	60%	80%
None	0.60%	2.75%	7.27%	16.28%
Wealth $\tau_w = 0.34\%$	0.67%	3.14%	8.38%	18.86%
Capital income $\tau_{ci} = 5.08\%$	0.73%	3.37%	8.94%	20.02%

Table 1.12: Tax policies in the infinite horizon setting (lognormal returns across households).

	Lorenz points			
Tax scheme	20%	40%	60%	80%
None	0.42%	2.05%	5.98%	16.73%
Wealth $\tau_w = 0.33\%$	0.43%	2.12%	6.14%	16.86%
Capital income $\tau_{ci} = 4.97\%$	0.46%	2.23%	6.46%	17.72%

Table 1.13: Tax policies in the life-cycle setting (lognormal returns across households).

For the welfare effects in this setting, I estimate the consumption-equivalent compensation required under the wealth tax to make a newborn indifferent to the capital income tax. The conclusion is the same as in the baseline specification: before learning their type, a newborn prefers the capital income tax to the wealth tax because expected lifetime utility is higher under the capital income tax regime.

Table 1.14: Expected Consumption-Equivalent Welfare Changes from Tax Reform (Lognormal Returns)

	Infinite horizon	**Life-cycle**
WT vs CIT	0.22%	0.18%
WT vs Original	0.78%	0.36%
CIT vs Original	0.56%	0.18%

Notes: Entries are consumption-equivalent (CE) welfare changes, Δ, expressed as percent changes under the assumption that individual returns follow a lognormal distribution across households. WT vs CIT is the percentage increase in consumption under the wealth tax required to make a newborn indifferent to the capital income tax. Positive values therefore indicate that the right-hand policy is preferred to the left-hand policy.

These welfare effects can again be decomposed by return type for the infinite-horizon case and by education-return type in the life-cycle scenario. The qualitative pattern is unchanged: the lower six return types prefer the capital income tax, while only the highest return type prefers the wealth tax. A notable difference across distributional assumptions is that the estimated lognormal distribution places fewer types below a gross return of one than the uniform distribution does. Despite this difference, the welfare ranking remains robust.

Table 1.15: Per-Type Consumption-Equivalent Welfare Change and Baseline Return (WT vs CIT, Lognormal Returns)

	Type 1	**Type 2**	**Type 3**	**Type 4**	**Type 5**	**Type 6**	**Type 7**
Baseline R (gross)	0.976	1.001	1.014	1.026	1.038	1.053	1.079
CE Δ (WT vs CIT, %)	0.267%	0.342%	0.329%	0.317%	0.308%	0.304%	-0.326%

Notes: CE entries are per-type consumption-equivalent welfare changes (pmv-weighted within type), expressed as percent. Positive values indicate that the capital income tax is preferred to the wealth tax for that return type. Baseline R values are pre-tax gross returns (low $\rightarrow$ high) under the lognormal returns specification.

Table 1.16: Per-Education Per-Type Consumption-Equivalent Welfare Change and Baseline Return (WT vs CIT, Lognormal Returns)

		Type 1	**Type 2**	**Type 3**	**Type 4**	**Type 5**	**Type 6**	**Type 7**
NoHS	Baseline R (gross)	0.9363	0.9711	0.9910	1.0084	1.0260	1.0471	1.0865
	CE Δ (WT vs CIT, %)	0.132%	0.176%	0.225%	0.275%	0.335%	0.292%	-0.114%
HS	Baseline R (gross)	0.9363	0.9711	0.9910	1.0084	1.0260	1.0471	1.0865
	CE Δ (WT vs CIT, %)	0.131%	0.177%	0.229%	0.283%	0.332%	0.250%	-0.098%
College	Baseline R (gross)	0.9363	0.9711	0.9910	1.0084	1.0260	1.0471	1.0865
	CE Δ (WT vs CIT, %)	0.131%	0.177%	0.228%	0.276%	0.311%	0.218%	-0.096%

Notes: CE entries are consumption-equivalent welfare changes (pmv-weighted within type), expressed as percent. Positive values indicate that capital income taxation is preferred to wealth taxation for that return type. Baseline R values are pre-tax gross returns by type (low $\rightarrow$ high) at $t = 0$ under the lognormal returns specification.

1.7.4 Mechanism for bank heterogeneity

Table 17 captures the estimated returns distribution that best matches the 2004 SCF data on net worth and the corresponding implied elasticities for both the infinite horizon and life-cycle settings.

Table 1.17: Estimated Returns and Implied Elasticities (Lognormal)

Infinite Horizon		Life-cycle	
Estimated returns	Implied elasticities	Estimated returns	Implied elasticities
0.976	8.231	0.936	5.898
1.001	10.615	0.971	7.838
1.014	12.602	0.991	9.532
1.026	14.963	1.008	11.640
1.038	18.366	1.026	14.874
1.053	24.922	1.047	21.857
1.079	68.242	1.086	127.378

1.8 Conclusion

This paper explores the ability of macroeconomic models to generate a distribution of wealth with substantial inequality by the estimation of a calibrated consumption-saving model that allows for heterogeneous returns. Consistent with other deviations from the representative agent framework, I find that the ex-ante heterogeneity in rates of return needed to match wealth moments compares well with empirical estimates of the returns to net worth found by recent work.

Unlike much of the existing literature linking persistent return heterogeneity to factors such as entrepreneurial ability or financial sophistication, I focus more on heterogeneity in deposit rates across banks. I incorporate related literature in the standard HA framework under a simple but realistic assumption that many households remain “stuck” with the bank in or around their neighborhood. That bank makes complex financial decisions that ultimately affect households through the channel of varying deposit rates offered.

Although I do not allow households to switch banks, this mechanism is similar to financial literacy explanations for returns heterogeneity, without the need to model portfolio risk. This exclusion is useful because (i) untangling how much of the persistent component of returns comes from risk preferences and from financial sophistication is not straightforward and (ii) there is significant heterogeneity in returns even when individuals hold no risky assets.

This model is a partial equilibrium analysis. The market interest rate is being taken as given. It is not determined by some market clearing condition. I view my model as the simplest implementation of a potential source of heterogeneity. In the simulation of the model and the resulting SMM estimation, I do not add banks as an agent type. Thus, the bank is not responding in every period to the level of deposits they receive after they set the optimal deposit rate based on the demand for deposits. Consequently, I avoid having to choose a particular scheme of allocating agents in the model to a particular bank. In this way, there are seven types of banks just as there are seven types of returns that an agent may receive.

The culmination of these simplfiying assumptions leaves us with a setting where we can consider the most stark role for returns in explaining wealth inequality. With more features in the model, like general equilibrium considerations, endogenous returns heterogeneity, portfolio choice, overlapping generations and bequests, etc., we leave more room for these features to explain the generated skewness in the model's distribution of wealth.

Chapter 2

Moderate Trust and Maximum Performance on Financial Investments

Abstract

This paper corroborates a nonlinear relationship between trust and economic performance; here, *economic performance* is defined as *returns to net wealth.* I use the RAND Longitudinal product version of the Health and Retirement Study to construct return measures over the period 2002–2022 from observed capital income, asset values across waves, debt, and transaction-related information. I estimate this hump-shaped relationship between trust and returns to net wealth in standard OLS regressions for average returns, pooled OLS, and panel OLS regression models by including both a linear and quadratic term for the trust variable in the set of regressors. The estimated level of trust that maximizes returns is consistently in the range of about 6 to 7 on a scale of 1–10, and this result is stable across the standard, pooled, and correlation random effects (CRE) specifications. Notably, the CRE model is the only specification that accomplishes both tasks of establishing this hump-shaped relationship between trust and returns while also characterizing the individual, persistent component of returns. These patterns persist even after controlling for risk exposure through portfolio composition, which is important since individuals are found to earn different returns on average even among riskless assets.

2.1 Introduction

There is much literature (Algan and Cahuc 2010) arguing that trust matters for aggregate economic outcomes such as long-term growth. Butler et al. 2016 shows that individual economic performance is not simply increasing in trust, but is maximized at intermediate levels. This paper contributes to the literature by asking whether a similar nonlinearity appears for realized respondent-level investment returns in the Household Retirement Survey (HRS).

The core argument of this paper is that if there is a link between trust and economic performance, then a similar (and possibly stronger) link should exist between trust and performance measures with respect to investment behavior. Inherent in financial environments where borrowing and lending are mediated through contracting between borrowers and lenders regarding a delayed payoff, is the *promise of future repayment*. If trust is too low, households may miss out on high-return opportunities. On the other hand, if trust is too high, investors may be too ambitious and get cheated more often. I argue that, if this "right amount of trust" mechanism results in a hump-shaped relationship between income and trust, then this fact about the nature of financial contracting should imply that the hump-shaped relationship is especially pronounced for returns.

Recent evidence using Norwegian administrative data documents persistent idiosyncratic return differences between households and scale dependence with wealth Fagereng et al. 2020. Related work using U.S. panel data also emphasizes a persistent return component and its relevance for inequality and life-cycle outcomes Daminato and

Pistaferri 2024. These findings motivate interest in documenting this persistent component and explaining it with observable household characteristics. In particular, a contribution of this paper would be to document this persistent component across respondents *holding constant this hump-shaped relationship between trust and returns.*

Replicating the Norwegian measurement environment in the U.S. is difficult. U.S. researchers typically combine survey data with partial administrative information, rather than observe a full administrative panel of wealth flows at the population level Kennickell 2017. In this paper, I use the publicly-available HRS RAND Longitudinal 2022 version of the data, which is unusually useful for this question because it contains a *measure of trust* as well as the core ingredients needed to construct returns over time: *capital income, asset values across waves, debt, and transaction-related information.*

I first verify that the moderate trust ranges are associated with the highest income levels in the HRS, both contemporaneously (2020) and for average income over the period (2002-2022). I then test for a comparable hump shape regarding trust and returns to net wealth under a number of specifications. Since trust is observed in one wave, I treat it as time-invariant in pooled and panel regression models.[1] The main results of the paper is that I not only recover the hump-shaped relationship between trust and returns to net wealth, but also control for it while characterizing the persistent component of returns across respondents.

The paper proceeds as follows. Section 2.2 reviews the relevant literature on trust, household returns, and panel methods for time-invariant regressors. Section 2.3

[1]Lesmeister et al. 2019 is an example of a paper which also has trust as a time-invariant regressor.

describes the construction of trust, income, wealth, portfolio shares, and return measures, and presents the main descriptive statistics using the RAND longitudinal HRS product. Section 2.4 presents the main empirical results, including the average-return and pooled specifications, and the panel evidence on persistent heterogeneity in returns. I also consider the distribution of trust responses compared to the estimated trust values which maximize returns to net wealth and report model-implied and realized mean returns by race relative to the sample average. Section 2.5 extends and repeats the analysis by controlling for risk exposure through portfolio shares and shows that the main trust results are robust to this adjustment. Section 2.6 concludes. The determinants of trust, the established hump-shaped relationship between trust and income in the HRS, and other results can be found in the Appendix A.

2.2 Literature Review

This work builds on the literature interested in measures of trust and its determinants, empirical estimates of the rate of return in broad and narrow asset classes, and the statistical relationship between trust and economic performance.

2.2.1 Trust

This paper draws on the literature on trust and its determinants. An example of this is Alesina and Ferrara 2000, who have the advantage of the standard demographic control variables (age, race, education, sex) as well as living in a community with significant heterogeneity in terms of race or income. My approach here is different in that I let the available survey responses in 2020 guide the set of demographic control variables used to explain the general trust measure, which can be found in the Appendix.

This main contribution of this paper to the literature is in relation to the empirical relationship between trust measures and economic performance. Guiso et al. 2008 finds that trust can lead to greater participation in the stock market. Guiso et al. 2004 finds that measures of social capital and trust lead to greater use of financial instruments such as checks, as well as diversification of the poor portfolio away from holding cash and towards holding stocks and other assets. The paper in this literature on trust and economic performance which serves as a key motivation is Butler et al. 2016. The authors find that an intermediate level of trust is associated with the maximum level of income. The key difference between what is done there and my paper is that the measure of economic performance here is the return to net wealth: measures of returns using this dataset can be interpreted as an investor's performance in their financial contracting behavior within a given asset/portfolio class. Because returns to net wealth are thought to differ persistently between individuals over time, I focus on the panel structure of the HRS to describe the individual fixed effect for the return measure.

Although I focus on the general trust measure in this paper, there are other trust measures available in the HRS 2020 wave. Haran Rosen et al. 2025 also uses this specially-designed trust module, although the authors focus on retirement security. They identify aggregate measures of trust; in particular, trust in financial institutions, and also make use of the set of financial literacy questions available in that same wave; each of these serve as potential directions to test the robustness of the main result for trust and returns to net wealth.

Alsan and Wanamaker 2018 find that the 1972 public revelation of the Tuskegee Syphilis Study led to lower trust in black males in medical institutions, ultimately leading to worse health outcomes over a lifetime. Algan and Cahuc 2010 find that aggregate trust has a large positive causal effect on GDP per capita by showing that a measure of inherited trust by second generation immigrants in the U.S. is strongly correlated with aggregate trust in the origin country (and is persistent across generations). These papers serve as inspiration for attempts at identifying the causal effects of trust on returns, although the HRS data was unfavorable in terms of results in this direction.

2.2.2 Returns

An important work in the literature on empirical estimates of the rate of return is Fagereng et al. 2020. The authors provide estimates of the distribution of returns within narrowly defined asset classes and also characterize the persistent component of returns across individuals. The HRS RAND product is publicly available survey data, which is much lower in quality than the Norwegian population-level tax data available to those authors for their project. That said, we are able to produce estimates of the distribution of returns (for asset classes and portfolios) and the individual fixed effect for returns to net wealth, which can be directly compared to those same objects. The major deviation of my paper from this work is that I complete these two tasks while controlling for the empirical relationship between trust and returns, which this paper makes no comment on.

Daminato and Pistaferri 2024 also documents the persistent component of returns in the Panel Survey of Income Dynamics (PSID), and then uses the data to calibrate a return-earnings process within a life cycle model of consumption-saving behavior. The usefulness of this paper for my own work is that the authors present a standard method for computing returns using the PSID. Due to the PSID's structure being very similar to the HRS, I use the authors' definition of returns as a template to guide the definition of return measures using the HRS data. The major deviation of my paper from this work is also my interest in the trust measure (which the PSID does not measure in any wave of its survey).

2.2.3 Panel regression techniques for time-invariant regressors

The fact that the returns can be computed for multiple waves is essential to describe the persistent component in the returns between respondents. This, combined with the fact that the HRS only asks respondents about trust for a single wave of the survey, drives the empirical techniques chosen in this paper. In particular, trust must be treated as time-invariant to make use of the HRS data. Said differently, a regression model that accommodates both time-varying and time-invariant regressors is required to make use of the HRS data. [2]

Moreover, when the regressor of interest is time-invariant, it is no longer sufficient to extend the pooled regression environment to allow for individual fixed effects.[3]

---

[2]Assuming trust is time-invariant is not unreasonable – there is empirical evidence to support this assumption.

[3]The education variable in the HRS (*Years of Schooling*) is also treated as time-invariant in the RAND version of the dataset. Since education is generally thought of to be important for returns (or co-variates) dealing with time-invariant regressors in the panel setting is likely unavoidable when

Indeed, estimating the individual fixed effect prevents the model from estimating coefficients on any of the time-invariant regressors, since the term absorbs all latent time variation. This would be at odds with establishing a hump-shaped relationship between trust and returns using the data. This is the central econometric issue of this paper, and the following papers help to address it. Baltagi and Liu 2026 offers an example of the general panel regression environment that fits the structure of HRS data well. The main panel regression models studied in this paper are the Fixed Effects (FE), Random Effects (RE), and Correlated Random Effects (CRE) models. Wooldridge 2019 offers a useful discussion on the Hausmann test and selecting among panel regression models. Bollen and Brand 2010 focuses on the CRE/Mundlak version of the model where time-varying regressors are allowed to co-vary with the individual fixed effect component.

2.3 Data

For the statistical analysis in this paper, I use the RAND Longitudinal version of the Household Retirement Survey (HRS) for the years 2002-2022. First, I define asset classes and portfolios, which will be guided by the unique structure of the HRS data set. Then, I provide descriptive statistics on the underlying return components, as well as returns by asset and portfolio. From there, I discuss some statistics on the demographic variables which form the set of control variables in the regression models. Finally, I look at the relationship between returns to net wealth and a number of variables (net wealth, age, education, trust) and describe portfolio compositions across

using the HRS.

respondents.[4]

2.3.1 Sample restrictions

After merging the RAND longitudinal extract with the constructed return and wealth fields, the analysis-ready wide file contains about 45,200 respondent identifiers; reshaping to a long person–year panel yields about 543,000 rows spanning the interview years used here. Most core RAND covariates are observed for a large share of those rows, but the key sample of interest is comprised of the respondents who have nonmissing values for the trust and returns to net wealth variables (along with the other variables used in the regression analysis). Wave-specific returns to net wealth (5% winsorized within survey year) are nonmissing for about 137,000 person-years: many observations drop out in a given wave when wealth, flows, or other return ingredients are incomplete. Generalized trust is measured only in the 2020 module; in the long panel, 900 respondents have a nonmissing 2020 trust response, and that value is carried on their other waves so that about 10,800 person-year rows carry nonmissing trust for estimation.

The descriptive statistics for which observations are pooled (histograms, bin-scatter figures, return-by-age and return-by-education tables, and the portfolio composition tables below) restrict person-years to the same listwise-complete sample as a benchmark pooled OLS of 5% winsorized returns to net wealth on quadratic trust with the baseline

[4] Unless noted otherwise, tables and figures in this document use the standalone chapter defaults for sample construction; the replication code also supports an optional unified dissertation sample (Stata `Code/Analysis/5_run_pipeline_ch2_dissertation.do` or `Code/Analysis/45_run_prep_and_exports.do`) that harmonizes exported objects with the thesis Chapter 2 submission when a perfect match is needed.

time-varying controls and education categories only—2,919 person-years. Of those, 449 rows are 2020 interviews and enter the trust histogram and the race-by-trust summary (all 449 have nonmissing generalized trust). The main regression tables in Section 2.4 use the specifications printed there (larger or smaller N by column); the extended Correlated Random Effects (CRE) model for net wealth uses about 2,900 person-years. When the dissertation build sets a single unified Chapter 2 sample (extended CRE person-years), the corresponding flag yields about 2,900 person-years for tables and figures tied to that mode.

2.3.2 Asset classes and portfolio definitions

The HRS asks a number of questions aimed at measuring household wealth and portfolio composition in the sample. I collect data on:

1. **Interest income and dividends**
2. **Capital gains**
3. **Net investment flows**
4. **Wealth holdings of the previous period**

All of which are required to define the desired measure of household returns. Since population-level administration data on these objects for individual taxpayers are not available in the United States, it is important to understand each component needed in the return calculation. This will help to determine whether or not our measured distribution of returns in this dataset is sensible. To understand what portfolios are accruing returns, we will let the structure of the HRS questions guide how we define returns to specific asset classes for which the survey collects data. From there, portfolios are defined as collections of these defined asset classes.

2.3.2.1 Interest income and dividends

The HRS RAND product has a distinct way of measuring the interest income and dividends received on the main asset classes, such as stocks and bonds, private business and real estate. In particular, there is a variable (`HwICAP`) in the survey that measures `Household Capital Income`:

- "...is the sum of household capital income received over the last calendar year, including business or farm income, self-employment earnings, gross rent, dividend and interest income, trust funds or royalties, and other asset income."

This particular feature of the data drives the main modeling assumptions of this paper regarding measuring returns: *asset classes are defined as narrowly as they can be given observation of interest income and dividends on those assets.* For example, although we see capital gains for stocks, business, bonds, and real estate, we do not observe interest income or dividends on these assets individually. Thus, returns to the narrowest asset class defined in this paper will be called **returns to core assets** and will consist of these assets.

The RAND product has variables (`RwIPENA`) and (`RwISRET`) which measure `Individual Income from Employer from Pension or Annuity` and `Individual Income from Social Security Retirement`, respectively. Together, these describe the interest income and dividends on retirement assets, so a measure of **returns to retirement assets** can be constructed. Moreover, I define the first broader notion of portfolio returns by considering **returns to core and retirement assets**.

A key assumption is that there are *no interest income or dividends earned on residential assets*.[5] With this assumption, we can define **returns to residential assets**.

This accounts for the interest income and dividends for every variable that accrues some capital gains: either their interest income goes to core assets, retirement assets, or residential assets. Thus, interest income and dividends on the entire portfolio of assets (i.e. on net wealth) is the same as interest income and dividends on the portfolio with just core and retirement assets. **Returns to net wealth** are computed by adding the remaining available capital gains (and other components) for those variables that receive no interest income. For example, returns to net wealth includes capital gains for the `Net value of vehicles` (`HwATRAN`) and the `Net value of all other savings` (`HwAOTHR`), although these variables earn no interest income and dividends. The capital gains to the `Net value of businesses` (`HwABSNS`), on the other hand, show up both in the returns to core assets and returns to net wealth.

Figure 2.1 shows the evolution of interest income and dividends over time within the relevant asset classes and for the overall portfolio.

[5] An assumption made after verifying that there is no "interest income and dividends" variable equivalent for the real estate asset class in the HRS dataset.

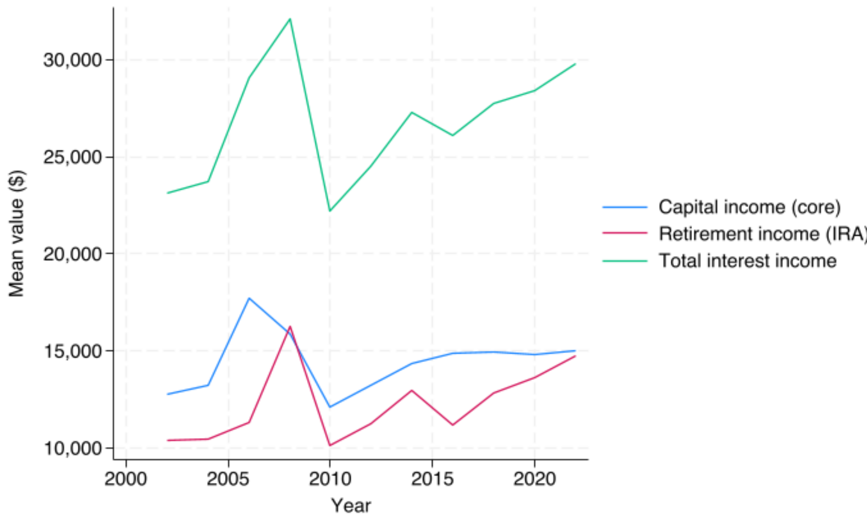


Figure 2.1: Interest income mean by year

2.3.2.2 Capital gains

Possible assets for which capital gains can be computed in the HRS are:

1. `Net value of primary residences` (`HwATOTH`)
2. `Net value of secondary residences` (`HwANETHB`)
3. `Net value of real estate (not primary residence)` (`HwARLES`)
4. `Net value of business` (`HwABSNS`)
5. `Net value of IRA, Keogh accounts` (`HwAIRA`)
6. `Net value of stocks, mutual funds, and investment trusts` (`HwASTCK`)

7. `Net value of bonds and bond funds` (`HwABOND`)

8. `Net value of checking, savings, or money market accounts` (`HwACHCK`)

9. `Net value of CD, government savings bonds, and T-bills` (`HwACD`)

10. `Net value vehicles` (`HwATRAN`)

11. `Net value of other savings` (`HwAOTHR`)

Possible liabilities are:

1. `Mortgage debt, primary residence` (`HwAMORT`)

2. Other `home loans` (`HwAHMLN`)

3. `Total other debt` (`HwADEBT`)

Capital gains for each asset is calculated as the estimated change in valuation across survey waves. Figure 2.2 shows the mean (blue) and median (red) for each wave of the survey for each possible asset.

Figure 2.2: Capital gains in core assets, by year

The `Net value of business` and `Net value of safe assets` each show a large decrease in average capital gains across respondents in the 2020 wave of the survey, likely an indication of the effects of the COVID-19 pandemic. The `Net value of real estate (not primary residence)` shows a similar decline in mean capital gains but in the 2010 wave of the survey, likely an indication of the effects of the Global Financial Crisis in 2008. Interestingly, `Net value of stocks, mutual funds, and investment trusts` shows no significant fall in average capital gains; notably, they begin at less than −\$10,000 and finish at around \$10,000 by the final wave of the survey. The `Net value of IRA, Keogh accounts` also starts out negative but ends well above \$20,000.

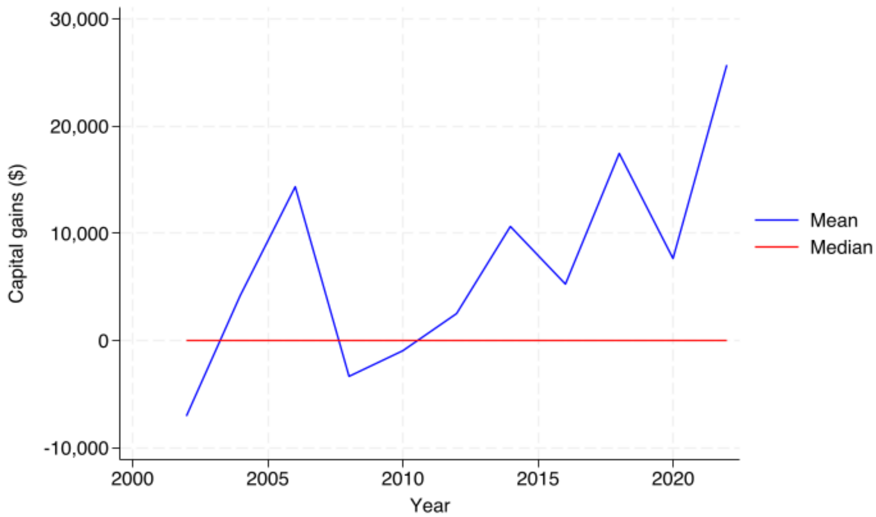


Figure 2.3: Capital gains in retirement assets, by year

An important finding from inspecting the capital gains data is that most of the respondents do not have assets. This can be seen by the red median line at 0 for most waves. This is another sign that the data are sensible so far – a large amount of non-participation despite evidence of returns is consistent with the empirically documented equity premium puzzle: the inability to rationalize the excess premium return earned by risky assets over riskless assets.[6]

[6]See Mehra and Prescott 2003 for a review of this literature.

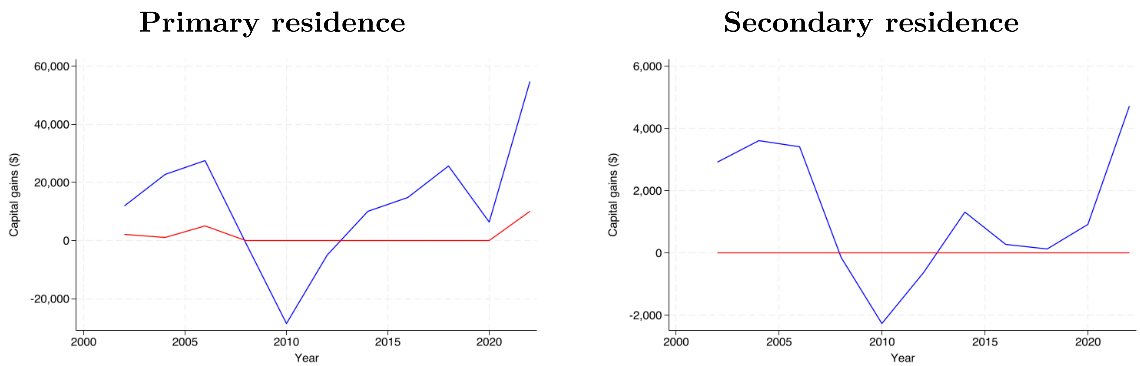


Figure 2.4: Capital gains in residential assets, by year

Notably, the residential panel (Figure 2.4) highlights that the mean of `Net value of primary residences` is much higher in level and volatility than the other assets, and is the only series with a clearly positive median in some years. The synchronized dip around 2008–2010 for the housing-related assets is another good sign regarding how reliably measured the data is.

2.3.2.3 Net investment flows

Net investment flows are vital in accurately computing returns because capital gains and interest income can miss other flows of value in a given asset class. Using the definitions of interest income and dividends from Subsubsection 2.3.2.1, I use these flows to construct net investment flows into three asset classes and two constructed portfolios:

1. **Core assets**

2. **Retirement assets**

3. **Residential assets**

4. **Core and residential assets**

5. **Net wealth**

An important concern regarding the RAND HRS Longitudinal product is that *net investment flows are the only variables relevant to the calculation of returns that are not offered as cleaned and processed fields.* Instead, the net investment flow data for each asset must be merged in using the RAND HRS Fat File for each survey wave. For this reason, there are substantially fewer observations for these variables by asset class than for the other return components.

To resolve this issue, I do two things. First, I process the net investment flow variables per asset class myself using the associated Section R flag for each amount. For example, `Personal funds into business amount` (`wR050`) and `Private business selling price amount` (`wR055`) pair with `Personal funds into business` (`wR049`) and `Sell any part interest in private business` (`wR054`), respectively; for primary and second homes, purchase and sale amounts (`wR007`, `wR013`) pair with `wR002`, and major improvement spending (`wR024`) pairs with `wR023`. When the gate item is "no" (`5`), the corresponding missing amount is set to zero.

For the second step in handling missing values for net investment flows in a given asset class, retirement assets provide a useful example. In each wave, the HRS has three separate variables for net investment flows into IRA/Keogh accounts (`wQ171_1`, `wQ171_2`, and `wQ171_3`). After using the accompanying flag variables (`wQ170_1`, wQ170_2, and wQ170_3) to set some missing amount variables to zero, I define the **net investment flow for retirement assets** as the sum of these three components. So, if only one or two of the three retirement flow variables are observed, the retirement flow is simply the sum of the nonmissing components and it remains missing only if all three retirement flow variables are missing.

Recall that net investment flows for business, stocks, and other real estate assets are combined into a **net investment flow for core assets**. The relevant net investment flow variables for primary and secondary residential assets are grouped to create **net investment flows for residential assets**. **Net investment flows to net wealth** are defined by combining the core, retirement, and residential asset classes. This second step towards handling missing values – requiring all components in a sum to be missing for the entire sum to be missing – holds for singular assets (the IRA/Keogh example), for asset classes (core assets would be missing if all subcomponents are missing), and for the broader portfolio definitions.

With this in mind, I present a figure of net investment flows into the assets available in the data set. Figure 2.5 shows the evolution of net investment flows, averaged between respondents, into each of the assets over time. In general, mean net investment flows into stocks and retirement assets are trending upward over the period. Net investment flows into businesses are particularly volatile and flows into residential assets are falling across the period.

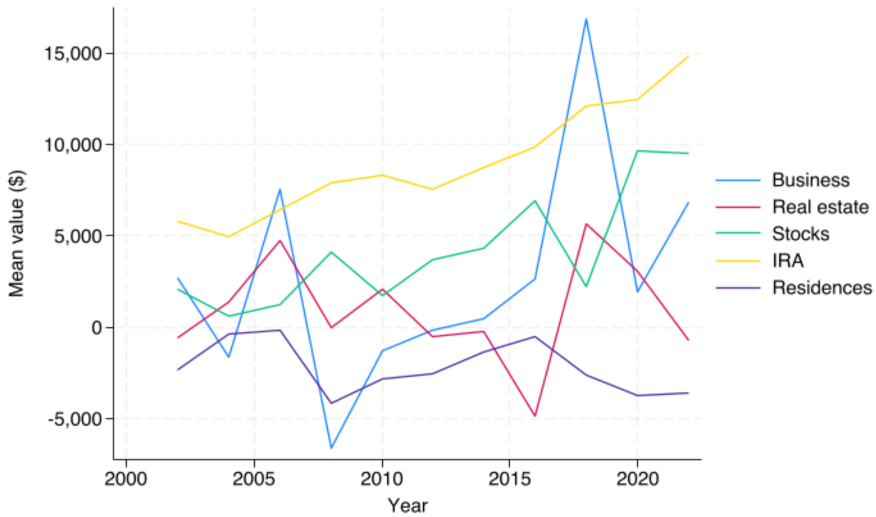


Figure 2.5: Flows by asset, mean by year

2.3.2.4 Defining wealth for assets and portfolios

The final component needed for the returns calculations is the wealth held by a respondent (in the previous period). This is measured as the total stock in the given asset classes (core, retirement, and residential), and summing among these creates the stock of wealth in the given portfolios (core and retirement assets, net wealth). For example, `Net value of primary residences` and `Net value of primary residences` from the previous period are summed together as a measure of **wealth held in residential assets at the beginning of the previous period**. Figure 2.6 shows the mean wealth within these narrowly defined asset classes for each year of the survey.

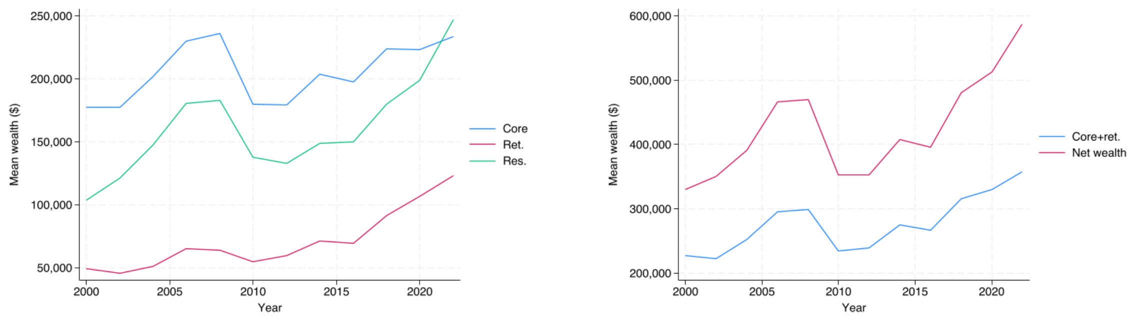


Figure 2.6: Mean wealth by year

Retirement assets seem to perform the worst in terms of average holdings. This may not be surprising, as the survey over samples older households and these households draw down their assets for consumption during retirement, in general. Core assets seem to offer the best performance, and all asset classes see a downward dive leading up to 2010 – historically accurate in the context of investment performance during the global financial crisis.

The following formula from Daminato and Pistaferri 2024 serves as a useful baseline for the return measures defined using HRS data:

$$r_t = \frac{y_t^c + cg_t - F_t - y_t^d}{A_{t-1} + .5F_t}$$

where y_t^c interest income and dividends, capital gains cg_t measured as the difference between the reported stocks across waves of the survey[7], F_t net investment flows, y_t^d payments on debt (in the RAND Longitudinal file, the variables mentioned are all in

[7]The capital gains to respondent i for an asset j is $V_{i,j,t} - V_{i,j,t-1}$, where $V_{i,j,t}$ is the value of the j-th asset for the i-th respondent in the t-th wave of the survey.

net terms, so this variable is equal to 0), and A_{t-1} total net wealth at the beginning of the previous period.

Although this return measure is useful, the RAND HRS Longitudinal product has a unique structure that will guide how I define the asset classes and portfolios for which returns are accrued. As mentioned before, core assets are the most narrowly defined asset class here because interest income and dividends do not disaggregate into the asset classes: stocks, bonds, private business, IRA/Keogh.

2.3.2.5 Narrow asset classes

Recall that we define the interest income and dividends to core assets as the RAND variable for `Household Capital Income` (`HwICAP`), which sums together the interest income and dividends to the following underlying assets of interest: stocks, bonds, business, and real estate. Since interest income is an important component of the return calculation, we define the returns to core assets in the following way. I find the variables capital gains and net investment flows to each of these assets and aggregate them to reach a measure of capital gains on core assets and net investment flows to core assets.

The returns to the core assets are given by

$$r_{it}^{core} = \frac{y_{it}^{core} + cg_{it}^{core} - F_{it}^{core}}{A_{i,t-1}^{core} + .5F_{it}^{core}}$$

where y_{it}^{core} is the interest income and dividends on the core assets, cg_{it}^{core} are the capital gains on the core assets, F_{it}^{core} are the net investment flows to core assets, and importantly, $A_{i,t-1}^{core}$ is the total wealth held in core assets in the previous period.

The rest of the return measures are defined similarly. In the HRS data, we also observe income on retirement assets. This allows us to define a return measure for retirement assets

$$r_{it}^{ret} = \frac{y_{it}^{ret} + cg_{it}^{ret} - F_{it}^{ret}}{A_{i,t-1}^{ret} + .5F_{it}^{ret}}$$

where y_{it}^{ret} is the interest income and dividends on the retirement assets, cg_{it}^{ret} are the capital gains on retirement assets, F_{it}^{ret} are the net investment flows to retirement assets, and $A_{i,t-1}^{ret}$ is the total wealth held in retirement assets in the previous period.

Based on the modules offered in a given survey wave, I assume that there are no interest income or dividends to primary and secondary residential assets, so that the interest income and dividends accrued to residential assets are 0 in a given period ($y_{it}^{res} = 0$). Thus, I define returns to residential assets as

$$r_{it}^{res} = \frac{cg_{it}^{res} - F_{it}^{res}}{A_{i,t-1}^{res} + .5F_{it}^{res}}$$

where $cg_{i,t}^{res}$ are the capital gains on the residential assets, $F_{i,t}^{res}$ are the net investment flows to residential assets (including improvement costs), and $A_{i,t-1}^{res}$ is the total value of the residential assets in the previous period.

2.3.2.6 Broad asset classes

After defining the returns for narrow asset classes, I also define the return measure for portfolios (returns on multiple assets). The first portfolio definition is for returns to core and retirement assets:

$$r_{it}^{coret} = \frac{y_{it}^{coret} + cg_{it}^{coret} - F_{it}^{coret}}{A_{i,t-1}^{coret} + .5F_{it}^{coret}}$$

where y_{it}^{coret} is the interest income and dividends on core and retirement assets, cg_{it}^{coret} are the capital gains on core and retirement assets, F_{it}^{coret} are the net investment flows to core and retirement assets, and $A_{i,t-1}^{coret}$ is the total wealth held in core and retirement assets in the previous period.

The broadest portfolio definition is for total net wealth holdings. Returns to net wealth are given by

$$r_{it}^{net} = \frac{y_{it}^{net} + cg_{it}^{net} - F_{it}^{net}}{A_{i,t-1}^{net} + .5F_{it}^{net}}$$

where y_{it}^{net} is the interest income and dividends on core and retirement assets, cg_{it}^{net} are the capital gains on all available asset classes in the survey, F_{it}^{net} are the net investment flows on assets for which it is relevant in the survey (stocks, bonds, business, IRA, residences, and real estate), and $A_{i,t-1}^{net}$ is the total value of the available assets minus liabilities in the previous period[8].

[8] `Value of all mortgages/land contracts (secondary residence)` (`HwAMRTB`) is an optional

2.3.2.7 The distribution of measured returns

I present the the evolution of the return measures, averaged over the respondents, over time in the following figure 2.7. I group them by asset classes (core, retirement, residential) and portfolio definitions (core and retirement, net wealth).

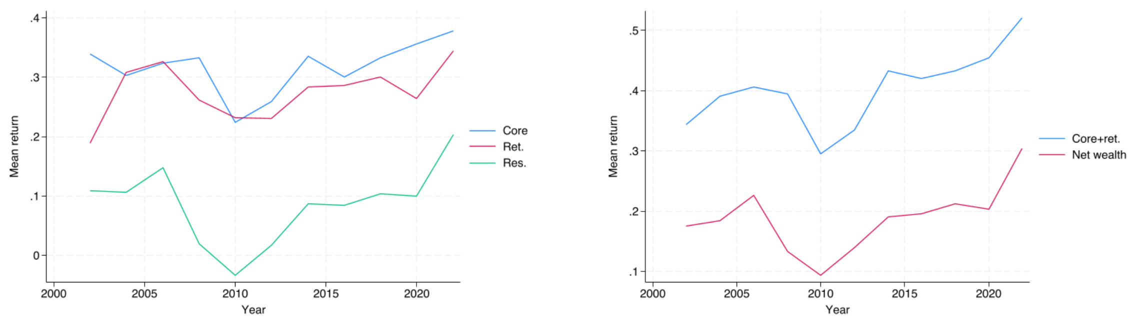


Figure 2.7: Mean returns by year

Mean returns to core assets seem to have the best performance in terms of asset classes, followed by retirement and then residential assets. For portfolios, returns to net wealth are lower than returns to core and retirement assets on average. Across asset classes (and thus portfolios), there is a decline in returns on average in 2010, which is likely an artifact of the economic downturn associated with the Global Financial Crisis of 2008.

Figure 2.8 shows the distribution of returns across person-years, pooled over survey waves (2002–2022), for the three asset classes: core, retirement, and residential.

question in some survey waves so it is omitted from the list of liabilities used to compute net wealth.

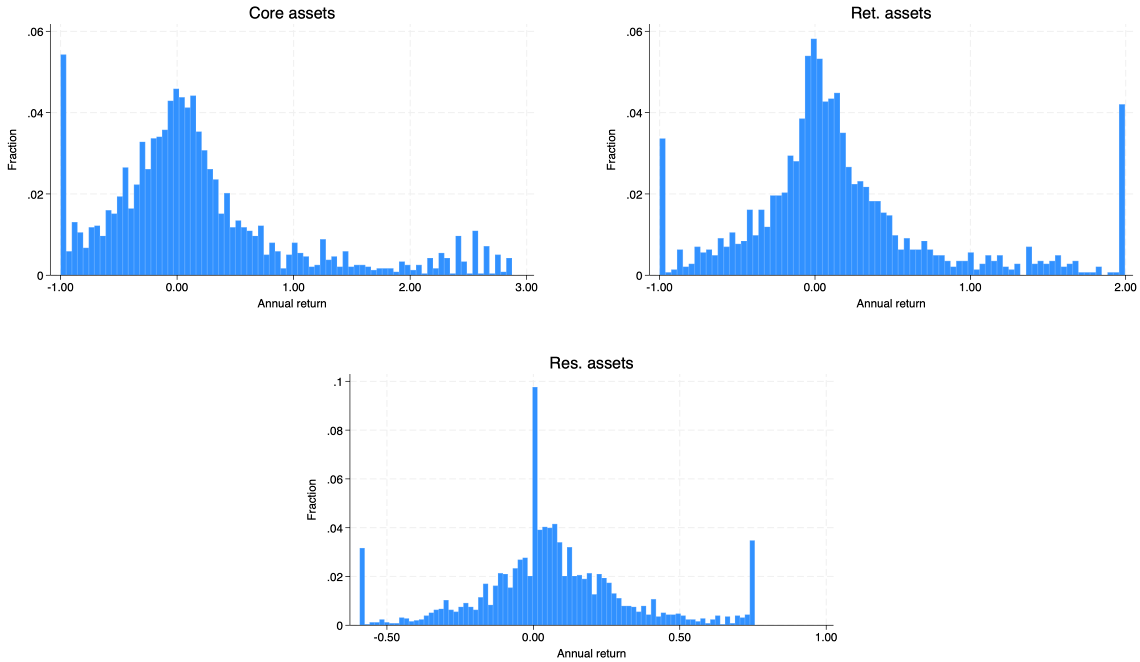


Figure 2.8: Returns histogram, components (pooled across survey years)

In particular, the distribution of returns to core assets has a right tail with extremely high values relative to the distribution of returns to retirement and residential assets. Each distribution seems to be centered near 0, which is a good sign. Residential assets feature a significant share of respondents earning a positive return; this is in line with the empirical fact that home ownership in the United States is relatively common compared to investment in other asset classes.

Figure 2.9 shows the distribution of returns for the constructed portfolio measures: core and retirement assets and net wealth. In general, these figures are encouraging, as its shape closely resembles other estimates of the empirical distribution of individual realized returns using other datasets.

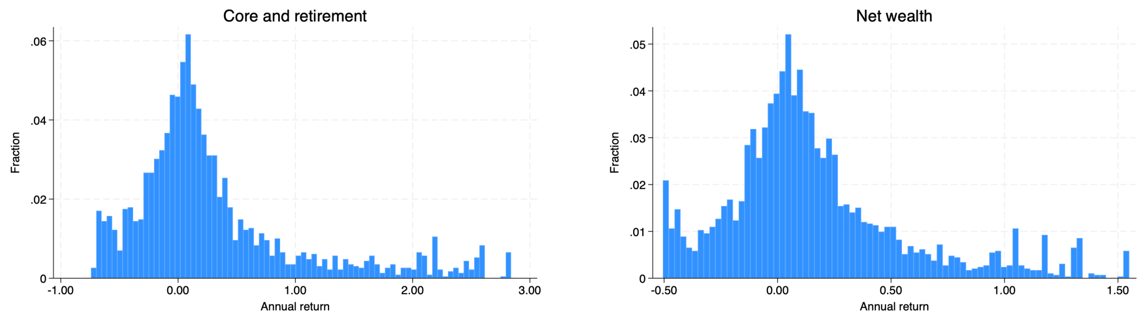


Figure 2.9: Returns histogram, core and retirement vs. net wealth (pooled across survey years)

Table 2.1 summarizes the distribution of returns (mean, standard deviation, skewness, kurtosis and selected percentiles) for each asset class and portfolio, pooling person-years in the same sample as Figure 2.8 and the return-by-age and return-by-education tables (pooled OLS sample for r^{net} on generalized trust with education categories and standard panel controls). A major takeaway from this table is that *the mean returns across both asset classes and portfolios are much higher than the median.* This suggests that there are a significant number of respondents with computed returns that are extremely large.

Table 2.1: Returns to assets/portfolios

Returns to	Mean	SD	Skewness	Kurtosis	P1	P50	P99
Core assets	0.3066	1.5804	7.45	97.65	-1.0000	0.0249	6.6292
Residential assets	0.0777	0.3935	3.38	41.27	-1.0000	0.0422	1.3805
Retirement assets	0.2666	1.1189	5.90	62.82	-1.0000	0.0575	5.3233
Core and retirement assets	0.3905	1.2199	7.41	107.65	-0.9413	0.1138	4.9135
Net wealth	0.2011	0.6305	4.48	38.68	-0.7085	0.0843	2.6968

2.3.3 Demographic variables

The HRS offers a collection of trust questions in "Section V: Modules" section of the 2020 survey. There are a total of 8 questions, and the one I focus on in this paper is `Trust: People in General` (`RV557`):

- "How much do you trust people in general?"

Respondents are asked to give an answer between 1 and 10. This set of trust questions was only a part of the questionnaire in that year. This is an issue, since the panel structure of the HRS allows me to estimate the persistent component of returns, variation in the trust measure would be ideal if it matters for trust. That said, in the panel setting, we can assume that trust, like education, is fixed over time.[9]

Figure 2.10 shows the distribution of responses to the trust question. Interestingly, most of the respondents report a trust level of 5 or 7. In addition, there are a significant number of respondents who report trusting other people at the most extreme values (0 or 10). This suggests that the sample has a significant number of individuals who are very trusting and also very distrusting. In general,

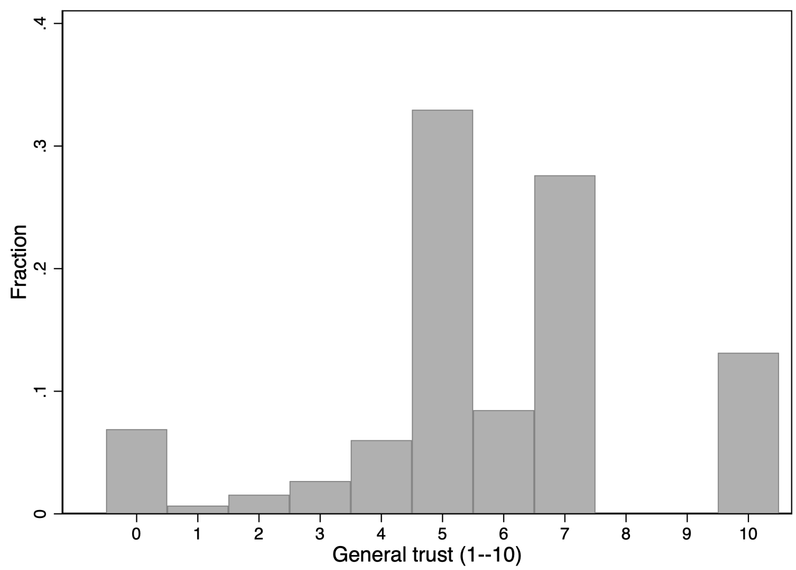


Figure 2.10: Distribution of generalized trust among sample respondents (2020)

Figure 2.11 summarizes the average trust responses by race and ethnicity. As can be seen in Appendix Table A.1 (*Determinants of General Trust*), the race variable (in particular, the `Black` racial group) is robustly found to be predictive of and negatively

[9]Literature on trust talks about how history of being cheated or treated fairly form individuals' trust over time. If this is true and at some point, one learns enough and forms their trust level, then a sample with an over-representation of older household is a reasonable environment to assume constant trust levels.

correlated with trust. Minority groups, especially `Black`, report lower trust levels on average than their white counterparts in the sample.

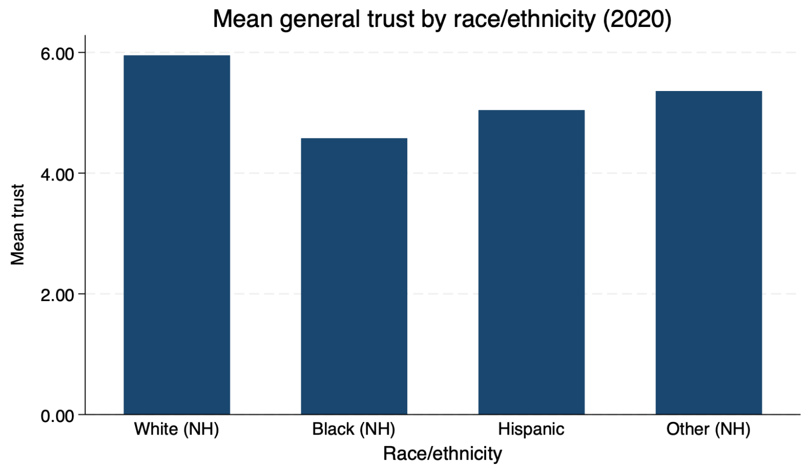


Figure 2.11: Mean generalized trust by race/ethnicity (2020)

Table 2.2 summarizes the relevant baseline control variables (`Years of Schooling` (`RAEDYRS`) and `Labor Force Status` (`RwLBRF`)) and the extended set of control variables (`Gender` (`RAGENDER`), `Race` (`RARACEM`), `Whether Hispanic` (`RAHISPAN`)[10], `Place of Birth` (`RABPLACE`), and `Census Region` (`RwCENREG`)) relevant for the regression models. Notice that the 2020 demographic summary statistics (or proportions, for the categorical variables) are very close to the pooled versions of those same variables. This is a good sign, since trust is only measured in 2020.

The data suggests that more than half of the sample has at least some college, and less than a third of the respondents are working full-time. Most respondents are married

[10] The two `Race, Ethnicity` variables are combined to create the racial categories: NH White, NH Black, Hispanic, NH Other.

and were born in the United States. In terms of geographic location, respondents seem to be concentrated in the South census region.

Table 2.2: Sample Demographics

	(1)	(2)
Panel A: Education categories and labor force		
Share: Less HS	0.183	0.192
Share: HS Grad	0.258	0.291
Share: Some College	0.247	0.242
Share: College	0.158	0.135
Share: Postgrad	0.154	0.141
Share in labor force	0.305	0.373
Panel B: Gender, marital status, race/ethnicity, birthplace, and census region		
Share female	0.537	0.554
Share married	0.630	0.681
Share: NH White	0.612	0.680
Share: NH Black	0.198	0.163
Share: Hispanic	0.149	0.121
Share: NH Other	0.038	0.035
Share born in U.S.	0.833	0.877
Share: Northeast	0.147	0.129
Share: Midwest	0.198	0.231
Share: South	0.448	0.451
Share: West	0.205	0.182

Column (1) reports shares for the 2020 cross-section; column (2) reports shares pooled across all survey years.

2.3.4 Descriptive statistics for returns to net wealth

Since regression analysis focuses on returns to net wealth, this section will offer a number of decompositions of returns to net wealth by wealth, age, employment, and education.

2.3.4.1 Return vs. net wealth

Figure 2.12 is the appropriate place to read the raw relationship between returns and wealth in the sample. For returns to net wealth, that descriptive relationship appears fairly weak and close to flat, with at most a slight positive tilt in the fitted line. That said the observations towards the ends of the net wealth distribution do seem to exhibit the *scale dependence* result found in the literature regarding returns.

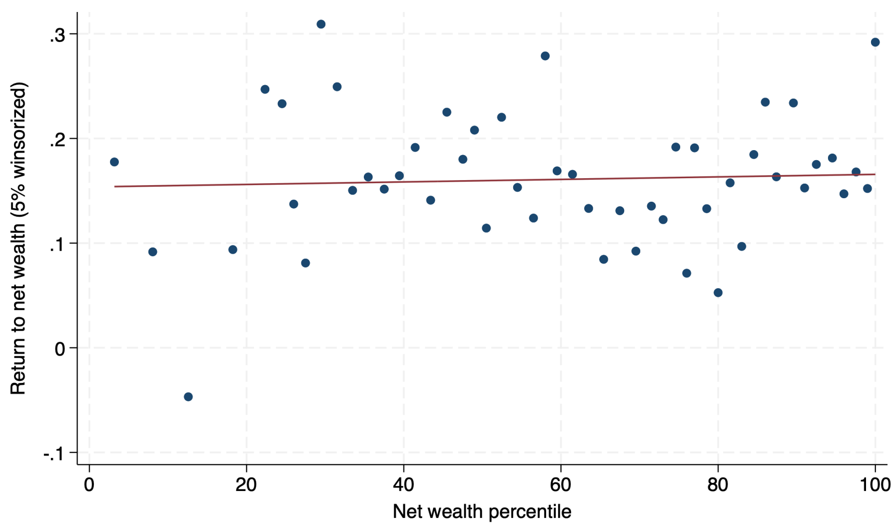


Figure 2.12: Return to net wealth vs. net wealth percentile

2.3.4.2 Returns by age and education

`Age at Interview` (`RwAGEY_B`) and `Years of Schooling` are control variables which are thought to matter empirically for both trust and returns. Since each of these play a unique role in the HRS data (education time-invariant, age is higher on average due to oversampling of retirees), a decomposition of returns by each of these should be considered. Table 2.3 reports moments of the returns to net wealth distribution conditional on age.

Table 2.3: Return to Net Wealth by Age Bin

Age bin (years)	N	Mean	SD
30–34	1	-0.4655	.
35–39	5	0.2799	0.6294
40–44	7	-0.0688	0.5094
45–49	14	0.0794	0.3828
50–54	47	0.1480	0.4688
55–59	178	0.1482	0.4011
60–64	660	0.1800	0.4331
65–69	756	0.1932	0.4089
70–74	575	0.1256	0.3594
75–79	394	0.1398	0.3682
80–84	213	0.1133	0.3619
85–89	65	0.2430	0.4537
90–94	4	0.5089	0.8286

Return to net wealth (5% winsorized), pooled person-years. Age bins are 5-year groups.

A very interesting feature of the sample is that there are less than twenty respondents between the ages of 30 and 44, and less than 40 respondents between the ages of 30 and 49. The majority of the respondents are between the ages of 60 and 79.

Table 2.4 reports moments of the returns to net wealth distribution conditional on education. In general, it appears that more years of schooling lead to lower average returns to net wealth, but also less volatility as measured by the standard deviation. Interestingly, the group of respondents that did not complete high school have the highest returns on average in the sample.

Table 2.4: Return to Net Wealth by Education Category

Education category	N	Mean	SD
Less HS	559	0.1788	0.4200
HS Grad	848	0.1546	0.3978
Some College	706	0.1673	0.4195
College	394	0.1497	0.3868
Postgrad	412	0.1479	0.3575

Return to net wealth (5% winsorized), pooled person-years. Education categories are constructed from RAND HRS years of education: Less HS (< 12), HS Grad (= 12), Some College (13–15), College (= 16), Postgrad (> 16).

2.3.4.3 Return vs. labor force participation status

Figure 2.13 shows the mean returns to net wealth conditional on labor foce paritipation status. Following the global financial crisis of 2008, it seems that returns are higher on average conditional on being employed in general.

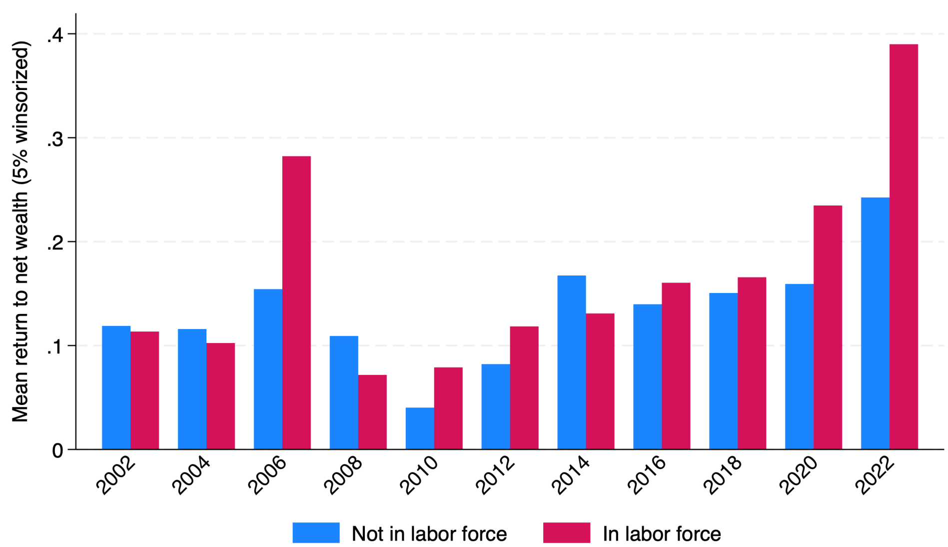


Figure 2.13: Mean return to net wealth and labor-force status by survey year

2.3.5 Trust and returns to net wealth

Establishing an empirical relationship between trust and returns is a major goal of this paper. To begin in this direction, I inspect the relationship between trust responses and computed returns to net wealth values for the sample. Figure 2.14 suggests some evidence of a hump-shaped relationship between trust and returns to net wealth, as the highest return values seem to be concentrated at the moderate trust ranges.

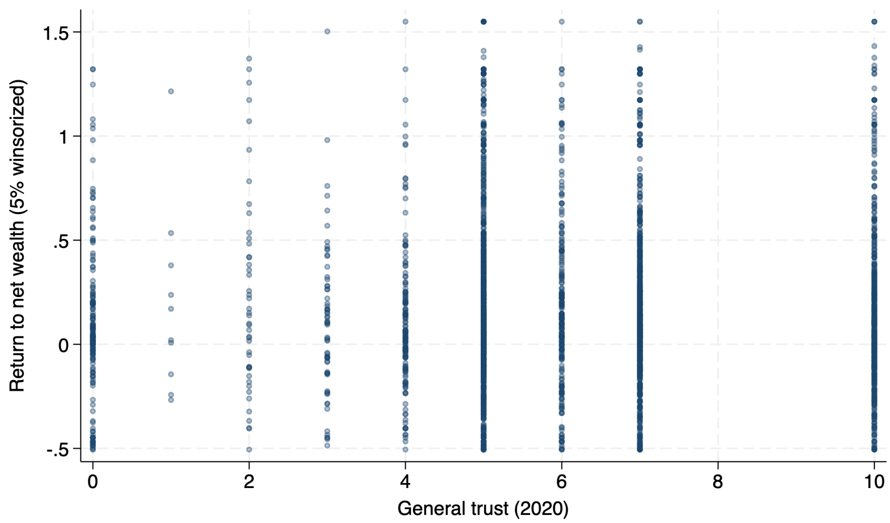


Figure 2.14: Returns to net wealth vs. trust pooled across survey years

2.3.6 Portfolio composition

Table 2.5 shows that participation is highest in safe assets, residential assets, and the broader core-asset grouping, while direct participation in business and other real estate is much less common. In terms of composition, residential assets account for the largest average share of household balance sheets, followed by core assets and then retirement assets. Debt is also economically important in the pooled sample: over half of person-years involve some debt, and the leverage measures show a long right tail, especially for long-term and total debt.

Table 2.5: Participation and Portfolio Composition

Statistic	Mean	SD	P1	P50	P99
Fraction with business assets	0.117	0.322	0.000	0.000	1.000
Fraction with stocks assets	0.297	0.457	0.000	0.000	1.000
Fraction with safe assets	0.828	0.378	0.000	1.000	1.000
Fraction with other real estate assets	0.173	0.378	0.000	0.000	1.000
Fraction with retirement assets	0.479	0.500	0.000	0.000	1.000
Fraction with residential assets	0.887	0.317	0.000	1.000	1.000
Fraction with core assets	0.858	0.349	0.000	1.000	1.000
Fraction with long-term debt	0.332	0.471	0.000	0.000	1.000
Fraction with other debt	0.308	0.462	0.000	0.000	1.000
Fraction with some debt	0.510	0.500	0.000	1.000	1.000
Share in private business	0.031	0.113	0.000	0.000	0.619
Share in stocks/mutual funds	0.060	0.140	0.000	0.000	0.659
Share in safe assets	0.095	0.167	0.000	0.031	0.949
Share in other real estate	0.040	0.124	0.000	0.000	0.613
Share in retirement assets	0.132	0.208	0.000	0.000	0.818
Share in residential assets	0.488	0.328	0.000	0.461	1.000
Share in core assets	0.245	0.269	0.000	0.135	1.000
Share in core and retirement assets	0.377	0.315	0.000	0.358	1.000
Leverage, Long-term debt	0.114	0.439	0.000	0.000	1.500
Leverage, Other debt	0.071	0.581	0.000	0.000	1.348
Leverage, Total debt	0.185	0.724	0.000	0.001	2.491

All shares are expressed as fractions of gross wealth. Participation is the fraction of person-years with a strictly positive holding. Person-years match the pooled regression sample used for Tables 2.3 and 2.4. Leverage ratios can exceed one when liabilities exceed gross assets. Person-year observations per statistic range from 2,886 to 2,886 (same pooled regression sample as return descriptives by age and education). The modest spread reflects missing values for specific portfolio components in some person-years.

Table 2.6 shows that asset composition shifts sharply across the wealth distribution. At the bottom, balance sheets are dominated by debt, with very little exposure to business, stocks, or other real estate. Moving up the distribution, risky assets become more important: stocks, business wealth, and other real estate all rise with wealth, and at the very top business ownership becomes especially prominent. This is consistent with the idea that observed portfolio exposure is highly heterogeneous across households.

Table 2.6: Asset Composition by Wealth (Core) Fractiles

Fractile	Real est.	Business	Stocks	Safe assets	Mortgage debt	Other home loans	Other debt
Bottom 10%	0.000	0.000	0.000	0.000	0.124	0.014	0.059
10–20%	0.000	0.000	0.000	0.032	0.119	0.013	0.515
20–50%	0.008	0.001	0.003	0.103	0.145	0.014	0.107
50–90%	0.062	0.038	0.090	0.130	0.082	0.007	0.008
90–95%	0.131	0.134	0.202	0.084	0.050	0.002	0.005
95–99%	0.127	0.165	0.220	0.122	0.020	0.004	0.001
99–99.9%	0.067	0.168	0.370	0.079	0.012	0.000	0.002
99.9–99.99%	0.000	0.158	0.178	0.304	0.019	0.002	0.000

Mean share of gross wealth in core assets in each of the asset/liability classes, computed as fractile of the core wealth distribution. Fractile boundaries use rank within the same pooled regression person-year sample as Tables 2.3 and 2.4. Safe assets = bonds + checking/saving.

The broader portfolio aggregates in Table 2.7 tell a similar story. Residential wealth is the dominant asset for much of the middle of the distribution, while core assets take over at the top, and retirement assets are largest in the upper-middle and upper-tail groups rather than at the very top. Debt shares fall rapidly with wealth, so leverage is concentrated in the lower part of the distribution.

Table 2.7: Portfolio Composition by Net-Wealth Fractiles

Fractile	Core	Residential	Retirement	Long debt	Other debt	Total debt
Bottom 10%	0.198	0.284	0.049	0.327	0.548	0.874
10–20%	0.152	0.544	0.033	0.219	0.105	0.324
20–50%	0.150	0.680	0.069	0.122	0.032	0.154
50–90%	0.299	0.433	0.199	0.057	0.004	0.061
90–95%	0.425	0.283	0.245	0.034	0.005	0.038
95–99%	0.431	0.241	0.230	0.020	0.002	0.021
99–99.9%	0.644	0.174	0.130	0.017	0.000	0.018
99.9–99.99%	0.621	0.150	0.063	0.014	0.000	0.014

Mean share of gross wealth in each broad asset/liability class, computed within net-wealth fractiles for the same pooled regression person-year sample as Tables 2.3 and 2.4. Core = business + stocks + bonds + checking/savings + other real estate. Total debt = long-term debt + other debt.

These composition patterns matter in Section 2.5, where I revisit the main specifications by using the portfolio shares as a proxy for risk exposure.

2.4 Results

The presence of hump-shaped empirical relationship between earnings and trust in the HRS data is important because it aligns with previous literature Butler et al. 2016. Given the oversampling of older and retired households by the HRS, this is an important first step towards using the dataset to argue a similar pattern trust and returns. Appendix Tables A.5 and A.6 show that, for total income measures, the linear trust term is positive and statistically significant while the quadratic term is negative and statistically significant, consistent with a concave (hump-shaped) pattern.

The main statistical analysis of this paper is focused on returns to net wealth as the dependent variable of interest. [11] I show that trust is predictive for the raw measure of returns to net wealth, that the hump-shaped relationship does persist, and that it is robust to different trimming and winsorization choices. Due to the number of outliers in the sample, I present the results for a top and bottom winsorization of 5% for returns to net wealth. [12]

[11] The results for the smaller portfolio compositions (core, retirement, residential, and core and retirement) are largely insignificant across specifications (though the results are more encouraging when retirement assets are excluded). Again, with more retirees in the sample, an analysis of retirement assets may take more creative efforts.

[12] This is important given the difficulty with computing returns in the U.S. due to lack of available data on flows in and out of narrowly defined asset classes.

2.4.1 Trust and average returns to net wealth

I begin the analysis with establishing the empirical relationship between the average returns to net wealth over the period 2002-2022 and the 2020 trust measure. This is captured by the regression equation

$$\bar{r}_i^{net} = X_i'\beta + \varepsilon_i.$$

Trust in this model is assumed to be an invariant characteristic about households (measured in a single wave, but the same across waves) that influences economic decisions over time so that it matters for long-term returns. [13]

[13] Lesmeister et al. 2019 is an example of work that also assumes that trust is time invariant.

Table 2.8: Trust and Average Returns to Net Wealth

	(1)	(2)	(3)	(4)
Trust	0.01**	0.07***	0.04***	0.04***
	(0.00)	(0.01)	(0.02)	(0.02)
Trust^2		−0.01***	−0.00**	−0.00**
		(0.00)	(0.00)	(0.00)
In labor force			0.09***	0.10***
			(0.03)	(0.03)
HS Grad			0.06	0.05
			(0.04)	(0.04)
Some College			0.07	0.05
			(0.04)	(0.04)
College			0.09*	0.07
			(0.05)	(0.05)
Postgrad			0.05	0.04
			(0.05)	(0.05)
Female				−0.04
				(0.02)
NH Black				0.03
				(0.04)
Hispanic				−0.10**
				(0.05)
NH Other				−0.02
				(0.08)
Born in U.S.				−0.02
				(0.04)
Midwest				−0.01
				(0.04)
South				0.03
				(0.04)
West				0.13***
				(0.04)
Married				−0.00
				(0.03)
_cons	0.15***	0.04	0.06	0.01
	(0.03)	(0.04)	(0.07)	(0.09)
N	750.00	750.00	595.00	592.00
Adj. R^2	0.01	0.03	0.07	0.09
Trust joint p	.	0.00	0.03	0.02
Age joint p	.	.	0.08	0.11
Wealth joint p	.	.	0.30	0.14
Race joint p	.	.	.	0.11
Region joint p	.	.	.	0.01
Educ. joint p	.	.	0.40	0.69

OLS. Heteroskedasticity-robust standard errors. Columns *(3)–(4)* add wealth deciles, age bins, labor force, and education; column *(4)* adds gender, race, region, immigration status, and marital status.

First, allowing for the hump-shaped relationship between trust and returns by adding the quadratic term for trust not only increases the predictive power of returns, but considerably increases the explained variation of the model. Second, as for other variables which reject the null hypothesis of no significant effect on returns, wealth and age add the most to explained variation, followed by labor force participation status and years of education. These controls represent the minimal, baseline specification of the model, captured by column 3 in table 2.8. A final specification, with additional demographic variables, is robust to the hump-shaped relationship, as well as the predictive power of labor force participation status for returns.[14]

Figure 2.15 shows the distribution of respondent-level average returns to net wealth in the final average-return estimation sample. Even after the 5% winsorization used throughout the main analysis, the cross section remains dispersed, which helps motivate the move from average returns to the person-year pooled specification.

[14] As we will see later, Race and Gender are variables which predict trust well, so they are additional demographic variables which may be of interest here.

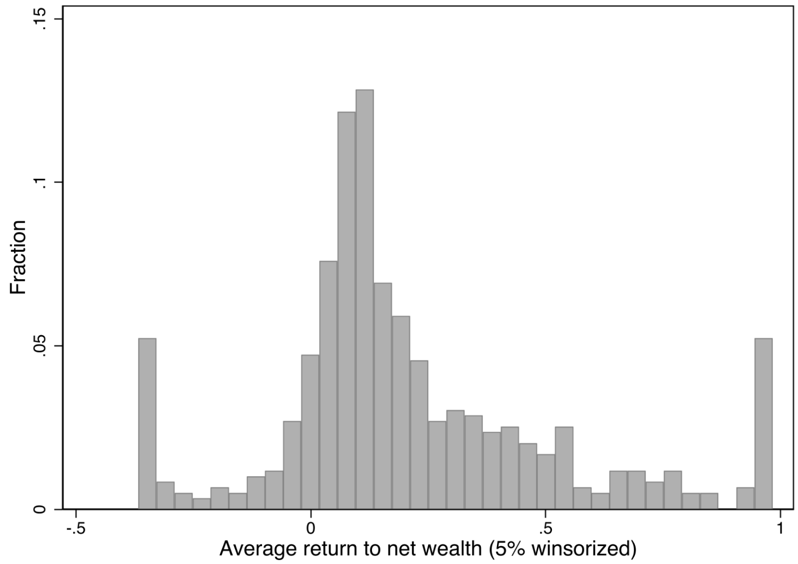


Figure 2.15: Distribution of average returns to net wealth in the final average-return sample.

2.4.2 Pooled regression model

With the implicit assumption that the 2020 trust measure is constant for respondents, then the hump-shaped relationship holds between trust and long-run, average returns to net wealth in the sample. A key object of analysis in the literature interested in measuring realized returns is the estimated persistent component of returns. To build toward this, we move to the panel regression model

$$r_{it}^{net} = \lambda_t + X_{it}'\beta + \varepsilon_{it}.$$

In table 2.9, I show that the hump-shaped relationship persists in the pooled model with the baseline set of controls and with the maximal set of controls.

Table 2.9: Pooled OLS of Returns to Net Wealth on Trust

	(1)	(2)
Trust	0.03***	0.03***
	(0.01)	(0.01)
Trust^2	−0.00**	−0.00**
	(0.00)	(0.00)
In labor force	0.03**	0.03**
	(0.02)	(0.02)
Married	−0.07***	−0.06***
	(0.02)	(0.02)
HS Grad	−0.06**	−0.06**
	(0.03)	(0.03)
Some College	−0.05**	−0.05**
	(0.03)	(0.03)
College	−0.10***	−0.11***
	(0.03)	(0.03)
Postgrad	−0.10***	−0.10***
	(0.03)	(0.03)
Female		−0.03*
		(0.02)
NH Black		0.05*
		(0.03)
Hispanic		−0.00
		(0.03)
NH Other		0.01
		(0.03)
Born in U.S.		0.02
		(0.02)
_cons	−0.57***	−0.61***
	(0.04)	(0.06)
N	2919.00	2899.00
Adj. R^2	0.04	0.04
Trust joint p	0.01	0.00
Age joint p	0.00	0.00
Wealth joint p	0.00	0.00
Race joint p	.	0.28
Region joint p	.	0.11
Year joint p	0.00	0.00
Educ. joint p	0.01	0.00

Person-year OLS. Standard errors clustered by household. Column *(1)*: baseline controls (wealth, age, labor force, education, year effects). Column *(2)*: adds gender, race, region, immigration status, and marital status.

In the pooled setting, the baseline control blocks remain strongly predictive: age, wealth, education, labor-force status, and survey-year effects are all jointly significant. Adding the extended demographic controls leaves the adjusted R^2 essentially unchanged at .04, but marital status remains strongly predictive and several individual demographic coefficients remain nonzero. The key point is that the hump-shaped trust relationship survives even after moving from the respondent-level average-return regression to the person-year pooled specification.

2.4.3 Panel regression models

2.4.3.1 The Fixed Effects (FE) model

The next step is to allow for individual fixed effects in the panel regression so that the error term ε_{it} can be written as

$$\varepsilon_{it} = \alpha_{it} + e_{it},$$

where e_{it} may possibly be serially correlated. Notably, *the predictive power of any the time-invariant regressors (trust, education, race, gender, region, immigration status) will be absorbed by the individual fixed effect term.*

To see this, first recall that the FE model is equivalent to demeaning within individuals:

$$r_{it}^{net} = \alpha_i + \lambda_t + X_{it}'\beta + \varepsilon_{it},$$

For each individual i, take the time average:

$$\bar{r}_i^{net} = \alpha_i + \bar{\lambda}_i + \bar{X}_i'\beta + \bar{\varepsilon}_i,$$

Then subtract from the original equation:

$$r_{it}^{net} - \bar{r}_i^{net} = \left(\lambda_t - \bar{\lambda}_i\right) + \left(X_{it} - \bar{X}_i\right)' \beta + \left(\varepsilon_{it} - \bar{\varepsilon}_i\right),$$

Notice that the individual fixed effect is eliminated by this transformation. As a result, any time-invariant regressors are also removed from the model.

Table 2.10 reports two within-household specifications that differ only in which *time-varying* controls are added on top of the same wealth-decile indicators, five-year age-bin fixed effects, and survey-year fixed effects (those blocks enter both columns but are summarized by joint tests in the lower panel rather than coefficient-by-coefficient). Column *(1)* isolates labor-force participation as the only time-varying demographic shown in the body of the table. Column *(2)* adds marital status and wave-specific census region (relative to Northeast), with the joint tests for marital status and region reported alongside the tests for age bins, wealth deciles, and years.

Table 2.10: Panel OLS with Individual Fixed Effects

	(1)	(2)
In labor force	0.03***	0.03***
	(0.00)	(0.00)
Married		−0.05***
		(0.01)
Census region: Midwest		−0.02
		(0.03)
Census region: South		−0.03
		(0.02)
Census region: West		−0.07**
		(0.03)
_cons	−0.25	−0.21
	(0.16)	(0.18)
N	102894.00	102348.00
Within R^2	0.18	0.18
ρ	0.66	0.65
σ_u	0.46	0.45
σ_e	0.33	0.33
Age joint p	0.00	0.00
Wealth joint p	0.00	0.00
Year joint p	0.00	0.00
Region joint p	.	0.14

Within (household) fixed-effects estimator. Standard errors clustered by household. Time-invariant regressors absorbed. Column *(1)*: wealth deciles, age-bins, labor force, and year FE. Column *(2)*: Additional controls include time-varying marital status and census region. Reported ρ, σ_u, and σ_e are variance-component estimates from the FE decomposition.

The parameters ρ, σ_u, and σ_e are of particular interest, as they summarize the variance decomposition of the error term in the panel regression. In this framework, σ_u denotes the standard deviation of the unobserved individual-specific component, α_i, capturing persistent differences in returns across individuals. The parameter σ_e is the standard deviation of the idiosyncratic error term, ε_{it}, reflecting within-individual variation over time that is not explained by the included covariates.

The parameter ρ represents the fraction of the total variance attributable to the individual-specific component, defined as

$$\rho = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_e^2}.$$

A high value of ρ indicates that a substantial share of the variation in returns arises from persistent differences across individuals, providing evidence of heterogeneity in returns across respondents.

The within-R^2 measures the proportion of within-individual variation in returns explained by the time-varying regressors—how much of the deviation of an individual's returns from their own mean is accounted for by observables.

Focus on the second column of Table 2.10. There remains substantial within-person variation not captured by time-varying regressors $\sigma_e^2 (= .33)$. Heterogeneity across individuals captured by $\sigma_u^2 (= .45)$ is larger than in the RE or CRE specifications reported below, which explains why ρ is much higher here ($\rho = .65$). The within-R^2 in that column is $(= .18)$. Because time-invariant regressors are absorbed by the household fixed effects, the estimated individual-specific component captures persistent heterogeneity across households, including variation that would otherwise load on time-invariant observables.

Figure 2.16 shows the distribution of estimated fixed effects from the column *(2)* first-stage FE specification (the same extended time-varying controls as in that column), analogous to how Figure 2.18 reports the CRE individual effect when time-invariant controls are included. In this setting, where time-invariant factors like education and trust are absorbed, the estimated distribution is particularly comparable to empirical estimates of the same object using Norwegian population data from 2004-2015 by Fagereng et al. 2020 and using the PSID from 1999-2019 by Daminato and Pistaferri 2024.

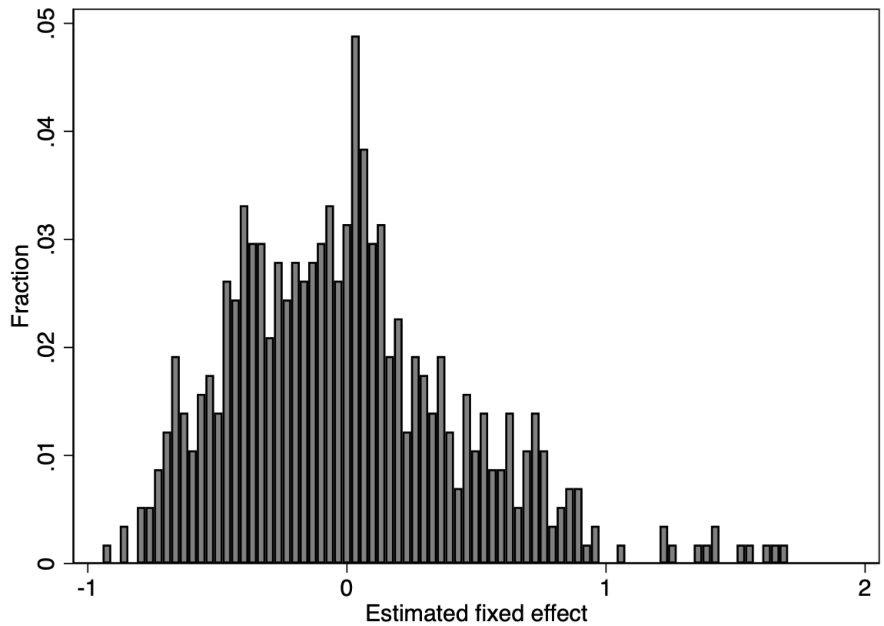


Figure 2.16: Distribution of estimated fixed effect (FE) with extended set of controls.

The fixed effect here is the residual household-specific intercept left after removing time-varying variation associated with wealth deciles, age bins, labor-force status, year effects, and in the extended specification marital status and region. Figure 2.17 shows that this estimated household fixed effect is strongly *downward* sloping in wealth percentile. [15]

[15] A direct check in Stata reproduces the same negative relationship in both correlation and simple-regression form, with the slope statistically different from zero at conventional levels.

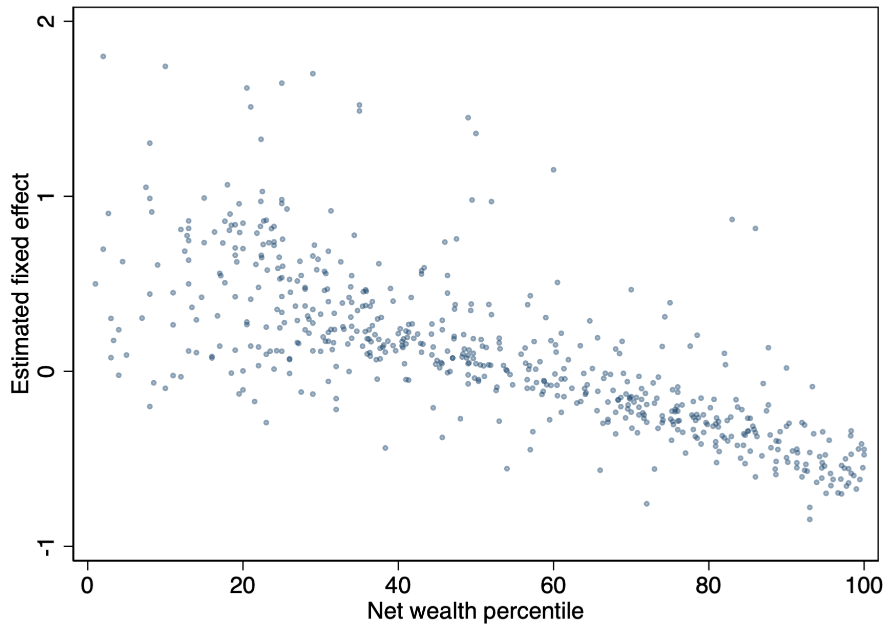


Figure 2.17: Estimated fixed effect vs. net wealth percentile (extended controls).

It is a bit unsavory to show that a host of time-invariant factors, including the hump-shaped trust relationship, matter for explaining the variation in trust, but then to exclude them when characterizing the persistent component of returns. Said differently, we would like to control for this variation (especially in trust) when describing how individuals differ in their returns on average.

To move in this direction, I begin by regressing the estimated fixed effect onto the time-invariant variables which were excluded from the regression once the indivdiual fixed effect was accounted for in the model. In table 2.11, race and education seem to have the most predictive power regarding variation in the predicted individual effects. The negative education coefficients are not a coding mistake: they are reproduced even in a simple cross-sectional regression of the predicted fixed effect on each education category, and they become more negative at higher education categories. At the same time, I hesitate to give these signs a strong structural interpretation. The left-hand side in this second-stage regression is itself a generated regressor from the first-stage FE model, so the coefficients should be read as descriptive correlations with the estimated residual household intercept after the first stage has already removed time-varying variation.

Table 2.11: Explaining Estimated Fixed Effect with Time-Invariant Controls

	(1)	(2)
Trust	-0.01	0.01
	(0.02)	(0.02)
Trust^2	0.00	-0.00
	(0.00)	(0.00)
HS Grad	-0.13^{**}	-0.05
	(0.05)	(0.06)
Some College	-0.13^{**}	-0.08
	(0.06)	(0.06)
College	-0.40^{***}	-0.30^{***}
	(0.07)	(0.08)
Postgrad	-0.49^{***}	-0.37^{***}
	(0.06)	(0.06)
Female		0.02
		(0.04)
NH Black		0.25^{***}
		(0.05)
Hispanic		0.26^{***}
		(0.06)
NH Other		0.05
		(0.07)
Born in U.S.		0.08
		(0.05)
_cons	0.36^{***}	-0.03
	(0.06)	(0.09)
N	575.00	573.00
Adj. R^2	0.12	0.17
Trust joint p	0.57	0.85
Race joint p	.	0.00
Educ. joint p	0.00	0.00

Cross-sectional OLS of predicted household fixed effects on time-invariant regressors. Heteroskedasticity-robust standard errors. The dependent variable in column *(1)* is the fixed effect from the Table 2.10 column *(1)* specification; in column *(2)* it is the fixed effect from column *(2)* of that table (so time-varying marital status and census region are already part of the first-stage FE when forming the left-hand side). Column *(1)*: education categories and trust; column *(2)*: adds gender, race/ethnicity, US-born, and census region.

2.4.3.2 The Random Effects (RE) model

The model with fixed effects is one specification of the general panel regression model, when the term capturing the latent time-invariant variation affecting returns for individuals α_i is possibly correlated with the observable, time-varying variables X_{it}. The Random Effects (RE) model allows for estimation and inference in a similar setting by assuming that the α_i's are normally distributed with some constant variance, but that $Cov(\alpha_i, X_{it}) = 0$.

Since it is not clear that this persistent component capturing differences in returns is uncorrelated with factors like wealth, age, and labor force particiaption, the RE orthogonality restriction is difficult to defend a priori. The specification evidence reported below and in the CRE discussion strongly rejects that restriction, so RE is not the preferred model for the trust-return relationship. I therefore treat Table 2.12 mainly as a benchmark for comparing variance decompositions across panel estimators.

That said, one can still describe the variance structure of the estimated fixed effect in the RE setting. Focus on the second column of Table 2.12. There is significant within-person variation not captured by time-varying regressors $\sigma_e^2 (= .36)$. However, the estimates of heterogeneity across individuals captured by σ_u^2 is much smaller here $(= .18)$. These two features combined explains why the value for ρ is much closer to 0 here than in the FE model. That said, a value of $\rho = .20$ still represents substaintial heterogeneity.

Table 2.12: The RE Model for Trust and Returns to Net Wealth

	(1)	(2)
Trust	0.02*	0.02*
	(0.01)	(0.01)
Trust^2	−0.00	−0.00
	(0.00)	(0.00)
In labor force	0.03	0.03
	(0.02)	(0.02)
Married	−0.09***	−0.08***
	(0.02)	(0.02)
HS Grad	−0.06*	−0.05
	(0.03)	(0.03)
Some College	−0.07*	−0.06*
	(0.03)	(0.03)
College	−0.16***	−0.15***
	(0.04)	(0.04)
Postgrad	−0.18***	−0.17***
	(0.04)	(0.04)
Female		−0.03
		(0.02)
NH Black		0.08**
		(0.03)
Hispanic		0.04
		(0.04)
NH Other		0.01
		(0.04)
Born in U.S.		0.03
		(0.03)
_cons	−0.56***	−0.63***
	(0.05)	(0.07)
N	2919.00	2899.00
Within R^2	0.10	0.11
ρ	0.21	0.20
σ_u	0.19	0.18
σ_e	0.36	0.36
Trust joint p	0.18	0.08
Age joint p	0.00	0.00
Wealth joint p	0.00	0.00
Race joint p	.	0.09
Region joint p	.	0.58
Year joint p	0.00	0.00
Educ. joint p	0.00	0.00

Random-effects GLS. Standard errors clustered by household. Within R^2 reported. ρ, σ_u, and σ_e are variance-component estimates from the RE decomposition. Joint tests are Wald tests for the indicated covariate blocks.

2.4.3.3 The Correlated Random Effects (CRE) model

The rejection of the RE model is informative, but it does not help with the issue that we would like to account for the variation of time-invariant factors (like education and trust) while still characterizing the persistent component of returns for individuals. To do so, we move to the Correlated Random Effects (CRE) model, where the assumption of is $Cov(\alpha_i, X_{it}) = 0$ relaxed; the persistent component and time-varying factors are allowed to be correlated.

The correlated random effects (CRE) model augments the regression with the individual-specific means of the time-varying regressors:

$$r_{it}^{net} = \alpha_i + \lambda_t + X_{it}'\beta + \bar{X}_i'\theta + \varepsilon_{it},$$

where $\bar{X}_i$ denotes the within-individual averages of the regressors. The term $\bar{X}_i'\theta$ explicitly captures the correlation between the unobserved individual component α_i and the observed variables $X_{i,t}$. In the manual implementation used here, the reported Mundlak test is the cluster-robust Wald test that the coefficients on all added respondent-level mean terms are jointly zero, which is the manual equivalent of Stata 19's `estat mundlak`[16]. That null is overwhelmingly rejected in both CRE

[16]In the empirical implementation below, the correlated random-effects specification is estimated manually using the traditional Mundlak device rather than Stata 19's dedicated `xtreg, cre` syntax. Concretely, I estimate `xtreg, re` and augment the right-hand side with respondent-level means of the time-varying regressors, which is the same econometric construction that earlier Stata workflows required users to code by hand before Stata 19 automated it and paired it with `estat mundlak`. The panel used in file 38 is strongly balanced, despite the legacy dataset name, so the within-respondent means of the survey-year indicators are constants across households. For that reason, I do not add averages of the year dummies separately in the Mundlak augmentation: in this balanced-panel setting they are collinear with the intercept and add no identifying variation.

specifications, with $p \approx 8.0 \times 10^{-44}$ in the baseline column and $p \approx 1.8 \times 10^{-43}$ in the extended column, strongly favoring the CRE specification over standard RE. The key finding of this paper is that, *the coefficeints on the trust coefficients are statistically significant in the CRE setting, where the persistent component of returns can be estimated as well*. This can be seen in Table 2.13.

With the estimated individual fixed effect being a key empirical object of this paper, the interpretation of the variance structure is important here. Focus on the second column of Table 2.13. Much like the FE and RE models, there is significant within-person variation not captured by time-varying regressors $\sigma_e^2 (= .36)$. Moreover, the estimates of heterogeneity across individuals captured by σ_u^2 are smallest here $(= .15)$. These two features combined explains why the value for $\rho = .15$, documenting significant cross-sectional variation.

Table 2.13: The CRE (Mundlak) Model for Trust and Returns to Net Wealth

	(1)	(2)
Trust	0.03***	0.03***
	(0.01)	(0.01)
Trust^2	−0.00***	−0.00***
	(0.00)	(0.00)
In labor force	0.01	0.01
	(0.02)	(0.02)
Married	−0.04	−0.03
	(0.04)	(0.04)
HS Grad	0.02	−0.00
	(0.02)	(0.03)
Some College	−0.00	−0.02
	(0.03)	(0.03)
College	0.01	−0.01
	(0.03)	(0.04)
Postgrad	−0.00	−0.01
	(0.03)	(0.03)
Female		−0.03**
		(0.02)
NH Black		−0.01
		(0.03)
Hispanic		−0.07**
		(0.03)
NH Other		−0.05
		(0.04)
Born in U.S.		0.02
		(0.02)
_cons	−0.67***	−0.62***
	(0.21)	(0.22)
N	2919.00	2899.00
Within R^2	0.15	0.15
ρ	0.16	0.15
σ_u	0.16	0.15
σ_e	0.36	0.36
Trust joint p	0.01	0.02
Age joint p	0.00	0.00
Wealth joint p	0.00	0.00
Race joint p	.	0.14
Region joint p	.	0.18
Year joint p	0.00	0.00
Educ. joint p	0.87	0.96
Mundlak test p	0.00	0.00

Random effects with respondent-level Mundlak means of the time-varying regressors. Standard errors clustered by household. Within R^2 reported. ρ, σ_u, and σ_e are variance-component estimates from the RE decomposition of the augmented CRE specification. "Mundlak test p" reports the cluster-robust Wald test of the null that the coefficients on all added panel means are jointly zero, the manual equivalent of Stata 19's `estat mundlak`.

Figure 2.18 shows the distribution of the estimated CRE individual effect from the specification when controls are included. The estimated distribution is comparable estimated the using Norwegian population data and using the PSID, though less so than the FE model (especially near the center of the distriution).

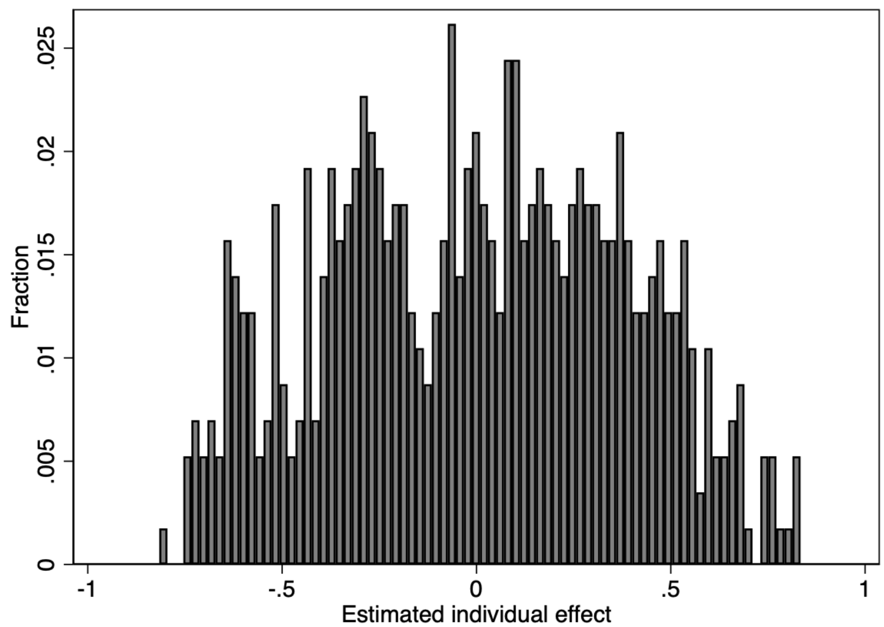


Figure 2.18: Distribution of Estimated Fixed Effect (CRE) with extended set of controls..

Figure 2.19 shows the same basic pattern for the CRE individual effect. The relationship is again downward sloping, though flatter than in the FE case. The interpretation is the same as above: this figure is about the estimated residual persistent component after accounting for observed time-varying covariates and, in the CRE specification, their respondent-level means. It should therefore not be read as the raw relationship between returns and wealth, which is shown earlier in Figure 2.12.

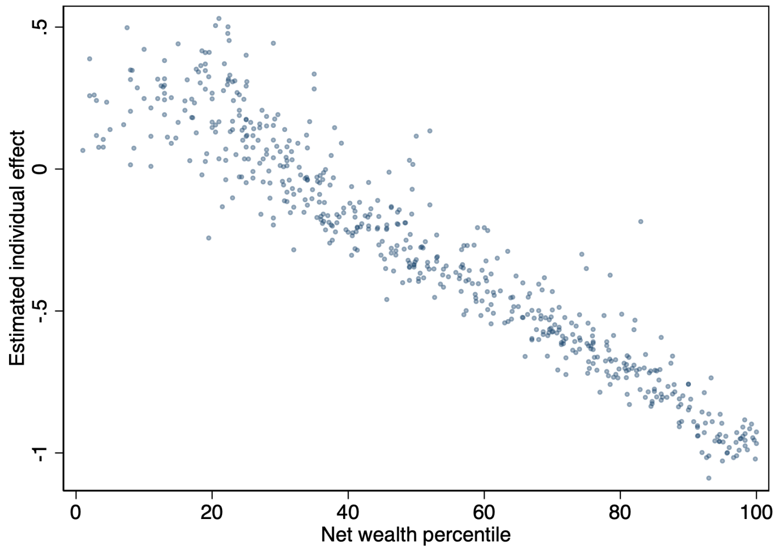


Figure 2.19: Estimated CRE individual effect vs. net wealth percentile (extended controls).

2.4.4 Which level of trust maximizes returns?

By choosing to model the hump-shaped relationship between trust and returns using a combination of a linear and quadratic terms, I can describe the level of trust which maximizes returns to net wealth in the following way. Starting from

$$r_{it}^{net} = \alpha_i + \lambda_t + X_{it}'\beta + \varepsilon_{it},$$

holding controls fixed, the implied marginal effect of trust is

$$\frac{\partial r_{it}^{net}}{\partial \, Trust_i} = \beta_1 + 2\beta_2 \, Trust_i.$$

Setting this derivative to zero gives the critical point $Trust^* = -\beta_1/(2\beta_2)$. When $\hat{\beta}_1 \geq 0$ and $\hat{\beta}_2 \leq 0$, the fitted relationship in trust is concave (hump-shaped), and the estimated trust level that maximizes fitted returns to net wealth is $Trust^* = -\hat{\beta}_1/(2\hat{\beta}_2)$, reported as $(Trust)^*$ in the Table 2.14.

Table 2.14: The Level of Trust which Maximizes Returns

	Avg returns	Pooled	RE	CRE
Panel A: Without controls				
N	595	2,919	2,919	2,919
Both sig?	Yes	Yes	No	Yes
Joint test	0.0251	0.0062	0.1808	0.0088
$(Trust)^*$	6.7002	6.8860	6.7574	6.3383
SE$(Trust)^*$	0.9311	0.8357	1.4156	0.5472
p$(Trust)^*$	0.0000	0.0000	0.0000	0.0000
Conf. Interv.	[4.8753, 8.5251]	[5.2482, 8.5239]	[3.9829, 9.5319]	[5.2659, 7.4107]
Panel B: With controls				
N	592	2,899	2,899	2,899
Both sig?	Yes	Yes	No	Yes
Joint test	0.0201	0.0018	0.0770	0.0181
$(Trust)^*$	7.0701	7.6088	7.5538	6.5934
SE$(Trust)^*$	1.1220	1.1067	1.7324	0.6601
p$(Trust)^*$	0.0000	0.0000	0.0000	0.0000
Conf. Interv.	[4.8711, 9.2691]	[5.4397, 9.7779]	[4.1583, 10.9493]	[5.2996, 7.8871]

Panel A: minimal time-invariant controls (education only in panel specifications). Panel B: full demographics. Trust* computed as $-\hat{\beta}_1/(2\hat{\beta}_2)$; standard errors, p-values for H_0: Trust$^* = 0$, and 95% confidence intervals use the delta method (`nlcom` in Stata). "Both sig?" indicates both trust terms significant at 10% in separate tests; joint test is Wald on both trust terms.

In general, the estimated trust level that maximizes fitted returns lies in a moderate range. In the average-return and CRE models it is around 6.3 to 7.1, while the pooled specification places the peak somewhat higher once controls are added, at about 7.6. When controls are added, the maximal level of trust is generally higher than when controls are excluded for each regression model tested. Because Trust^* is a nonlinear function of $\hat{\beta}_1$ and $\hat{\beta}_2$, Stata uses the delta method to approximate its standard error. I use this to report the standard error on the estimated maximal level of trust, as well as 95% confidence intervals, and p-values which test $H_0 : \mathit{Trust}^* = 0$.

2.4.4.1 Distribution of observed trust around the estimated maximum

Figure 2.20 overlays the observed distribution of responses to the generalized-trust question in the 2020 HRS survey with the three extended-specification maxima. Those three models are average returns, pooled OLS, and CRE. This makes the comparison easier to read substantively, because it shows whether the implied maxima lie in the thick part of the observed support rather than at the extreme tails of the trust distribution.

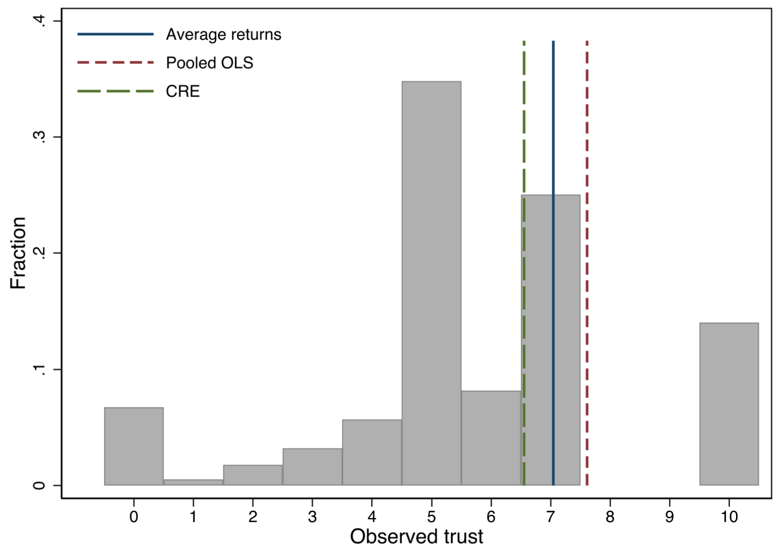


Figure 2.20: Observed responses to the generalized-trust question in the 2020 HRS survey, with estimated return-maximizing trust levels from the extended average-return, pooled, and CRE specifications.

The visual pattern is reassuring. The fitted maxima all fall in the same moderate range already emphasized by Table 2.14, and they do so in a part of the trust distribution with substantial empirical support. I omit the FE model from this comparison because trust and trust^2 are absorbed by the within transformation and therefore are not estimated directly in the FE return equation. I also omit RE because the specification tests reported earlier reject it in favor of CRE.

2.4.5 Predicted vs Actual returns by race

An important finding in the HRS data is that i) the `NH Black` racial category predicts trust well, ii) regressions of returns to net wealth on race without trust show nontrivial racial differences, and iii) once trust (and trust^2) enter the return regressions, those differences attenuate. Table 2.15 summarizes the no-trust race coefficients for the average-return and pooled specifications using the reduced and full control sets.

Table 2.15: Race Coefficients in No-Trust Average-Return and Pooled Specifications

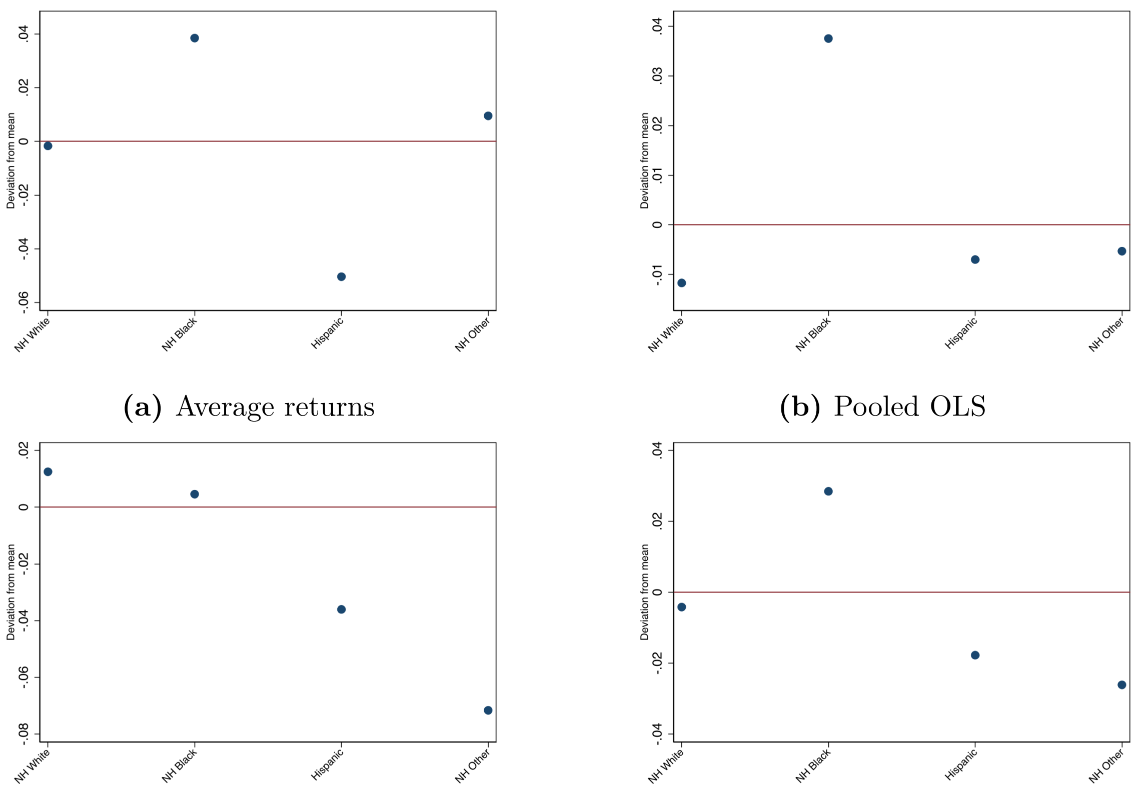
	(1) Avg. red.	(2) Avg. full	(3) Pooled red.	(4) Pooled full
NH Black	0.01	0.01	0.07***	0.06***
	(0.01)	(0.01)	(0.00)	(0.00)
Hispanic	0.01	−0.01	0.03***	0.03***
	(0.01)	(0.01)	(0.00)	(0.01)
NH Other	−0.00	−0.02	0.02**	0.02**
	(0.02)	(0.02)	(0.01)	(0.01)
N	10678.00	10613.00	102711.00	102091.00
Adj. R^2	0.04	0.04	0.03	0.03
Race joint p	0.44	0.32	0.00	0.00

Columns *(1)–(2)* are cross-sectional OLS of 5% winsorized average returns to net wealth; columns *(3)–(4)* are pooled person-year OLS of 5% winsorized returns to net wealth with standard errors clustered by household. In each model family, the reduced column includes race together with the baseline return controls, while the full column adds the extended demographic controls. Only race coefficients are displayed.

2.4.5.1 Predicted returns by race

Figure 2.21 does this for the final displayed average-return, pooled OLS, FE, and CRE models. For the panel specifications, I first predict returns at the person-year level on the estimation sample and then average those fitted values within respondent, so each respondent contributes one fitted-return value before taking race-specific means. Within each panel, a point above zero indicates that the group's average predicted return exceeds the model-wide average, while a point below zero indicates the opposite.

In each of the specifications, the mean predicted returns are higher on average for `NH Black` respondents. In the pooled OLS and CRE versions of the model, the trust terms are estimated directly, and these are the same figures that show the clearest positive deviation in mean predicted returns for Black respondents. The FE model, which does not estimate the effect of time-invariant trust on returns, produces a deviation closer to zero (though still positive). No other racial categories show a robust, positive deviation from mean predicted returns.



(a) Average returns

(b) Pooled OLS

(c) FE

(d) CRE

Figure 2.21: Race-group deviations from the overall mean predicted return by specification.

The comparison is useful because it compresses the fitted racial differences into a single object that can be compared across model families. The FE panel should be interpreted more cautiously than the other three. In the FE model, the predictions are formed after absorbing the time-invariant regressors into the household effect, so they do not separately identify the roles of trust, trust^2, education, gender, race/ethnicity, or U.S.-born status. Put differently, the FE predicted values summarize fitted returns once all time-invariant heterogeneity has been loaded into the fixed effect term, whereas the average-return, pooled, and CRE panels retain direct roles for those observables in the fitted outcome.

2.4.5.2 Actual returns by race

Figure 2.22 makes the realized-return comparison in the most direct way: I first average 5% winsorized returns to net wealth within respondent over the observed panel, then estimate the race-specific means of those respondent-level averages and subtract the overall respondent-level mean. The confidence intervals therefore summarize the sampling uncertainty around the race-specific respondent means. Because this object is built directly from realized returns rather than from any fitted model, it is common across the specifications above.

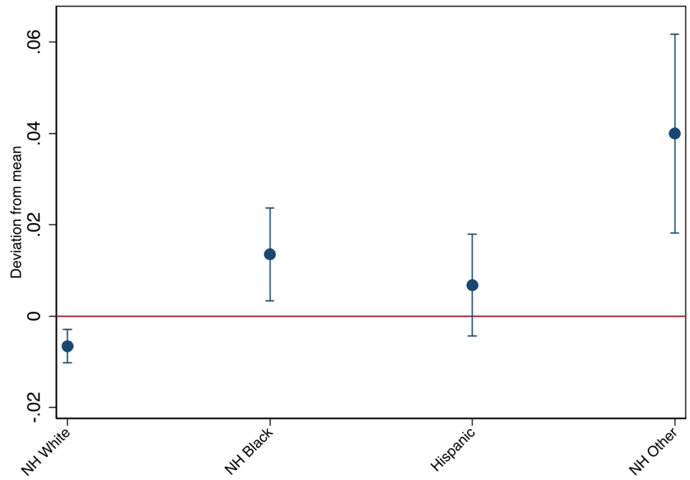


Figure 2.22: Race-group deviations from the overall mean realized return, with 95% confidence intervals for the respondent-level race means.

The realized-return comparison is useful as a check on the fitted-return evidence because it removes the model-specific machinery and looks directly at the respondent-level return averages in the data. The point estimates still place `NH White` slightly below the overall respondent-level mean, while the other three groups lie above it, with the largest positive deviation for `NH Other`. The confidence intervals also make clear how much noise surrounds those respondent-level race means.

2.5 Extension: controlling for risk exposure

The key point about the documented persistent component to returns is that, even when controlling for risk preferences and exposure, individuals systematic earn different returns on the same assets[17]. Moreover, the descriptive statistics for mean returns

[17]This point is further corroborated by evidence of heterogeneity in returns on safe assets like deposit accounts. An example from the academic literature is Deuflhard et al. 2014, but one can compare

show that returns do not seem stable over the time period (ex. Global Financial Crisis). To address this concern, I re-estimate the cross-section (average returns), pooled person-year OLS, and panel (FE and CRE) specifications after augmenting the set of controls with portfolio-share variables (core assets, IRA assets, residential assets, long-term debt, and other debt) fully interacted with year dummies.

2.5.1 Trust and average returns

I begin with the analysis of trust and average returns, now adding 2020 portfolio shares as an attempt at controlling for risk exposure by estimating the cross-sectional regression equation

$$\bar{r}_i^{net} = X_i' \beta + \sum_{j=1}^{J} \gamma_j s_{ji} + \varepsilon_i.$$

Table 2.16 shows that the hump-shaped relationship between trust and average returns survives this extension in both the baseline and extended specifications: the trust coefficients remain individually significant, and the joint test on the linear and quadratic trust terms is rejected in both columns.

Relative to the corresponding average-return specifications in Table 2.8, adding shares increases model fit substantially: the adjusted R^2 in the baseline model rises from .07 to .14 and from .09 to .17 in the extended model. This suggests that portfolio composition explains a meaningful share of the cross-sectional variation in average

rates offered on deposit account at commercial banks like Bank of America versus local banks and see similar discrepancies in everyday life.

Table 2.16: Trust and Average Returns to Net Wealth with Portfolio-Share Controls

	(1)	(2)
Trust	0.04***	0.04***
	(0.01)	(0.01)
Trust^2	−0.00***	−0.00***
	(0.00)	(0.00)
Share core	−0.18***	−0.16**
	(0.07)	(0.07)
Share IRA	−0.12	−0.07
	(0.08)	(0.08)
Share residential	−0.13**	−0.11**
	(0.06)	(0.05)
Share long-term debt	−0.05	−0.05
	(0.03)	(0.03)
Share other debt	0.00	−0.00
	(0.00)	(0.01)
In labor force	0.05*	0.05*
	(0.03)	(0.02)
HS Grad	−0.02	−0.02
	(0.03)	(0.03)
Some College	−0.02	−0.03
	(0.03)	(0.03)
College	−0.05	−0.06*
	(0.03)	(0.03)
Postgrad	−0.06	−0.07**
	(0.03)	(0.03)
Female		−0.03*
		(0.02)
NH Black		0.02
		(0.03)
Hispanic		−0.05
		(0.03)
NH Other		0.06
		(0.06)
Married		−0.00
		(0.02)
Born in U.S.		0.01
		(0.03)
Midwest		−0.04
		(0.03)
South		0.03
		(0.03)
West		0.05
		(0.03)
_cons	0.03	0.04
	(0.07)	(0.08)
N	439.00	437.00
Adj. R^2	0.14	0.17
Trust joint p	0.00	0.00
Age joint p	0.00	0.02
Wealth joint p	0.03	0.01
Region joint p	.	0.00
Share (asset) joint p	0.03	0.05
Share (debt) joint p	0.32	0.27

OLS. Heteroskedasticity-robust standard errors. Trust and portfolio shares are measured in 2020; dependent variable is average returns to net wealth (winsorized). Column *(1)* uses the baseline controls from the main text plus asset and debt shares; column *(2)* adds the full demographic controls. For readability, footer rows for joint tests with $p \geq 0.10$ are omitted from the final presentation, though the underlying regressions include the full control set.

returns. At the same time, the trust coefficients are essentially unchanged in sign and remain jointly significant, which is important for interpretation: controlling for observed portfolio shares improves fit, but it does not eliminate the hump-shaped relationship between trust and long-run returns. Furthermore, the Wald test on the coefficeints for the asset classes being jointly zero is rejected, suggesting that the proxies for risk exposure are predictive of returns to net wealth.

2.5.2 Pooled OLS with portfolio-share controls

Table 2.17 reports person-year OLS estimates that mirror the pooled specification in the main text (Table 2.9), augmented with the portfolio-share controls and their interactions with survey year. As in the main results, standard errors are clustered by household. Column *(1)* includes the same baseline time-varying controls as in the pooled baseline column of Table 2.9, together with the share blocks; column *(2)* adds the full time-invariant demographic controls. The trust terms remain jointly significant in both columns. In this pooled setting, the share levels themselves are not jointly significant, but several of the share-by-year interaction blocks are, so the time-varying portfolio structure still loads on returns.

Table 2.17: Pooled OLS of Returns to Net Wealth on Trust with Portfolio-Share Controls

	(1)	(2)
Trust	0.03***	0.03**
	(0.01)	(0.01)
Trust^2	−0.00**	−0.00**
	(0.00)	(0.00)
Share core	0.23	0.24
	(0.23)	(0.24)
Share IRA	−0.09	−0.09
	(0.24)	(0.24)
Share residential	0.02	0.00
	(0.22)	(0.23)
Share long-term debt	−0.16	−0.18
	(0.28)	(0.27)
Share other debt	0.15	0.16
	(0.16)	(0.16)
In labor force	0.02	0.02
	(0.02)	(0.02)
Married	−0.07***	−0.07***
	(0.02)	(0.02)
HS Grad	−0.06**	−0.06**
	(0.02)	(0.03)
Some College	−0.05*	−0.05*
	(0.03)	(0.03)
College	−0.10***	−0.10***
	(0.03)	(0.03)
Postgrad	−0.10***	−0.10***
	(0.03)	(0.03)
Female		−0.02
		(0.02)
NH Black		0.04
		(0.02)
Hispanic		−0.00
		(0.03)
NH Other		0.03
		(0.04)
Born in U.S.		0.02
		(0.02)
Midwest		0.00
		(0.02)
South		0.04**
		(0.02)
West		0.04*
		(0.03)
_cons	−0.47**	−0.51**
	(0.21)	(0.22)
N	2886.00	2866.00
Adj. R^2	0.06	0.06
Trust joint p	0.02	0.01
Age joint p	0.00	0.00
Wealth joint p	0.01	0.01
Race joint p	.	0.43
Region joint p	.	0.06
Year joint p	0.00	0.00
Educ. joint p	0.01	0.01
Share (asset) joint p	0.38	0.31
Share (debt) joint p	0.57	0.52
Share core×year p	0.00	0.01
Share IRA×year p	0.01	0.01
Share res×year p	0.00	0.01
Share debt-long×year p	0.10	0.13
Share debt-other×year p	0.00	0.00

Person-year OLS. Standard errors clustered by household. Portfolio shares enter as levels with share×year interactions (parallel to the CRE specification). Column *(1)*: baseline time-varying controls plus education; column *(2)* adds gender, race/ethnicity, U.S.-born status, and census region. Footer rows report Wald tests for the three asset-share levels and for the two debt-share levels.

2.5.3 Augmenting the FE model

I next add the portfolio-share controls to the panel regression model with respondent-level fixed effects. Since the FE model absorbs the time-invariant trust measure, the point here is not to estimate trust itself, but to see how much these proxies for risk exposure help explain within-respondent variation in returns.

Let j index asset classes.

$$r_{it}^{net} = \alpha_i + \lambda_t + X_{it}'\beta + \sum_{j=1}^{J} \left[\gamma_j s_{jit} + \sum_{t' \in \mathcal{T} \setminus \{t_0\}} \delta_{jt'} \left(s_{jit} \cdot 1\{t = t'\} \right) \right].$$

Table 2.18 shows the results of the estimation after adding the additional controls relevant to risk exposure. Relative to Table 2.10, adding the share controls increases the within-R^2 from .18 to .20 in both columns. The variance structure changes only modestly: ρ falls from .66 and .65 to .64 and .63, while σ_u stays around .46–.47 and σ_e rises slightly from about .33 to .35. Thus, *controlling for risk exposure does not effect the estimated distribution of fixed effects much.* This is useful information, since the indivdiual fixed effect is estimated after removing all time-varying variation from returns.

The joint tests on the remaining control variables also show a robustness to the risk controls. Age, wealth, and year effects remain strongly significant. The asset-share levels are not jointly significant, but the debt-share block is. Over time, the interaction terms for core shares, IRA shares, residential shares, and other debt are jointly significant, while the interaction for long-term debt is not.

Table 2.18: Panel OLS with Individual Fixed Effects and Portfolio-Share Controls

	(1)	(2)
In labor force	0.02	0.01
	(0.02)	(0.02)
Share core	0.36	0.36
	(0.22)	(0.23)
Share IRA	−0.05	−0.05
	(0.22)	(0.22)
Share residential	0.17	0.18
	(0.20)	(0.20)
Share long-term debt	−0.43	−0.42
	(0.26)	(0.26)
Share other debt	−0.74**	−0.74**
	(0.35)	(0.34)
Married		−0.04
		(0.05)
Midwest		0.01
		(0.11)
South		−0.06
		(0.07)
West		−0.06
		(0.10)
_cons	−0.80**	−0.76**
	(0.34)	(0.35)
N	2886.00	2868.00
Within R^2	0.20	0.20
ρ	0.64	0.63
σ_u	0.47	0.46
σ_e	0.35	0.35
Age joint p	0.00	0.00
Wealth joint p	0.00	0.00
Year joint p	0.01	0.01
Share (asset) joint p	0.13	0.14
Share (debt) joint p	0.01	0.01
Share core×year p	0.02	0.03
Share IRA×year p	0.00	0.00
Share res×year p	0.01	0.02
Share debt-long×year p	0.17	0.15
Share debt-other×year p	0.00	0.00

Within (household) fixed-effects estimator. Standard errors clustered by household. Column *(1)* adds time-varying portfolio shares and share x year interactions to the baseline FE specification; column *(2)* adds the extended time-varying controls. Reported ρ, σ_u, and σ_e are variance-component estimates from the FE decomposition. In the final table presentation, joint-test rows with $p \geq 0.10$ are omitted for readability, but all share and interaction terms remain in the regression.

2.5.3.1 Augmenting the CRE model

The correlated random effects specification is the key panel model of interest here, since it retains the time-invariant trust measure while still allowing the persistent component to be correlated with the observed time-varying regressors.

Table 2.19: The CRE (Mundlak) Model with Portfolio-Share Controls

	(1)	(2)
Trust	0.03***	0.03***
	(0.01)	(0.01)
Trust^2	−0.00***	−0.00***
	(0.00)	(0.00)
Share core	0.23	0.24
	(0.20)	(0.20)
Share IRA	−0.11	−0.11
	(0.19)	(0.19)
Share residential	0.12	0.12
	(0.18)	(0.18)
Share long-term debt	−0.45*	−0.46*
	(0.26)	(0.25)
Share other debt	−0.26*	−0.26
	(0.16)	(0.16)
HS Grad	−0.00	−0.01
	(0.02)	(0.02)
Some College	−0.00	−0.01
	(0.03)	(0.03)
College	0.01	−0.01
	(0.03)	(0.03)
Postgrad	−0.00	−0.01
	(0.03)	(0.03)
In labor force	0.01	0.01
	(0.02)	(0.02)
Married	−0.03	−0.02
	(0.04)	(0.04)
Female		−0.03**
		(0.02)
NH Black		−0.01
		(0.03)
Hispanic		−0.06**
		(0.03)
NH Other		−0.02
		(0.03)
Born in U.S.		0.01
		(0.02)
Midwest		−0.00
		(0.02)
South		0.02
		(0.02)
West		0.05*
		(0.03)
_cons	−0.33	−0.28
	(0.26)	(0.27)
N	2886.00	2866.00
Within R^2	0.19	0.19
ρ	0.11	0.11
σ_u	0.12	0.12
σ_e	0.35	0.35
Trust joint p	0.00	0.00
Age joint p	0.00	0.00
Wealth joint p	0.00	0.00
Year joint p	0.00	0.00
Share (asset) joint p	0.20	0.19
Share (debt) joint p	0.04	0.04
Share core×year p	0.00	0.01
Share IRA×year p	0.00	0.00
Share res×year p	0.01	0.01
Share debt-long×year p	0.00	0.01
Share debt-other×year p	0.00	0.00
Mundlak test p	0.00	0.00

Random effects with respondent-level Mundlak means of the time-varying regressors, augmented with portfolio-share levels and share x year interactions. Standard errors clustered by household. Column *(1)* uses the baseline time-invariant controls; column *(2)* adds the full demographic controls. Reported ρ, σ_u, and σ_e are variance-component estimates from the augmented CRE specification. The footer includes separate joint tests for asset-share levels (core, IRA, residential) and debt-share levels (long-term, other), then joint tests for each share×year block. "Mundlak test p" reports the cluster-robust Wald test of the null that the coefficients on all added panel means are jointly zero, the manual equivalent of Stata 19's `estat mundlak`.

The trust results survive this extension. The coefficients on trust remain positive on the linear term and negative on the quadratic term, and the joint test on the trust terms is still rejected in both columns. The corresponding Mundlak specification test is again overwhelmingly rejected, with $p \approx 3.7 \times 10^{-43}$ in the baseline share-augmented CRE and $p \approx 7.0 \times 10^{-43}$ in the extended specification, so the evidence against standard RE remains just as strong once portfolio composition is introduced. Relative to Table 2.13, adding shares raises the within-R^2 from .15 to .19 in both columns. At the same time, ρ falls from .16 and .15 to .11, which suggest a smaller share of the total variance is attributed to the persistent component after controlling for risk. Furthermore, σ_u declines from about .15–.16 to .12, which implies that less unexplained variation is being attributed to persistent respondent-level differences. In other words, some of the previous variation across respondents has now been explained by risk exposure. Lastly, σ_e remains around .35–.36.

The joint tests again show that the interaction structure does most of the work. The debt-share block is jointly significant, while the asset-share levels are not. But the interactions for core shares, IRA shares, residential shares, long-term debt, and other debt are all jointly significant. By contrast, the broader time-invariant blocks for education, race, and census region are not.

The key takeaway is that *controlling for risk exposure in the CRE model increases explained variability, while retaining the hump-shaped relationship between trust and returns. The estiamted persistent component is estimated more accurately in terms of a smaller stndard deviation across respondents and a lower share of the total variance attributed to the persistent component.*

2.5.4 Maximizing returns with trust

As in the main text, I summarize the implied interior turning point of the quadratic trust specification by reporting the trust level that maximizes fitted returns. Because the FE model absorbs the time-invariant trust measure, the natural comparison across models that estimate trust in levels is among average returns, pooled OLS, and CRE. Table 2.20 reports turning points for those three model families in both baseline and extended specifications.

The estimated trust-maximizing values remain in the same moderate range emphasized in the main results. In the average-return regressions, the turning point lies around 6.3 in the baseline specification and 6.7 in the extended one; in the pooled OLS regressions with shares, it lies in the same neighborhood as the person-year pooled estimates in Table 2.14; in the CRE regressions, it lies around 6.3 and 6.6. Compared with Table 2.14, these values move only modestly once portfolio shares are added, with the CRE turning point especially stable. The confidence intervals remain fairly tight, and all reported turning points are statistically different from zero. Substantively, this reinforces the main conclusion: even after conditioning on detailed portfolio composition and its time-varying association with returns, the fitted relationship still suggests that households with moderate trust levels earn the highest returns to net wealth.

2.5.4.1 Distribution of observed trust around the estimated maximum

Figure 2.23 repeats this comparison for the share-augmented specifications, using the observed responses to the generalized-trust question in the 2020 HRS survey. The

Table 2.20: The Level of Trust which Maximizes Returns with Portfolio-Share Controls

	Avg returns	Pooled OLS	CRE
Panel A: Baseline controls			
N	439	2,886	2,886
Both sig?	Yes	Yes	Yes
Joint test	0.0050	0.0194	0.0021
$(\mathit{Trust})^*$	6.2593	6.8850	6.3272
SE$(\mathit{Trust})^*$	0.5366	0.8970	0.4895
p$(\mathit{Trust})^*$	0.0000	0.0000	0.0000
Conf. Interv.	[5.2076, 7.3110]	[5.1270, 8.6431]	[5.3678, 7.2867]
Panel B: Extended controls			
N	437	2,866	2,866
Both sig?	Yes	Yes	Yes
Joint test	0.0018	0.0108	0.0050
$(\mathit{Trust})^*$	6.6659	7.5678	6.5955
SE$(\mathit{Trust})^*$	0.6099	1.1935	0.6103
p$(\mathit{Trust})^*$	0.0000	0.0000	0.0000
Conf. Interv.	[5.4707, 7.8612]	[5.2286, 9.9071]	[5.3992, 7.7917]

Trust* is computed as $-\hat{\beta}_1/(2\hat{\beta}_2)$ from the share-augmented quadratic specifications. Standard errors, p-values for $H_0:$ Trust* $= 0$, and 95% confidence intervals use the delta method. "Both sig?" indicates whether the linear and quadratic trust terms are each significant at 10%, while "Joint test" reports the Wald test on the two trust terms.

vertical lines mark the estimated maxima from the extended average-return, extended pooled OLS, and extended CRE specifications (Table 2.20). I omit FE because trust and trust^2 are absorbed in the within specification.

The figure shows that the implied maxima remain in the same interior part of the trust distribution as in the main results. This matters for interpretation: once portfolio

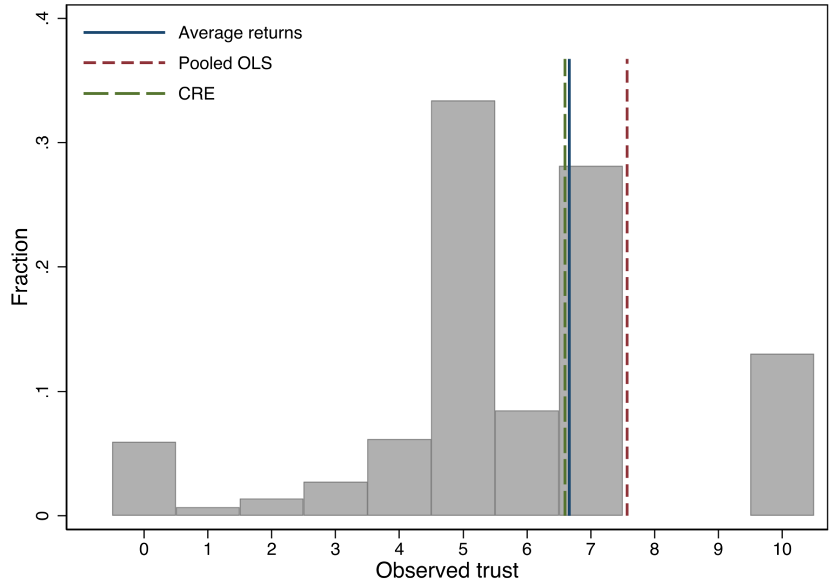
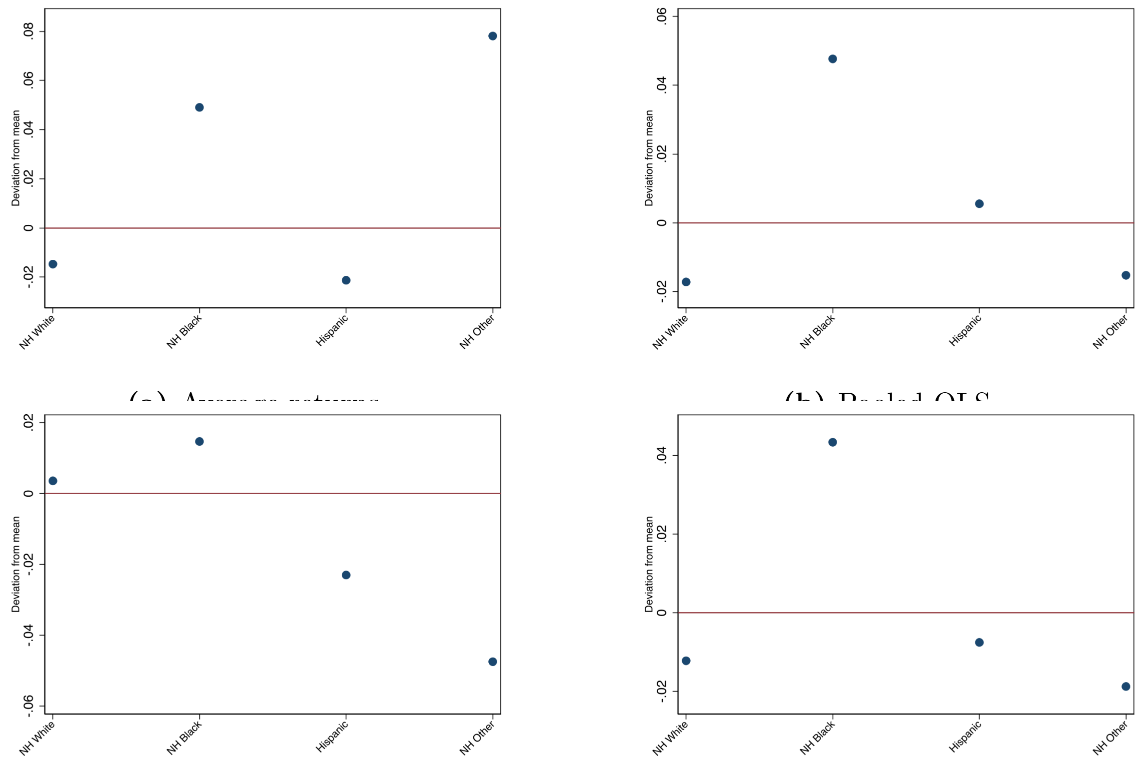


Figure 2.23: Observed responses to the generalized-trust question in the 2020 HRS survey, with estimated return-maximizing trust levels from the extended average-return, pooled OLS, and CRE specifications with portfolio-share controls.

shares are added, the estimated peak of the fitted trust-return profile still lies in a region of trust with substantial empirical support rather than at the boundary of the observed data.

2.5.4.2 Predicted returns by race

Figure 2.24 extends the race-based comparison of fitted returns to the portfolio-share specifications. As in the main results, each point reports a race group's mean predicted return relative to the overall mean predicted return from that model. For pooled OLS, FE, and CRE, I first predict returns at the person-year level on the estimation sample and then average those fitted values within respondent, so each respondent contributes one fitted-return value before taking race-specific means.



(c) FE

(d) CRE

Figure 2.24: Race-specific mean predicted return minus the overall mean predicted return for each specification with portfolio-share controls (subfigure titles). The vertical axis in the exported figures is abbreviated "Deviation from mean"; see Figure 2.21 for the parallel main-results construction.

The extension is informative because it asks whether the fitted racial pattern is materially changed once portfolio composition is added to the specification. As before, the FE panel requires a narrower interpretation: these fitted values are constructed after absorbing the time-invariant regressors into the respondent effect, so they do not separately identify trust, trust^2, education, gender, race/ethnicity, or U.S.-born status. The average-return, pooled OLS, and CRE panels, by contrast, retain direct fitted roles for those observables while also conditioning on the portfolio-share controls.

The realized-return comparison in Figure 2.22 is model-free, so I do not repeat it here. The useful comparison in this extension is therefore whether adding portfolio shares materially changes the fitted race pattern relative to that common realized-return benchmark.

2.5.5 The estimated distribution of fixed effects

A final, but important comparison, is to take a look at the distribution of estimated fixed effects from the FE and CRE models with the full set of control variables extended to control for risk exposure. Figure 2.25 shows the final estimates of this persistent component of returns to net wealth across respondents in the survey. Interestingly, the estimated distribution from the CRE model seems to be less skewed than other empirical estimates suggest.

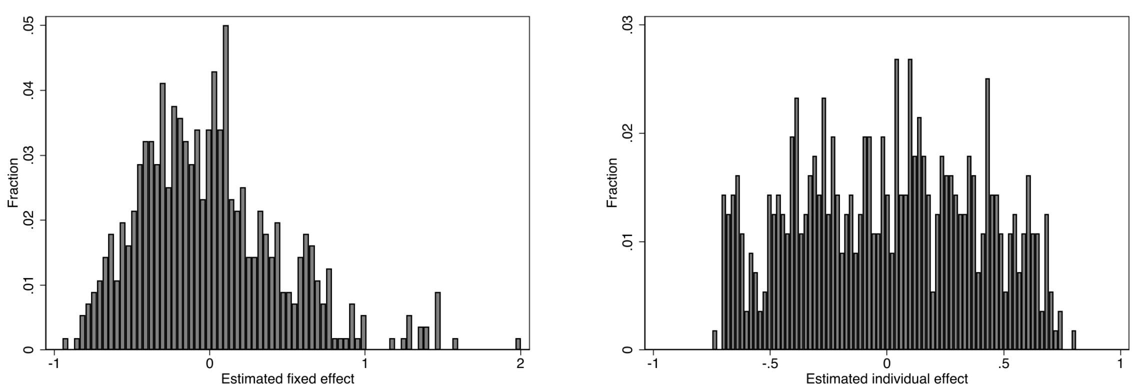


Figure 2.25: Distributions of estimated FE (left) and CRE (right) effects with portfolio-share controls included.

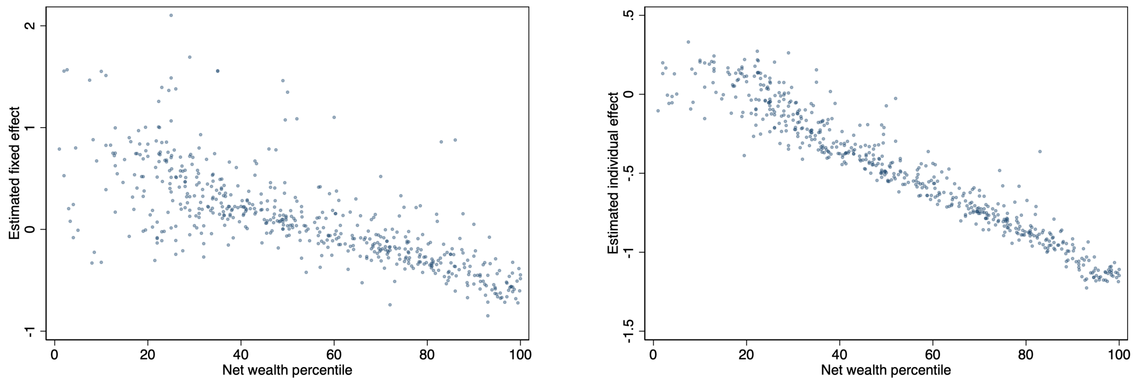


Figure 2.26: Estimated FE (left) and CRE (right) individual effects vs. net wealth percentile, with portfolio-share controls.

Figure 2.26 shows the same qualitative pattern after controlling for portfolio shares. In both the FE and CRE versions, the estimated individual effect remains downward sloping in wealth percentile. As in the main results, this should be interpreted as a feature of the residual persistent component after partialling out time-varying covariates and the share-by-year controls, not as a statement about the raw relationship between returns and wealth.

2.6 Conclusion

There is much empirical evidence of a positive relationship between trust and economic performance, at the aggregate and individual level. The results of this paper are in line with this literature. In particular, the RAND longitudinal product version of the HRS contains enough information to create a measure of returns to net wealth. Respondent-level returns exhibit significant heterogeneity, and the persistent component of individual returns is estimated and comparable to the distribution across indivdiuals using the Norwegian population data Fagereng et al. 2020. Moreover, the returns to net wealth exhibit strong evidence of a moderate amount of trust being associated with the maximal level of returns, as Butler et al. 2016 shows for household income and trust.

The remaining sections of this paper discuss findings relevant to the use of this HRS RAND product that are excluded from the main analysis but can be found along with the rest of the supplementary material for this project.

2.6.1 Drawbacks and limitations

A first limitation is that the HRS variables used in this paper do not all come from the same source. Most of the demographic, income, and wealth variables come from the cleaned and harmonized RAND HRS Longitudinal File, but net investment flows must be merged in wave-by-wave from the RAND HRS fat files. This creates a discrepancy in coverage between the harmonized RAND fields and the flow items. To partly address this, I pair each flow amount with its Section R gate so that an explicit no-activity response sets the corresponding missing amount to zero when appropriate, as described in Section 2.3. This increases the usable flow sample, but flow variables remain much thinner than the core RAND measures.

A second limitation is that the trust variables are only observed in 2020. This cannot be overstated. For general trust, there are 900 unique respondents with nonmissing values in the cross section, which is much smaller than the sample available for the standard RAND demographic and wealth variables. In the panel setting, the overlap between nonmissing returns to net wealth and nonmissing general trust is 4,115 person-years in total, ranging from 176 observations in 2002 to 565 in 2020. As a result, the trust analysis is much more sample-constrained than the analogous descriptive and regression exercises for returns, wealth, or demographics.

A third limitation concerns the implementation of the correlated random effects (CRE) specification. In this paper, the CRE model is implemented manually by augmenting the random-effects estimator (`xtreg ..., re` in Stata) with respondent-level means of the time-varying regressors—the same Mundlak device that Stata 19 now automates through its `xtreg, cre` option and the `estat mundlak` post-estimation command. In earlier Stata workflows, the researcher generated these panel means by hand and tested their joint significance directly; the implementation in this paper follows that convention and is econometrically equivalent in substance. Although survey-year fixed effects enter the model in levels, respondent-level averages of the year indicators are not included in the Mundlak augmentation because the panel is strongly balanced: the within-household means of year dummies are therefore constant across households and collinear with the intercept. The Mundlak means included are consequently limited to time-varying regressors that exhibit meaningful cross-household variation over time, such as wealth deciles, labor-force participation, and marital status.

2.6.2 Robustness checks

Labor income does not exhibit the same clean pattern as returns to net wealth. In both the 2020 cross section and the average labor-income specification, the raw quadratic trust terms are significant, but once controls are added the evidence weakens. The fully controlled quadratic remains only marginal in the 2020 cross section and is significant but less stable in the average-income regression. This weaker pattern is consistent with the HRS sample itself, which oversamples older and retired households and is therefore less naturally suited to labor-income variation than to broader measures of household resources.

Within narrower asset classes and portfolio definitions, the trust relationship is not robust in the way it is for returns to net wealth. In the cross section, the uncontrolled specifications for core, IRA, residential, and core-plus-IRA returns sometimes show a hump-shaped pattern, but those results disappear once the baseline controls are added. In that sense, the evidence outside net wealth is best viewed as suggestive at most rather than stable across specifications.

Even for returns to net wealth, the other trust measures generally do not reproduce the main result. The individual domain measures, the government-trust aggregates, and the simple average of trust in financial institutions are not jointly significant in the corresponding average-return regressions. The only partial exception is the first principal component for trust in financial institutions, which is weakly significant in the baseline specification, but that evidence disappears once the extended controls are included.

Financial literacy adds some information, but it does not overturn the trust results. In the recent extensions, the inflation item matters in the average-return specification, and the literacy score is significant in the FE second-stage regressions for the persistent component. In the CRE model, adding the literacy items one at a time still leaves the trust terms significant for the interest and inflation questions, although broader literacy bundles are less stable. Taken together, financial literacy explains some additional variation in returns, but it does not subsume the main trust relationship.

I also explored an instrumental-variables interpretation using depression-based measures and an indicator for non-Hispanic Black as excluded instruments for trust. The first stages are relevant in the sense that these variables predict trust, and the Kleibergen-Paap underidentification tests reject the null that the excluded instruments are irrelevant for trust. However, the endogeneity tests do not reject exogeneity of trust, the weak-identification diagnostics are only borderline, and the weak-instrument-robust Anderson-Rubin tests remain insignificant across the main specifications, meaning the IV exercises do not provide robust evidence that instrumented trust has a causal effect on returns. When the model is overidentified, the overidentification restrictions are not rejected, but the second-stage trust coefficients remain imprecise and unstable. Taken together, these exercises do not provide persuasive evidence for a causal 2SLS interpretation of the effect of trust on returns to net wealth.

Chapter 3

Wealth Inequality Under Ambiguity

Abstract

The inequality literature is notable in its ability to bring together four distinct concepts: (i) measures of inequality, (ii) social welfare functions, (iii) mean values, and (iv) models of choice under uncertainty, as noted by Dalton 1920. This interplay between statistical and theoretical analysis has been commonly applied in the context of inequality in the distribution of *income*. It is well-known that income and wealth inequality are markedly distinct phenomena, and this distinction becomes more pronounced when racial considerations are central. The literature has long documented that wealth is more strongly skewed than income, especially in the upper tail (Benhabib and Bisin 2018). Notably, as recent as 2019, it has been reported that "the median white household held \$188,200 in wealth – 7.8 times that of the typical Black household". This paper provides a theoretical foundation for wealth inequality that incorporates the first three of the four distinct concepts. The ranking over wealth distributions is established through a choice under ambiguity model, and the *social welfare approach* to inequality measures in Atkinson 1971 is applied in this setting to derive a measure of wealth inequality aversion. Using the triennial data on wealth in the Survey of Consumer Finances (SCF) from 1989–2022, I show that the choice under objective uncertainty inequality measure underestimates wealth inequality compared to the measure proposed via choice under ambiguity. Both theoretical measures of inequality

aversion track the general trend in the Black-White median wealth gap over the same time period.

3.1 Introduction

The distribution of outcomes such as income and wealth among individuals in a society are of interest not only to economists, but to those interested in social welfare and inequality. As Hong 1983 notes, the literature on inequality notably connects four distinct concepts; (i) measures of inequality, (ii) social welfare functions, (iii) mean values, and (iv) models of choice under uncertainty.

Measures of inequality, such as the variance and the coefficient of variation, represent the traditionally statistical approach to characterizing inequality in a society. Almost a century ago, Dalton 1920 suggested that a given measure of inequality would correspond to a social welfare function. This can be gleaned from the fact that different measures of inequality produce different rankings over distributions.

The contribution of Atkinson 1970 is two-fold. First, he provides a theoretical foundation for ranking distributions and a notion of income inequality aversion by connecting results from the choice under risk literature with Dalton's observation of underlying social welfare functions. Although not explicitly stated, he later proposes an inequality measure which highlights the connection between measures and quasilinear mean values. This is done through the use of an analogy to certainty equivalence from the choice under risk literature.

This literature is almost exclusively interested in characterizing inequality in the distribution of income. In the papers that claim to refer to wealth inequality, there is no "true" distinction, since the underlying choice under uncertainty model used only incorporates objective uncertainty. Since the literature is solid on the differences between skewness in the distribution of income and the distribution of wealth, one may wish to produce an inequality measure which makes a similar distinction. This measure can provide an alternative lens to view the evolution of racial disparities over time, much like summary statistics such as the racial wealth gap.

The goal of this paper is to produce an alternative measure inequality in the distribution of wealth using this *social welfare approach* to be compared to conventional summary statistics of racial wealth inequality, such as the black-white median wealth gap. This is achieved by exploiting connections between the aforementioned distinct concepts. I make use of an analogy between uncertainty aversion and wealth inequality aversion through specifications on the underlying social welfare function in a choice under ambiguity model, as well as the familiar *equally distributed measure* of an outcome to reach a measure of wealth inequality aversion, in the sense that it takes into consideration particular features of wealth accumulation and distribution which make it different from other distributional outcomes (like income).

The paper proceeds as follows. Section 3.2 presents arguments for distinguishing income and wealth inequality in this framework. Sections 3.3–3.5 then apply the approach to wealth, in the following order: choice under objective–subjective uncertainty, the social welfare approach, and the resulting inequality measure. Subsection 3.6 provides the SCF-based computational implementation and comparison to the Black-White median wealth gap. Section 3.7 concludes. The appendix B collects proofs and provides an explicit formulation of the construction towards the measure of income inequality aversion from the literature.

3.2 Motivation

The goal of this section is to argue that *there are fundamental features about wealth that make it different from income when it comes to ranking distributions*. The arguments I provide are a blend of empirical facts as well as a thought experiment and logical argument. Then the core argument of this paper becomes: **if i) there is a serious difference between ranking income distributions and ranking wealth distributions, and ii) economic theory can be used to produce a measure of income inequality using the choice under uncertainty literature, then the same approach can be used to produce a measure of wealth inequality using the choice under ambiguity literature**.

3.2.1 Empirical facts about income and wealth

It is well-known that income data is more reliably collected than wealth data. Consider a quote from Saez and Zucman 2014 which provides an estimate for wealth inequality using available administrative data:

> Because of the lack of administrative data on wealth, none of the existing sources offer a definitive estimate. We see our paper as an attempt at using the most comprehensive administrative data currently available, but one that ought to be improved in at least two ways: (i) by using additional information already available at the Statistics of Income division of the IRS,

> and (ii) new data that the US Treasury could collect at low cost. A modest data collection effort would make it possible to obtain a better picture of the joint distributions of wealth, income, and saving, a necessary piece of information to evaluate proposals for consumption or wealth taxation.

So it is clear that here are differences in the collection of income data and wealth data. But is it enough to justify incorporating modeling choices from the choice under subjective uncertainty framework in the exercise of comparing wealth distributions?[1]

Consider next a quote from Bricker et al. 2016 which provides a justification for their use of administrative data and their measurement of income and wealth inequality:

> In general, administrative data should provide better estimates of top income and wealth shares, because traditional random household surveys suffer from under- representation of wealthy families. Unlike most other household surveys, the SCF is designed to overcome the underrepresentation problem, because administrative data are used to select the sample, and rigorous targeting and accounting for wealthy family participation assures those families are properly represented in the survey data.

Not only is "good" data on income more accessible, but the estimation of wealth shares typically employs some combination of administrative and survey data. But survey data is by, quite literally, a subjective measure: it implicitly relies on individual's truthful and accurate reporting of their own levels of wealth.

[1] in combination with the choice under objective uncertainty is what is referred to as "choice under ambiguity" framework.

3.2.2 A thought experiment

Country A has 60 individuals; there are two possible races so that each individual is either black or white. Suppose there are 10 black and 50 white individuals. There is an initial wealth distribution $f(w)$, so that $w = (w_1, \dots, w_{60})$ and $f(w_i)$ gives the proportion of individuals with that level of wealth. One can consider total wealth, $\bar{w} = \sum_{i=1}^{60} w_i f(w_i)$, as well as group wealth totals, $\bar{w}_b$ and $\bar{w}_w$.

Partition each racial group into rich and poor individuals:

$$B = B^R \cup B^P, \quad W = W^R \cup W^P,$$

with $|B^R| = 5,\ |B^P| = 5,\ |W^R| = 25,\ |W^P| = 25.$

A policymaker compares two budget-balanced redistributive policies:

> **Policy A:** decrease wealth of each individual in B^R by 20 units, and increase wealth of each individual in W^P by 4 units.
>
> **Policy B:** decrease wealth of each individual in W^R by 4 units, and increase wealth of each individual in B^P by 20 units.

Both policies transfer the same total amount of wealth:

$$5 \times 20 = 25 \times 4 = 100.$$

Assuming that, aside from racial labels, the post-policy wealth profile is the same under both policies, the choice under objective uncertainty framework would treat Policies A and B as equivalent, since they induce the same resulting wealth distribution. In other words, they have the *same effect on wealth inequality under this framework*.

However, it is not unreasonable to suspect that these two policies would not receive the same support in practice, especially in the U.S. context where race is tied to historically persistent barriers to wealth accumulation. This is the sense in which the choice under objective uncertainty framework, while useful for income distributions, may miss relevant features when ranking wealth distributions. If this is the case, then a notion of subjective uncertainty can be incorporated in a straightforward way: denote the finite state space $S = \{\text{black}, \text{white}\}$. From here, using a model that allows for uncertainty aversion is a reasonable deviation from the objective case.

3.2.3 A logical argument

An immediate counterargument to the previous thought experiment is that the factors which make it more difficult for blacks to accumulate wealth in the U.S. (which is the assumed reasoning behind not being indifferent between the two policies) should also apply to blacks earning income. The first rebuttal to this point is the aforementioned observable data on differences between income and wealth inequality.

However, I believe an appeal to the underlying meaning of the word "wealth" statements of the late philosopher John Rawls may convince readers that income inequality and wealth inequality truly are different phenomena, and that the latter has a objective-subjective uncertainty structure that makes it suited for the choice under ambiguity model to produce an accompanying measure of wealth inequality aversion.

Rawls 1971 is interested in the problem of getting a group of people with different circumstances and motives to agree on a *social contract*; that is, an agreement on a system of governing for which all members of the group must abide by.

He proposes that this problem should be approached as if those deciding on the governance of society are behind a *veil of ignorance*: decision-makers are ignorant of their own circumstances. He then proposes two principles that may accompany this veil of ignorance in the delivery of justice for institutions which govern our society. Consider Rawls' final statement of the two principles of justice for institutions:

1. Each person is to have an equal right to the most extensive total system of equal basic liberties compatible with a similar system of liberty for all.

2. Social and economic inequalities are to be arranged so that they are both:

   (a) to the greatest benefit of the least advantaged, consistent with the just savings principle, and

   (b) attached to offices and positions open to all under conditions of fair equality of opportunity.

But how is this related to the social planner's decision problem of ranking wealth distributions? The link is that there is some **underlying notion of a social contract implicit in any concept of wealth, since wealth requires enforceable ownership rights.** In modern economies, those rights are defined and protected by a government (or governing legal authority): individuals must agree on what it means to own something, and must respect that some individuals own more and/or less than they do. By contrast, income is often generated through a worker–employer contract, which the employer may legally modify or terminate (subject to labor law and contract terms). This makes wealth inequality more directly tied to legal and political institutions than labor income[2].

Consider another quote from Rawls, which is almost exactly the intuitive counterpart to the modeling choices I make later in the characterization of a comparison over wealth distributions:

> The maximin rule tells us to rank alternatives by their worst possible outcomes: we are to adopt the alternative the worst outcome of which is superior to the worst outcomes of the others[3].

3.3 Ranking wealth distributions using choice under ambiguity

Consider the excerpt from Gilboa 2009 about the choice under ambiguity model, which motivates the modeling choices that will be made to characterize a ranking over wealth distributions[4]:

> To make sure that we understand the structure, observe that there are two

[2]Further arguments could be made that income suffers from legal and political contamination as well.

[3]Coincidentally, Rawls' second principle of justice for institutions is often considered by economists to be interchangeable with the maximin principle, despite his belief that it is "undesirable to use the same name for two things that are so distinct.".

[4]I move away from the notation used there in favor of the "choice over lotteries" notation to tie into the birth lottery analogy made earlier in the paper

sources of uncertainty: the choice of the state s, which is sometimes referred to as "subjective uncertainty", because no objective probabilities are given on it, and the choice of y, which is done with objective probabilities once you chose your act and Nature chose a state. Specifically, if you choose $p \in L$ and Nature chooses $s \in S$, a roulette wheel is spun, with distribution p_s over the outcomes Y, so that your probability to get outcome y is $p_s(y)$.

3.3.1 Analytical framework

Denote the set of outcomes $Y = [0, \bar{y}]$. Then, the choice set is given by:

$$L = \left\{ p : 2^{[0,\bar{y}]} \to \mathbb{R} \mid p(\cdot) \text{ is an income frequency distribution} \right\}.$$

We may define a binary relation over both sets Y, L :

$$\succsim_y \subseteq [0, \bar{y}] \times [0, \bar{y}] \subseteq \mathbb{R} \times \mathbb{R}$$

$$\succsim \subseteq L \times L.$$

Notice that $\succsim_y$ may be represented by a real-valued utility function. This is the social welfare $U(y)$ which is a key object of analysis in this paper.

Preliminary remarks

We are concerned with the comparison of two frequency distributions $f(w)$ of an outcome w which we refer to as wealth. We seek to use the notion of *uncertainty aversion* as an analogy to the use of the notion of risk aversion in the characterization of a ranking over income distributions.

The presence of both objective and subjective uncertainty is at the heart of this analysis. Section 2 covered the analysis for objective uncertainty. Thus, the wealth frequency distribution $f(w)$ must be formalized in this abstract setting so that it explicitly captures both forms of uncertainty. Namely, each wealth distribution is associated with some relevant, underlying state space S (the source of subjective uncertainty), and its objective component $p(y) \in L$.

This suggests that the objects of interest (rankable wealth distributions $f(w)$) are a family of distributions $p(y) \in L$, indexed by elements of the state space $s \in S$, or:

$$\{p_s(y)\}_{s \in S}.$$

Thus, we may view each act $f \in F$ as a "state-contingent frequency distribution". Moreover, upon fixing a state $s' \in S$, the analysis will become identical to the use of the choice under objective uncertainty model that was used to reach a ranking over income frequency distributions.

With this in mind, we work under the following assumption on the functional form of wealth distributions for the remainder of this paper:

Assumption 1. *Each wealth distribution $f(w)$ can be written as $p_s(y)$.*

3.3.2 Axiomatization

AA 1 (Weak Order)**.** $\succsim$ *is complete and transitive.*

AA 2 (Continuity)**.** *For every $f, g, h \in F$, if $f \succ g \succ h$, there exists $\alpha, \beta \in (0,1)$ such that*

$$\alpha f + (1-\alpha)h \succ g \succ \beta f + (1-\beta)h.$$

C-Independence (C-Independence)**.** *For every $f, g \in F$, every constant $h \in F$ and every $\alpha \in (0,1)$,*

$$f \succsim g \iff \alpha f + (1-\alpha)h \succsim \alpha g + (1-\alpha)h.$$

AA 3 (Monotonicity)**.** *For every $f, g \in F$, $f(s) \geq g(s)$ for all $s \in S$ implies $f \geq g$.*

AA 4 (Non-trivality)**.** *There exists $f, g \in F$ such that $f \succ g$.*

Uncertainty Aversion (Uncertainty Aversion)**.** *For every $f, g \in F$, if $f \sim g$, then, for every $\alpha \in (0,1)$,*

$$\alpha f + (1-\alpha)g \succsim f.$$

3.3.3 Expected utility representation of the ranking over wealth distributions

Finally, we have the representation theorem by Gilboa and Schmeidler 1989.

Theorem 1. $\succsim$ *satisfies AA1, AA2, C-Independence, AA4, AA5, and Uncertainty Aversion if and only if there exists a closed and convex set of probabilities on S,*

$C \subseteq \Delta(S)$, and a non-constant function $U : Y \to \mathbb{R}$ such that, for every $f, f^* \in F$,

$$f \succsim f^* \iff \min_{\lambda \in C} \int_S \Big(\mathbb{E}_{p_s}[U]\Big)\, d\lambda \geq \min_{\lambda \in C} \int_S \Big(\mathbb{E}_{p_s^*}[U]\Big)\, d\lambda.$$

From here on out, we assume that wealth distributions will be ranked according to:

$$W^A \equiv \min_{\lambda \in C} \int_S (\mathbb{E}_{p_s}[U]) d\lambda$$

$$= \min_{\lambda \in C} \int_S \int_0^{\bar{y}} p_s(y) U(y) dy d\lambda.$$

Notation. Superscript A denotes the ambiguity specification; this is the first appearance of W^A in the main text.

3.4 The social welfare approach to ranking wealth distributions

3.4.1 Partial ranking over wealth distributions

From here, I assume that the function $U : Y \mapsto \mathbb{R}$ is an increasing and concave function. Again, we seek the conditions for which a ranking over wealth distributions can be achieved without any further specifications on $U(y)$.

Recall that the second-order stochastic dominance result was necessary and sufficient to reach a partial ordering over income distributions. The proposition below provides a sufficient condition that extends this logic to the ambiguity setting with maxmin aggregation over priors.

Proposition 1. *Define*

$$W^A(\{p_s\}; U) \equiv \min_{\lambda \in C} \int_S \Big(\mathbb{E}_{p_s}[U]\Big)\, d\lambda, \quad C \subseteq \Delta(S).$$

Notation. The notation W^A was introduced in Section 3.3. If, for every $s \in S$,

$$\int_0^z [P_s(y) - P_s^*(y)]\, dy \leq 0 \quad \textit{for all } z,\ 0 \leq z \leq \bar{y},$$

where

$$P_s(y) = \int_0^y p_s(t)\, dt \quad \textit{and} \quad P_s^*(y) = \int_0^y p_s^*(t)\, dt,$$

then p is weakly preferred to p^* according to W^A for all $U(y)$ with $U' > 0$ and $U'' \leq 0$.

The proof can be found in the Appendix. In general, the converse need not hold under a maxmin criterion unless additional assumptions are imposed on C.

3.4.2 Complete ranking over wealth distributions

To reach our complete ordering over wealth distribution, we must guarantee that $U(y)$ is specified up to a linear (monotonic) transformation. Work done by Kihlstrom and Mirman 1981 extending the notion of relative risk aversion results to the case of "multidimensional commodities" implies that the restriction we are after is the class of social welfare functions representing homothetic preferences. That is,

$$U(y) = \begin{cases} \alpha + \beta \frac{y^{1-\epsilon}}{1-\epsilon}, & \epsilon \neq 1 \\ \alpha + \beta \ln(y), & \epsilon = 1 \end{cases} \tag{3.1}$$

For strict concavity in this class, take $\beta > 0$, $y > 0$, and $\epsilon > 0$ (with $\epsilon = 1$ giving the log case); note that $\epsilon = 0$ yields linear utility.

It is well-known that, in the risk and risk aversion literature, this functional form is associated with the class of utility functions representing constant relative risk averse (CRRA) and decreasing absolute risk averse (DARA) preferences.

3.5 Proposing a measure of wealth inequality aversion

The ranking over wealth distributions via specifications on $U(y)$ will allow us to propose measures of inequality. To see the novelty in this "social welfare approach", notice that measures of inequality are generally statistical objects. Namely, the variance, coefficient of variation, and mean deviation are each calculated using collected data on the distribution of an outcome such as income or wealth.

Consider one plausible inequality measure: given some present distribution $f(w) = \{p_s(y)\}_{s \in S}$, the ratio between social welfare at the current distribution and social welfare at equal wealth μ. Using the robust welfare index

$$W^A \equiv \min_{\lambda \in C} \int_S \int_0^{\bar{y}} U(y)\, p_s(y)\, dy\, d\lambda, \qquad C \subseteq \Delta(S),$$

we may write

$$D^A = \frac{W^A}{U(\mu)}.$$

Notation. Superscript A denotes ambiguity objects; this is the first appearance of D^A in the main text.

However, this ratio is not invariant to positive affine transformations of U, so different cardinal representations can produce different values of D^A even when they induce the same ranking.[5]

3.5.1 The equally distributed equivalent measure of inequality

To avoid this undesirable property, define the equally distributed equivalent wealth level as follows.

Definition 1. *The equally distributed equivalent level of wealth w_{EDE} is defined by*

$$U(w_{EDE}) = W^A.$$

Equivalently,

$$U(w_{EDE}) = \min_{\lambda \in C} \int_S \int_0^{\bar{y}} U(y)\, p_s(y)\, dy\, d\lambda.$$

Since each p_s integrates to one on Y and each $\lambda \in C$ is a probability on S, the normalization factor is one, so no extra scale term is needed in the definition.

[5]As shown by Dalton 1920 and others.

The corresponding equally distributed equivalent inequality measure is

$$I^A = 1 - \frac{w_{EDE}}{\mu}.$$

Notation. This is the first appearance of I^A in the main text.

This measure is cardinally robust (with respect to positive affine transformations of U) and lies in $[0, 1]$ under the usual ordering conditions.

3.5.2 Closed-form expression under homothetic preferences

For computation, the key identity is

$$w_{EDE} = U^{-1}(W^A).$$

For the homothetic class,

$$U(y) = \begin{cases} \alpha + \beta \dfrac{y^{1-\epsilon}}{1-\epsilon}, & \epsilon \neq 1, \\ \alpha + \beta \ln y, & \epsilon = 1, \end{cases} \qquad \beta > 0,$$

with $y > 0$ and $\epsilon > 0$ for strict concavity (note: $\epsilon = 0$ is linear and therefore not strictly concave).

If $\epsilon \neq 1$,

$$w_{EDE} = \left[\frac{1-\epsilon}{\beta}(W^A - \alpha)\right]^{\frac{1}{1-\epsilon}}, \qquad I^A = 1 - \frac{1}{\mu}\left[\frac{1-\epsilon}{\beta}(W^A - \alpha)\right]^{\frac{1}{1-\epsilon}}.$$

If $\epsilon = 1$,

$$w_{EDE} = \exp\left(\frac{W^A - \alpha}{\beta}\right), \qquad I^A = 1 - \frac{1}{\mu}\exp\left(\frac{W^A - \alpha}{\beta}\right).$$

Appendix B.3 gives the same derivation for the objective-uncertainty benchmark and then shows that the ambiguity case is obtained by replacing W^O with W^A.

3.6 Comparing inequality aversion measures to the racial wealth gap

3.6.1 Data

I use the triennial Survey of Consumer Finances (SCF), 1989–2022, and restrict the sample to Black and White households in each wave. For each year, I compute weighted medians for net wealth and gross assets, and then compute I^O and I^A under two treatments: (i) a restricted sample that keeps only strictly positive wealth values, and (ii) the full sample which adds a year-specific constant so that all observations shifted to be strictly positive.[6]

The *ambiguity* aggregator used for I^A follows the Gilboa–Schmeidler construction with $C = \Delta(S)$ and $S = \{\text{Black}, \text{White}\}$. Because the criterion is linear in probabilities over the two states, the inner minimum is attained by putting all weight on whichever racial group has the lower within-group expected utility that year, so computationally $W_t^A = \min\{\mathbb{E}_{B,t}[U(y)], \mathbb{E}_{W,t}[U(y)]\}$. The wedge $I_t^A > I_t^O$ arises relative to the objective benchmark W^O defined in Appendix B.3.

[6] Appendix Figure B.2 plots the shift constant and the weighted share with non-positive wealth.

With these modeling objects fixed, the theoretical welfare expressions map directly into computable objects in the data: for each year and wealth concept, we compute W^O and W^A, invert U to recover the equally distributed equivalent wealth level, and then construct I^O and I^A. The remaining step is to set values for ϵ, which governs the curvature of $U(y)$ and therefore the degree of inequality aversion imposed in the computation.

The parameter ϵ indexes the curvature of the social welfare function $U(y)$ and therefore the strength of inequality aversion: larger ϵ means stronger concavity, a higher penalty on low-wealth outcomes, and thus higher implied inequality aversion in both I^O and I^A.

For orientation, read the CRRA index in four regions: (i) $\epsilon = 0$ is the affine (linear) benchmark with no inequality aversion in the objective aggregator, reported separately in Appendix B.6; (ii) $0 < \epsilon < 1$ yields concavity that is still milder than the logarithmic case; (iii) $\epsilon = 1$ is the log boundary; (iv) $\epsilon > 1$ imposes tail sensitivity strictly stronger than the log case. The subsections below walk through $\epsilon \in \{0.25, 0.5, 1, 1.5, 2\}$ so that each reported value sits in a clearly labeled part of this map.

3.6.2 The Black-White gap in net wealth and gross assets

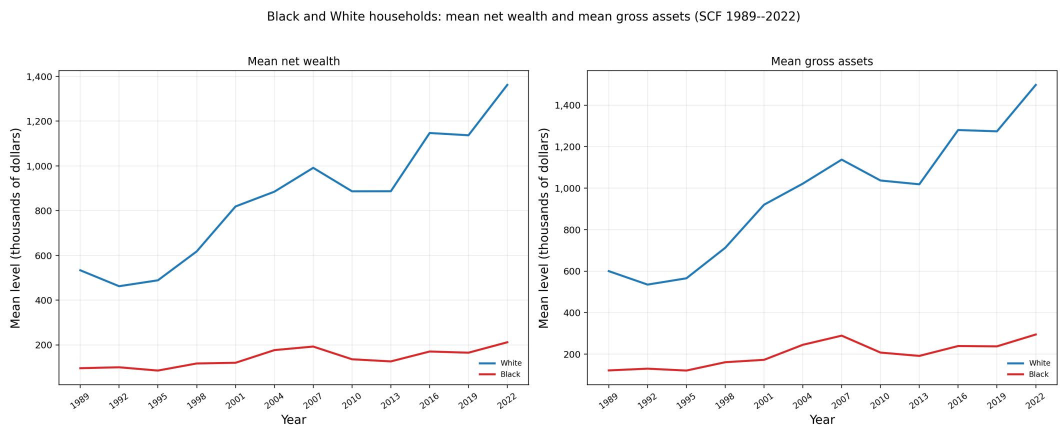


Figure 3.1: Mean net wealth (left panel) and mean gross assets (right panel) for Black and White households, SCF 1989–2022.

Figure 3.1 plots race-specific means before differencing: White household means lie above Black means in both net wealth and gross assets throughout the sample.

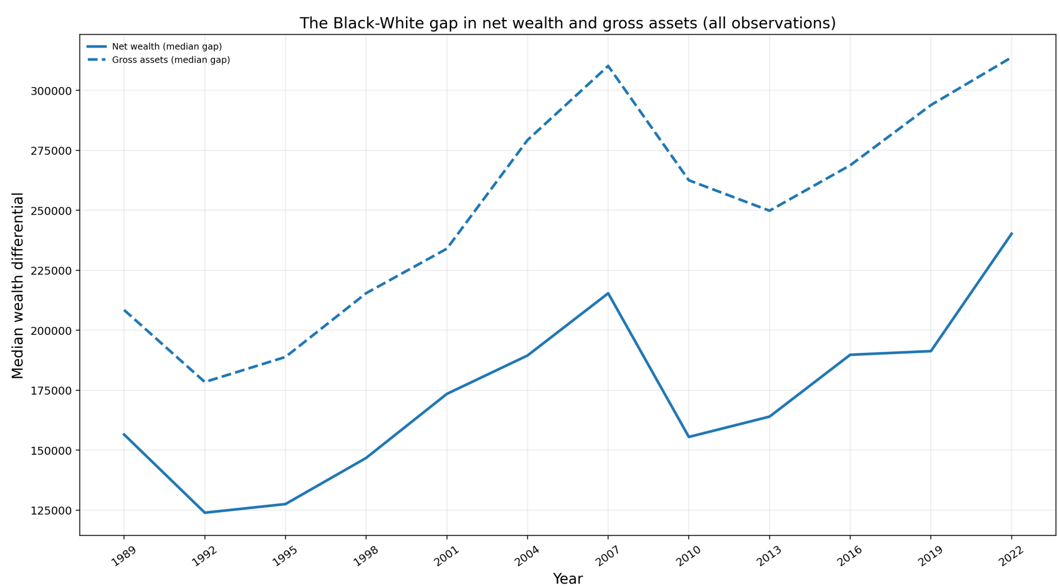


Figure 3.2: The Black-White gap in net wealth and gross assets: median wealth differential, SCF 1989–2022.

Figure 3.2 shows that the median Black-White differential is positive and persistent in both series, with gross-asset gaps generally larger than net-wealth gaps over the sample.

Checking gross assets alongside net wealth is useful because it verifies that the empirical trend is not an artifact of liability measurement alone: both series move with similar broad dynamics over time. At the same time, the gross-asset gap is higher in levels than the net-wealth gap, which is consistent with definition since net wealth subtracts debts while gross assets do not.

3.6.3 Low concavity: $\epsilon = 0.25$

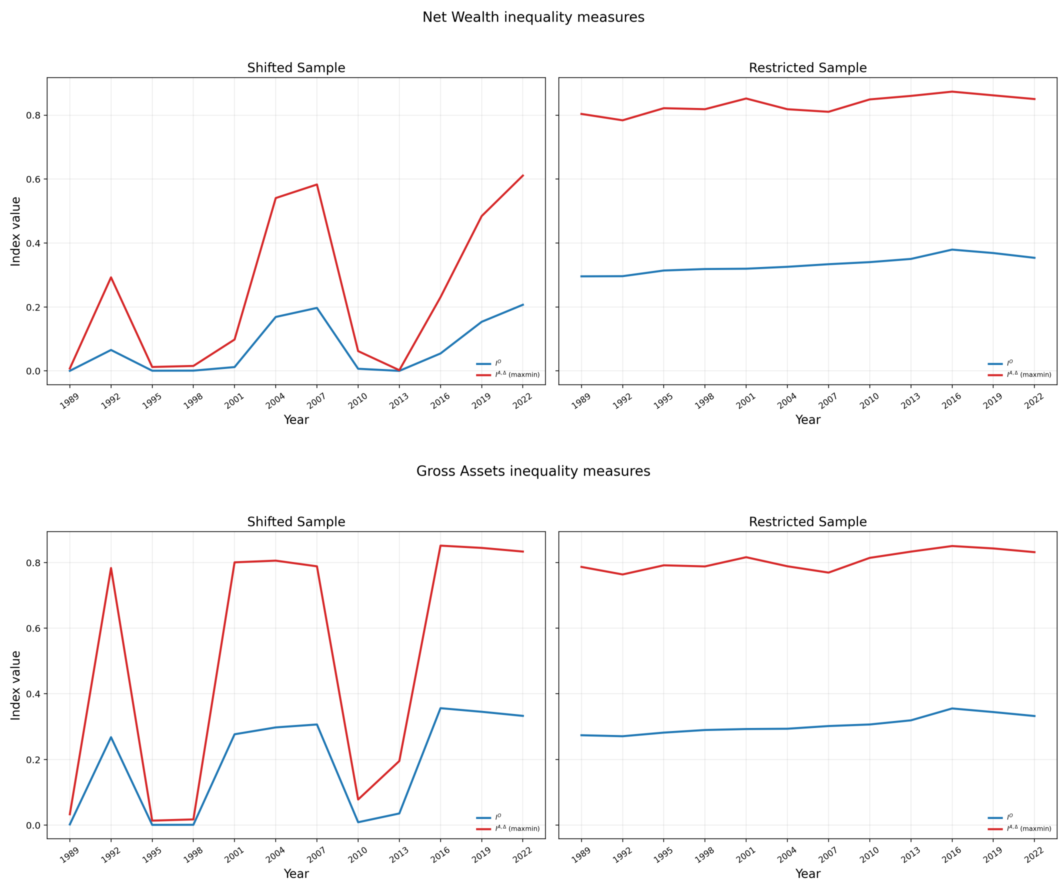


Figure 3.3: Inequality version measures for $\epsilon = 0.25$ (top: net wealth, bottom: gross assets, left: shifted sample, right restricted sample).

Even with a low level of inequality aversion (curvature near zero), the ambiguity ranking already yields higher inequality levels than the objective ranking in both wealth concepts ($I^A > I^O$). When I restrict the sample to respondents with positive wealth only, inequality is extremely high and stable across time for the wealth inequality aversion measure. For the full sample (shifted so that everyone has positive wealth), there is more variation over the time period, with some years showing significantly low levels of inequality. The key takeaway is that *the inequality aversion measure reached using the choice under objective uncertainty framework consistently reports that inequality is lower than it is under the ambiguity aversion measure.*

3.6.4 Mild concavity: $\epsilon = 0.5$

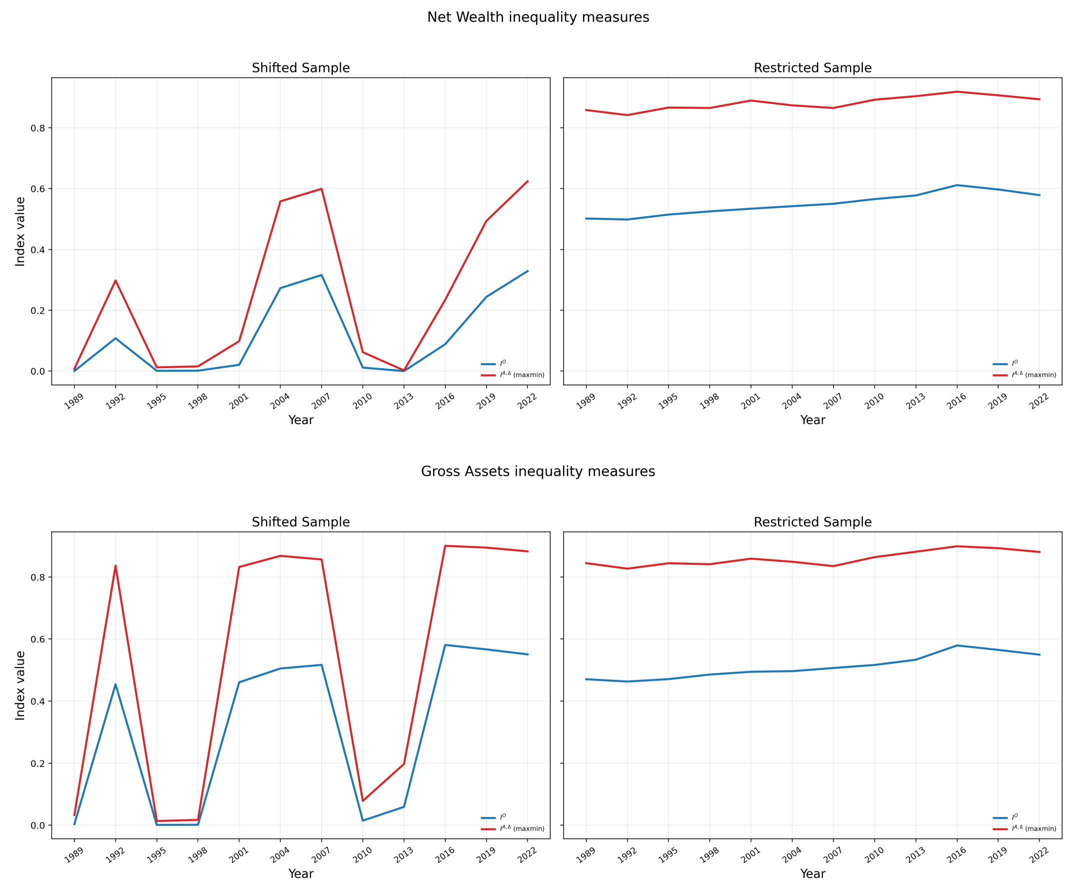


Figure 3.4: Inequality aversion measures for $\epsilon = 0.5$.

For $\epsilon = 0.5$, the performance of the wealth inequality aversion measure I^A in the restricted sample is the same: high and comparatively stable inequality levels over time. This time, the inequality aversion measure I^O is now higher for this higher value of epsilon. Again, the shifted sample shows more variation across year, while retaining the feature that I^A is consistently above I^O; so the finding that the objective measure understates inequality relative to the ambiguity measure is preserved.

3.6.5 Logarithmic case: $\epsilon = 1$

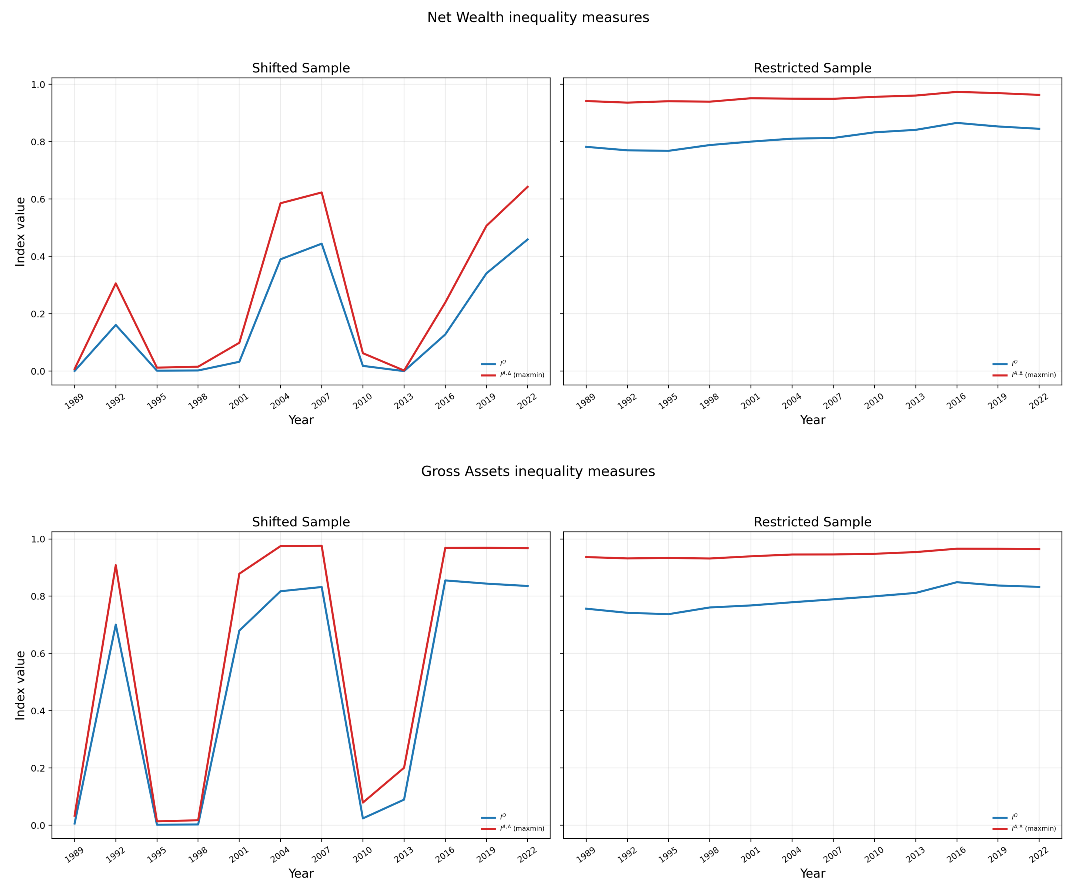


Figure 3.5: Inequality aversion measures for $\epsilon = 1$.

The same pattern holds $\epsilon = 1$. For both samples, $I^A > I^O$, and in the restricted-sample, the objective measure I^O has moved even closer to I^A. An interesting observation throughout these figures is that, for the shifted sample, both measures consistently report higher inequality levels in gross assets than in net wealth, especially between the years 1998-2010. This pattern is robust to values of ϵ.

3.6.6 Strong concavity: $\epsilon = 1.5$

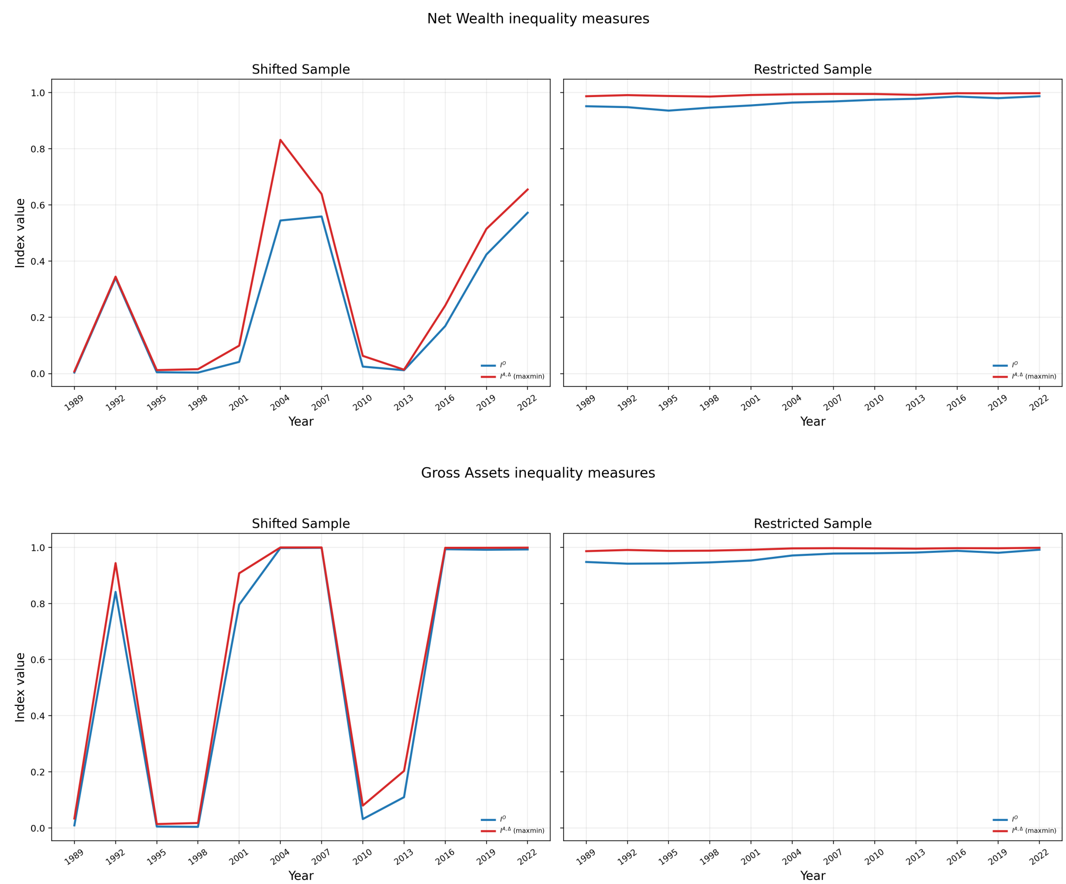


Figure 3.6: Inequality aversion measures for $\epsilon = 1.5$.

At this level of concavity in $U(y)$, the restricted sample shows that the measures I^A and I^O track each other very closely. Both are extremely high and stable over time. The pattern persists for the shifted sample, but reported levels of inequality are significantly larger here (for gross assets, there are multiple periods where inequality is essentially 1.). Still, the finding that the objective uncertainty aversion measure I^O underestimates inequality compared to the ambiguity aversion measure I^A is preserved.

3.6.7 Strongest curvature: $\epsilon = 2$

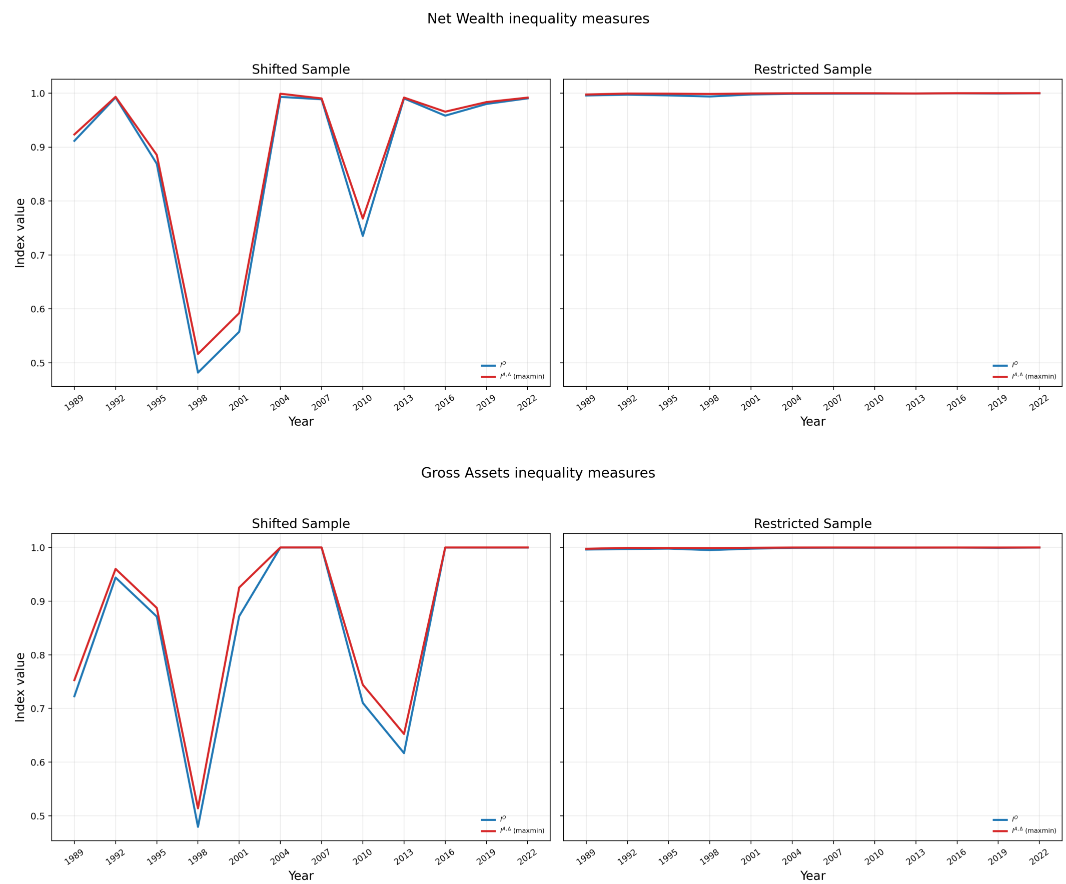


Figure 3.7: Inequality aversion measures for $\epsilon = 2$.

This is the largest value of ϵ reported in the main text; it is the most averse to inequality in the sense that the social welfare functional penalizes low wealth sharply.

As ϵ rises from 1.5 to 2, the restricted sample essentially shows no difference between

I^A and I^O. The variation in inequality over time in the shifted sample is the most pronounced here, and are almost identical for net wealth and gross assets.

3.6.7.1 Final takeaways

Taken together, the inequality-measure series tell the same within-sample story across specifications (from $\epsilon = 0.25$ through $\epsilon = 2$): rising (or at the very least, extremely high) inequality over the time period. Interestingly, when we restrict the sample to positive wealth values, the inequality aversion measures I^A and I^O are stable over time. However, Figure B.1 shows that net-wealth and gross-assets median gaps still vary over time and trend upward. This suggests that the theoretical indices are unusually stable when applied to the restricted sample. Additionally, in the restricted sample, the wealth inequality aversion measure I^A is especially robust; across reported ϵ values, it is stable over time and very close to 1.

By contrast, in the shifted sample, treatment both I^O and I^A are more variable over time and show episodes of comparatively low measured inequality at low and moderate ϵ; they begin to pull upward noticeably at $\epsilon = 1.5$, and become frequently near one only in the highest-curvature case $\epsilon = 2$. This higher time variation in the shifted series tracks the median-gap dynamics in Figure 3.2 much more closely than the restricted-sample series.

3.7 Conclusion

We have reached an alternative measure of wealth inequality, I^A. The measure both utilizes the notion of representative wealth levels and preserves the mean-independence property that many statistical measures of inequality have. The goal of this exercise is to apply it in the context of inequality in racial wealth. There, the story is generally told using the aforementioned statistical measures of inequality (most notable, the *black-white median wealth gap*).

With the arguments developed in Sections 3.2 and 3.3, it seems reasonable to suspect that a state space such as $S = \{\text{black, white}\}$ should largely inform our measure of wealth inequality via some ranking over wealth distributions. Historical episodes such as enslavement and the era of Jim Crow laws suggest that race may be a socially relevant feature in characterizing the level of inequality in a given wealth distribution. The main issue is whether the effects of these historical episodes persist over time, as well as how explanatory they are for racial differences in wealth.

The SCF-based computational exercise in Section 3.6 delivers a clear result: the objective benchmark understates wealth inequality relative to the ambiguity-based measure. This comparison is robust for both net wealth and gross assets, and the model-based series follow the same broad time pattern as the Black-White median wealth-gap diagnostics; the restricted sample is higher and smoother, while the shifted sample is more time-varying.

References

Adrien d'Avernas, Andrea L Eisfeldt, Can Huang, Richard Stanton, and Nancy Wallace (2024). *The Deposit Business at Large vs. Small Banks.*

Aiyagari, S Rao (1994). "Uninsured idiosyncratic risk and aggregate saving." In: *The Quarterly Journal of Economics* 109.3, pp. 659–684.

Alesina, Alberto and Eliana La Ferrara (2000). *The determinants of trust.* Tech. rep. w7621. Cambridge, MA: National Bureau of Economic Research. DOI: 10.3386/w7621.

Algan, Yann and Pierre Cahuc (2010). "Inherited Trust and Growth." In: *Am. Econ. Rev.* 100.5, pp. 2060–2092. DOI: 10.1257/aer.100.5.2060.

Alsan, Marcella and Marianne Wanamaker (2018). "Tuskegee and the health of black men." en. In: *Q. J. Econ.* 133.1, pp. 407–455. DOI: 10.1093/qje/qjx029.

Altmejd, Adam, Thomas Jansson, and Yigitcan Karabulut (2024). *Business education and portfolio returns.* IZA-Institute of Labor Economics.

Atkinson, A. B. (1971). "The distribution of wealth and the individual life-cycle." In: *Oxford Economic Papers* 23.2, pp. 239–254.

Atkinson, Anthony (1970). "On the measurement of inequality." In: *Journal of Economic Theory* 2.3, pp. 244–263.

Bach, Laurent, Laurent E. Calvet, and Paolo Sodini (2018). "Rich pickings? risk, return, and skill in household wealth." In: *American Economic Review* 110.9, pp. 2703–47. DOI: 10.1257/aer.20170666.

Baltagi, Badi H. and Long Liu (2026). *Time Invariant Variables in the Mundlak and Hausman-Taylor Panel Data Models.* Tech. rep. 288. CPR Working Paper No. 288. Center for Policy Research, Syracuse University.

Benhabib, Jess and Alberto Bisin (2018). "Skewed wealth distributions: theory and empirics." In: *Journal of Economic Literature* 56.4, pp. 1261–91. DOI: 10.1257/jel.20161390.

Benhabib, Jess, Alberto Bisin, and Mi Luo (2017). “Earnings inequality and other determinants of wealth inequality.” In: *American Economic Review* 107.5, pp. 593–97. DOI: 10.1257/aer.p20171005.

— (2019). “Wealth Distribution and Social Mobility in the US: A Quantitative Approach.” In: *Am. Econ. Rev.* 109.5, pp. 1623–1647. DOI: 10.1257/aer.20151684.

Bewley, Truman (1983). “A difficulty with the optimum quantity of money.” In: *Econometrica: Journal of the Econometric Society*, pp. 1485–1504.

Blundell, Richard, Luigi Pistaferri, and Ian Preston (2008). “Consumption inequality and partial insurance.” In: *American Economic Review* 98.5, pp. 1887–1921. DOI: 10.1257/aer.98.5.1887.

Bollen, Kenneth A. and Jennie E. Brand (2010). “A General Panel Model with Random and Fixed Effects: A Structural Equations Approach.” In: *Social Forces* 89.1, pp. 1–34. DOI: 10.1353/sof.2010.0072.

Bricker, Jesse, Alice Henriques, Jacob Krimmel, and John Sabelhaus (2016). “Measuring income and wealth at the top using administrative and survey data.” In: *Brookings Papers on Economic Activity* 47.1 (Spring), pp. 261–331.

Brown, Jeffrey R, Jeffrey B Liebman, and Joshua Pollet (2007). “Appendix: Estimating Life Tables That Reflect Socioeconomic Differences in Mortality.” en. In: *The Distributional Aspects of Social Security and Social Security Reform*. University of Chicago Press, pp. 447–458.

Butler, Jeffrey V, Paola Giuliano, and Luigi Guiso (2016). “The right amount of trust.” In: *Journal of the European Economic Association* 14.5, pp. 1155–1180.

Cagetti, Marco (2003). “Wealth Accumulation over the Life Cycle and Precautionary Savings.” In: *J. Bus. Econ. Stat.* 21.3, pp. 339–353.

Cagetti, Marco and Mariacristina De Nardi (2006). “Entrepreneurship, Frictions, and Wealth.” In: *J. Polit. Econ.* 114.5, pp. 835–870. DOI: 10.1086/508032.

— (2009). “Estate Taxation, Entrepreneurship, and Wealth.” In: *Am. Econ. Rev.* 99.1, pp. 85–111. DOI: 10.1257/aer.99.1.85.

Campbell, John Y, Tarun Ramadorai, and Benjamin Ranish (2019). "Do the Rich Get Richer in the Stock Market? Evidence from India." In: *American Economic Review: Insights* 1.2, pp. 225–240. DOI: 10.1257/aeri.20180158.

Carroll, Christopher, Jiri Slacalek, Kiichi Tokuoka, and Matthew N. White (2017). "The distribution of wealth and the marginal propensity to consume." In: *Quantitative Economics* 8.3, pp. 977–1020. DOI: https://doi.org/10.3982/QE694. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.3982/QE694.

Carroll, Christopher D (1992). "The Buffer-Stock Theory of Saving: Some Macroeconomic Evidence." In: *Brookings Pap. Econ. Act.* 1992.2, pp. 61–156.

Carroll, Christopher D, Jiri Slacalek, and Kiichi Tokuoka (2015). "Buffer-stock saving in a Krusell–Smith world." In: *Econ. Lett.* 132, pp. 97–100. DOI: 10.1016/j.econlet.2015.04.021.

Dalton, Hugh (1920). "The measurement of the inequality of incomes." In: *The Economic Journal* 30.119, pp. 348–361.

Daminato, Claudio and Luigi Pistaferri (2024). "Returns Heterogeneity and Consumption Inequality Over the Life Cycle." DOI: 10.3386/w32490.

De Nardi, Mariacristina and Giulio Fella (2017). "Saving and wealth inequality." In: *Rev. Econ. Dyn.* 26, pp. 280–300. DOI: 10.1016/j.red.2017.06.002.

Debacker, Jason, Bradley Heim, Vasia Panousi, Shanthi Ramnath, and Ivan Vidangos (2013). "Rising Inequality: Transitory or Persistent? New Evidence from a Panel of U.S. Tax Returns." In: *Brookings Pap. Econ. Act.*, pp. 67–122.

Den Haan, Wouter J, Kenneth L Judd, and Michel Juillard (2010). "Computational suite of models with heterogeneous agents: Incomplete markets and aggregate uncertainty." In: *J. Econ. Dyn. Control* 34.1, pp. 1–3. DOI: 10.1016/j.jedc.2009.07.001.

Deuflhard, Florian, Dimitris Georgarakos, and Roman Inderst (2014). *Financial Literacy and Savings Account Returns*. MPRA Paper 53857. University Library of Munich, Germany.

— (2018). "Financial Literacy and Savings Account Returns." en. In: *J. Eur. Econ. Assoc.* 17.1, pp. 131–164. DOI: 10.1093/jeea/jvy003.

Drechsler, Itamar, Alexi Savov, and Philipp Schnabl (2017). "The deposits channel of monetary policy." en. In: *Q. J. Econ.* 132.4, pp. 1819–1876. DOI: 10.1093/qje/qjx019.

Fagereng, Andreas, Luigi Guiso, Davide Malacrino, and Luigi Pistaferri (2020). "Heterogeneity and persistence in returns to wealth." In: *Econometrica* 88.1, pp. 115–170. DOI: https://doi.org/10.3982/ECTA14835. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.3982/ECTA14835.

Fagereng, Andreas, Martin B. Holm, and Gisle J. Natvik (2021). "MPC heterogeneity and household balance sheets." In: *American Economic Journal: Macroeconomics* 13.4, pp. 1–54. DOI: 10.1257/mac.20190211.

Friedman, Milton (1957). *Theory of the Consumption Function*. Princeton University Press.

Gabaix, X, J M Lasry, P L Lions, and B Moll (2016). "The dynamics of inequality." In: *Econometrica*. DOI: 10.3982/ECTA13569.

Genay, Hesna and Darrin R Halcomb (2004). *Rising Interest Rates, Bank Loans and Deposits - Federal Reserve Bank of Chicago*. en. https://www.chicagofed.org/publications/chicago-fed-letter/2004/november-208. Accessed: 2025-8-4.

Gilboa, Itzhak (2009). *Theory of Decision under Uncertainty*. Econometric Society Monographs. Cambridge University Press. DOI: 10.1017/CBO9780511840203.

Gilboa, Itzhak and David Schmeidler (1989). "Maxmin expected utility with non-unique prior." In: *Journal of Mathematical Economics* 18.2, pp. 141–153.

Guiso, Luigi, Paola Sapienza, and Luigi Zingales (2004). "The Role of Social Capital in Financial Development." In: *Am. Econ. Rev.* 94.3, pp. 526–556. DOI: 10.1257/0002828041464498.

— (2008). "Trusting the stock market." In: *the Journal of Finance* 63.6, pp. 2557–2600.

Guler, Bulent, Burhan Kuruscu, and Baxter Robinson (2022). *The composition and distribution of wealth and aggregate consumption dynamics*. https://events.bse.eu/live/files/4096-gkrdraftv31submitpdf. Accessed: 2023-9-6.

Guvenen, Fatih, Gueorgui Kambourov, Burhan Kuruscu, Sergio Ocampo, and Daphne Chen (2023). "Use It or Lose It: Efficiency and Redistributional Effects of Wealth Taxation." In: *Q. J. Econ.* 138.2, pp. 835–894. DOI: 10.1093/qje/qjac047.

Haran Rosen, Maya, Annamaria Lusardi, and Olivia S Mitchell (2025). *Trust, financial literacy, and financial behavior driving retirement security.* en.

Havránek, Tomáš and Anna Sokolova (2020). "Do consumers really follow a rule of thumb? three thousand estimates from 144 studies say "probably not"." In: *Review of Economic Dynamics* 35, pp. 97–122. DOI: 10.1016/j.red.2019.05.004.

Hong, Chew Soo (1983). "A generalization of the quasilinear mean with applications to the measurement of income inequality and decision theory resolving the allais paradox." In: *Econometrica* 51.4, pp. 1065–1092.

Huggett, Mark (1993). "The risk-free rate in heterogeneous-agent incomplete-insurance economies." In: *Journal of economic Dynamics and Control* 17.5-6, pp. 953–969.

Jappelli, Tullio and Luigi Pistaferri (2014a). "Fiscal policy and MPC heterogeneity." In: *American Economic Journal: Macroeconomics* 6.4, pp. 107–136. DOI: 10.1257/mac.6.4.107.

— (2014b). "Fiscal policy and mpc heterogeneity." In: *American Economic Journal: Macroeconomics* 6.4, pp. 107–36. DOI: 10.1257/mac.6.4.107.

Johnson, David S., Jonathan A. Parker, and Nicholas S. Souleles (2006). "Household expenditure and the income tax rebates of 2001." In: *American Economic Review* 96.5, pp. 1589–1610. DOI: 10.1257/aer.96.5.1589.

Kaplan, Greg and Giovanni L Violante (2022). *The Marginal Propensity to Consume in Heterogeneous Agent Models*. Working Paper 30013. National Bureau of Economic Research. DOI: 10.3386/w30013.

Kennickell, Arthur B (2017). "Lining up: Survey and administrative data estimates of wealth concentration." In: *Stat. J. IAOS* 33.1, pp. 59–79. DOI: 10.3233/sji-170349.

Kihlstrom, Richard E. and Leonard J. Mirman (1981). "Constant, increasing and decreasing risk aversion with many commodities." In: *The Review of Economic Studies* 48.2, pp. 271–280.

Krusell, Per and Anthony Smith (1998). "Income and Wealth Heterogeneity in the Macroeconomy." In: *Journal of Political Economy* 106.5, pp. 867–896. DOI: 10.1086/250034.

Lee, Seungcheol, Ralph Luetticke, and Morten O. Ravn (2021). *Financial frictions: micro vs macro volatility*. ECB Working Paper Series 2622. European Central Bank. DOI: 10.2866/13857.

Lesmeister, Simon, Peter Limbach, and Marc Goergen (2019). *Trust and shareholder voting*. en. Tech. rep.

Lusardi, Annamaria, Pierre-Carl Michaud, and Olivia S. Mitchell (2017). "Optimal financial knowledge and wealth inequality." In: *Journal of Political Economy* 125.2, pp. 431–477. DOI: 10.1086/690950. eprint: https://doi.org/10.1086/690950.

Lusardi, Annamaria and Olivia S Mitchell (2014). "The Economic Importance of Financial Literacy: Theory and Evidence." en. In: *J. Econ. Lit.* 52.1, pp. 5–44. DOI: 10.1257/jel.52.1.5.

Mehra, Rajnish and Edward C Prescott (2003). "Chapter 14 The equity premium in retrospect." In: *Handbook of the Economics of Finance*. Vol. 1. Handbook of the Economics of Finance. Elsevier, pp. 889–938. DOI: 10.1016/S1574-0102(03)01023-9.

Menzio, Guido and Saverio Spinella (2025). *A Quantitative Theory of Heterogeneous Returns to Wealth*.

Parker, Jonathan A., Nicholas S. Souleles, David S. Johnson, and Robert McClelland (2013). "Consumer spending and the economic stimulus payments of 2008." In: *American Economic Review* 103.6, pp. 2530–2553. DOI: 10.1257/aer.103.6.2530.

Paul, Pascal and Mauricio Ulate (2024). "A macroeconomic model of central bank digital currency." In: *Federal Reserve Bank of San Francisco, Working Paper Series* 2024.11, pp. 01–83. DOI: 10.24148/wp2024-11.

Rawls, John (1971). *A Theory of Justice: Original Edition*. Harvard University Press.

Sabelhaus, John and Jae Song (2010). "The great moderation in micro labor earnings." In: *J. Monet. Econ.* 57.4, pp. 391–403. DOI: 10.1016/j.jmoneco.2010.04.003.

Saez, Emmanuel and Gabriel Zucman (2014). *Wealth Inequality in the United States since 1913: Evidence from Capitalized Income Tax Data*. Working Paper 20625. National Bureau of Economic Research. DOI: 10.3386/w20625.

Sarkisyan, Sergey and Tasaneeya Viratyosin (2021). "The impact of the deposit channel on the international transmission of monetary shocks." en. In: *SSRN Electron. J.* DOI: 10.2139/ssrn.3938284.

Suen, Richard M. H. (2012). *Time Preference and the Distributions of Wealth and Income*. Working Paper 2012-01. University of Connecticut.

Wooldridge, Jeffrey M. (2019). "Correlated Random Effects Models with Unbalanced Panels." In: *Journal of Econometrics* 211.1, pp. 137–150. DOI: 10.1016/j.jeconom.2018.12.010.

Appendix A

Supplementary Appendix to Chapter 2

A.1 Determinants of trust

The focus of this paper is the general trust measure. Table A.1 describes the set of controls and determinants of general trust. The first column includes standard demographics (gender, education, marital status, and race/ethnicity) while the second column adds variables more unique to the 2020 wave of the HRS questionnaire such as hometown population size and the "Potential for Deception" measures for social security, medicare/medicaid, banks, financial advisors, mutual funds, and insurance companies. The third column adds health conditions, insurance coverage, life insurance, and marital history. The fourth column repeats that specification and adds an indicator for leaving any bequest. Columns *(2)–(4)* include census region; columns *(1)–(4)* include five-year age-bin fixed effects (coefficients omitted to save space). Below the main fit statistics, the table reports Wald tests for the "Potential for Deception" block and joint tests for age bins, race/ethnicity, and region where those terms appear in the column.

Table A.1: Determinants of General Trust

	(1)	(2)	(3)	(4)
Female	0.24	0.60***	0.61***	0.57
	(0.17)	(0.22)	(0.23)	(0.38)
Years of education	0.05	0.00	−0.00	0.00
	(0.03)	(0.05)	(0.05)	(0.08)
Married	0.34*	−0.01	−0.13	0.24
	(0.18)	(0.24)	(0.26)	(0.41)
NH Black	−1.08***	−1.19***	−1.25***	−0.62
	(0.22)	(0.31)	(0.31)	(0.51)
Hispanic	−0.44*	−0.38	−0.39	−0.51
	(0.27)	(0.40)	(0.40)	(0.63)
NH Other	−0.21	−1.00	−1.26*	−2.16*
	(0.31)	(0.63)	(0.64)	(1.21)
In labor force	0.20	0.14	−0.03	−0.33
	(0.19)	(0.23)	(0.23)	(0.37)
Depression	−0.16***	−0.14**	−0.12*	0.09
	(0.04)	(0.06)	(0.06)	(0.10)
Small/med city (10k-100k)		0.00	−0.10	0.02
		(0.29)	(0.29)	(0.47)
Large metro (100k+)		0.10	0.08	−0.09
		(0.29)	(0.29)	(0.48)
Deception: Social Security		−0.00	0.00	0.01
		(0.00)	(0.00)	(0.01)
Deception: Medicare/Medicaid		0.00	0.00	−0.01
		(0.00)	(0.00)	(0.01)
Deception: Banks		−0.00	−0.00*	−0.00*
		(0.00)	(0.00)	(0.00)
Deception: Financial advisors		−0.00	−0.00	−0.00
		(0.00)	(0.00)	(0.00)
Deception: Mutual funds		0.00	0.00	0.00
		(0.00)	(0.00)	(0.00)
Deception: Insurance companies		−0.00	−0.00*	−0.00
		(0.00)	(0.00)	(0.00)
Health conditions			−0.06	−0.17
			(0.09)	(0.16)
Covered by Medicare			−0.01	−0.59
			(0.35)	(0.48)
Covered by Medicaid			−0.25	0.24
			(0.37)	(0.64)
Has life insurance			0.39*	0.21
			(0.23)	(0.37)
Number of reported divorces			−0.07	−0.08
			(0.13)	(0.22)
Number of reported times being widowed			−0.11	0.28
			(0.32)	(0.37)
Leaves any bequest				0.01*
				(0.01)
Constant	2.72	1.02	1.54	−0.40
	(1.76)	(0.83)	(0.94)	(1.80)
Observations	890.00	502.00	493.00	236.00
Adj. R^2	0.11	0.14	0.15	0.17
Deception joint p	.	0.62	0.40	0.48
Age joint p	0.00	0.00	0.00	0.00
Race joint p	0.00	0.00	0.00	0.20
Region joint p	.	0.98	0.99	0.55

OLS. Heteroskedasticity-robust standard errors. Column *(1)*: demographics, depression, and labor force. Column *(2)*: adds population size, wealth deciles, region, born in U.S., and six "Potential for Deception" items. Column *(3)*: adds health conditions, Medicare/Medicaid, life

As Alesina and Ferrara 2000 suggests, demographic variables tend to determine trust. In the HRS data, the estimated coefficients for non-Hispanic Black and for marital status have the strongest significance; the race coefficient is negative, consistent with Alsan and Wanamaker 2018. Depression is negative and significant across specifications.

A.2 Race coefficients without trust in panel models

Table A.2 extends the no-trust race comparison to the panel setting. The FE columns report second-stage cross-sectional regressions of the estimated household fixed effect from no-trust FE first stages, while the CRE columns report the corresponding no-trust CRE return regressions. The point is not to force a single structural interpretation across these estimators, but to check whether the sign and rough magnitude of the race coefficients remain similar once the model class changes.

Table A.2: Race Coefficients in No-Trust FE and CRE Specifications

	(1) FE red.	(2) FE full	(3) CRE red.	(4) CRE full
NH Black	0.29***	0.27***	0.01*	0.01**
	(0.01)	(0.01)	(0.00)	(0.00)
Hispanic	0.18***	0.22***	−0.01**	−0.00
	(0.01)	(0.01)	(0.01)	(0.01)
NH Other	0.10***	0.14***	−0.01	−0.00
	(0.02)	(0.02)	(0.01)	(0.01)
N	21983.00	21891.00	102711.00	102091.00
Race joint p	0.00	0.00	0.02	0.14

FE columns report second-stage OLS of the estimated household fixed effect from no-trust FE specifications; CRE columns report no-trust CRE regressions of 5% winsorized returns to net wealth. In each model family, the reduced column includes race with the baseline return controls, while the full column adds the extended demographic controls. Only race coefficients are displayed.

A.3 Measures of income

Labor income is a narrow measure only capturing earnings and unemployment income. Total income is a more broad measure including retirement and capital income. Figure A.1 shows that, for the time period, mean incomes are relatively flat over the period. Moreover, the discrepancy between mean labor and total income is likely driven by the oversampling of older and retired households by the HRS.

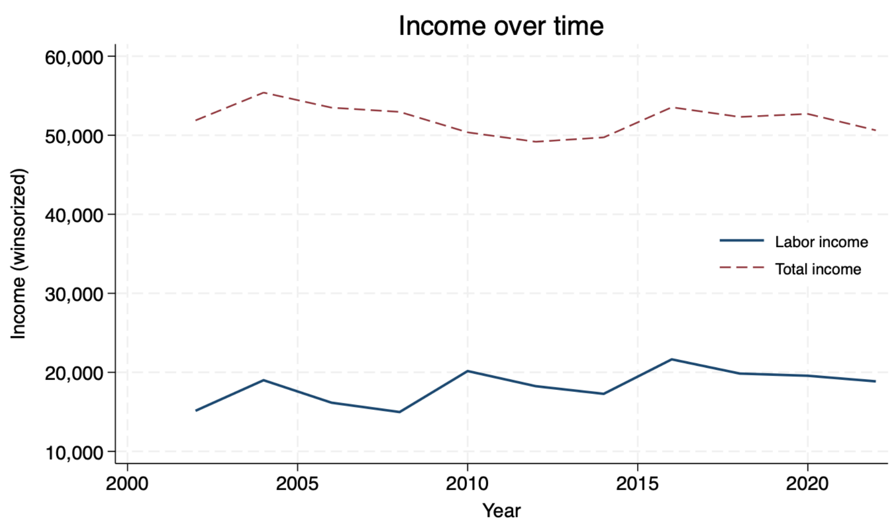


Figure A.1: Income over time

In particular, I pooled the observations of the survey together to get a sense of the distribution of incomes measured in the survey. This is captured by Table A.3.

Table A.3: Income Distribution Summary Statistics

Variable	Obs	Mean	SD	P50	P95	Min	Max
Labor income	2,919	17,777	39,181	0	109,894	0	215,261
Total income	2,919	62,671	83,430	33,048	239,107	0	457,548

Real USD, winsorized; person-years restricted to the same pooled regression sample used for return descriptives in the main text (Tables 2.3 and 2.4).

An important finding in the literature on earnings in the U.S. is that it is generally hump-shaped over the lifecycle. It is a good sign then that Figure A.2 displays this pattern.

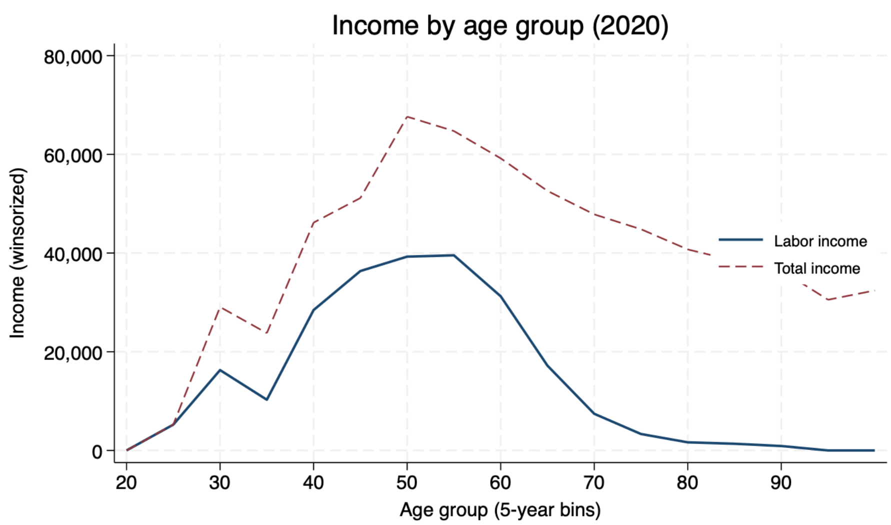


Figure A.2: Income by age group (2020)

Another empirical finding in the literature on earnings is that on average individuals with more education earn higher income. The trend captured in Table A.4 suggests that the earnings data is in line with what one would expect regarding earnings for a representative survey in the U.S.[1]

[1]Although the HRS oversamples older households, I use the provided respondent-level weights for

Table A.4: Mean Income by Education Group

Education	Labor income	Total income	Obs
No hs	4,849	21,847	559
Hs	11,090	42,552	848
Some college	17,306	56,183	706
4yr degree	31,089	107,494	394
Grad	37,157	127,723	412

Real USD, winsorized; same person-year sample as Table A.3. No hs = <12y; Hs = 12y; Some college = 13–15y; 4yr = 16y; Grad = 17+y.

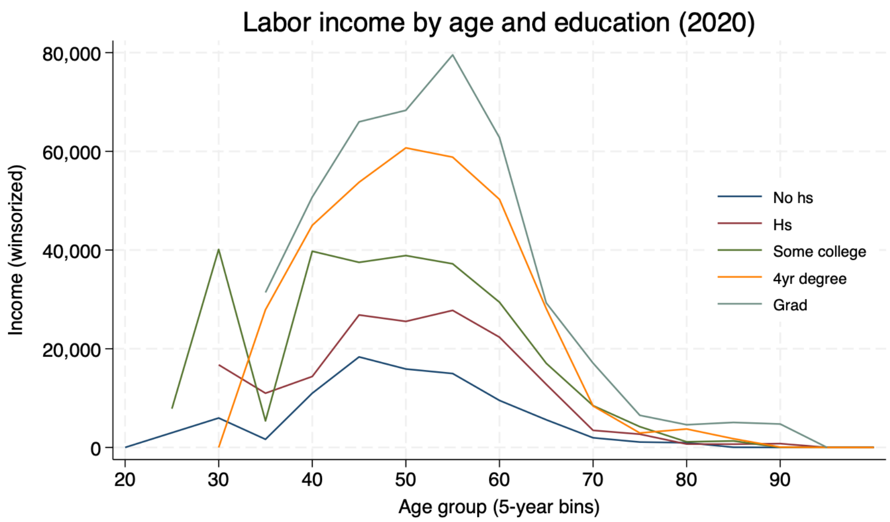


Figure A.3: Labor income by age and education (2020)

this interpretation of the summary statistics of the data.

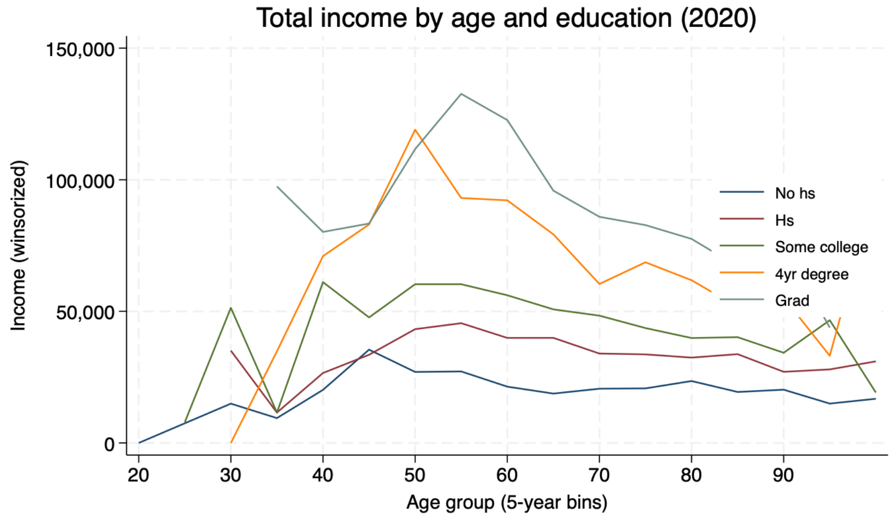


Figure A.4: Total income by age and education (2020)

A.4 Trust and total income

This section is an attempt to test whether the hump-shaped relationship between trust and income in Butler et al. 2016 holds for the 2020 HRS cohort. I focus the analysis of the empirical relationship between both income and returns and trust on a subset of measures. For income, I omit the analysis for labor income. I also exclude the analysis of the log of total income. Labor income results are not statistically significant. Given the oversampling of older, retired households, lack of predictive power for trust on labor income is not alarming. Total income results are robust to both log and IHS transformations.

In Table A.5, we see the coefficients for both the linear and quadratic terms are significant; the predictive power survives even when the additional controls important for explaining income are included. Since the goal is to uncover whether there is a maximal level of trust regarding economic performance, then we care about the significance of both trust coefficients. Here, we reject the hypothesis that both coefficients are 0.[2]

Table A.5: Trust and Total Income (2020)

	(1)	(2)	(3)	(4)
Trust	0.03***	0.19***	0.00	0.05**
	(0.01)	(0.02)	(0.01)	(0.02)
Trust^2		−0.02***		−0.00**
		(0.00)		(0.00)
Female			−0.27***	−0.27***
			(0.04)	(0.04)
Years of education			0.07***	0.07***
			(0.01)	(0.01)
Married			0.14***	0.14***
			(0.04)	(0.04)
Born in U.S.			0.02	0.02
			(0.07)	(0.07)
In labor force			0.50***	0.49***
			(0.05)	(0.05)
NH Black			−0.24***	−0.23***
			(0.05)	(0.05)
Hispanic			−0.24***	−0.23***
			(0.07)	(0.07)
NH Other			−0.27***	−0.27***
			(0.09)	(0.09)
Constant	0.83***	0.53***	−0.48***	−0.51***
	(0.05)	(0.06)	(0.14)	(0.16)
N	900.00	900.00	890.00	890.00
Adj. R^2	0.01	0.05	0.35	0.35
Trust joint p	.	0.00	.	0.08
Age joint p	.	.	0.00	0.00
Race joint p	.	.	0.00	0.00

Cross-sectional OLS. Heteroskedasticity-robust standard errors. Dependent variable is the inverse hyperbolic sine of deflated, winsorized 2020 total income. Columns *(1)–(2)*: trust only (linear and quadratic). Columns *(3)–(4)*: adds age-bin FE, gender, education, labor force, marital status, US birthplace, and race/ethnicity. Age-bin coefficients omitted from display.

I also conduct the joint Wald test for the null hypotheses that the coefficients on

[2]This rejection of the Wald test for the linear and quadratic trust terms (i.e. reject both coefficients are zero) will be true for most of the models analyzed in this paper.

the age bins and race categories are jointly equal to zero. In both cases, the null hypothesis is rejected at conventional significance levels.

With income data available from 2002–2022, I consider the statistical relationship between average total income and trust. This can be seen in Table A.6. Not only does the pattern persist, but the significance is stronger not only for trust coefficients, but for the other explanatory variables as well. Again, the Wald tests on the coefficients for race and age bins are rejected, meaning each matter for long-run total income.

Table A.6: Trust and Average Income

	(1)	(2)	(3)	(4)
Trust	0.03***	0.21***	0.00	0.05***
	(0.01)	(0.02)	(0.01)	(0.02)
Trust^2		−0.02***		−0.00***
		(0.00)		(0.00)
Female			−0.25***	−0.25***
			(0.04)	(0.04)
Years of education			0.08***	0.08***
			(0.01)	(0.01)
Married			0.20***	0.19***
			(0.04)	(0.04)
Born in U.S.			−0.03	−0.03
			(0.06)	(0.06)
In labor force			0.37***	0.37***
			(0.04)	(0.04)
NH Black			−0.27***	−0.26***
			(0.05)	(0.05)
Hispanic			−0.29***	−0.28***
			(0.06)	(0.06)
NH Other			−0.28***	−0.29***
			(0.08)	(0.08)
Constant	0.90***	0.58***	−0.62***	−0.66***
	(0.05)	(0.05)	(0.15)	(0.17)
N	900.00	900.00	890.00	890.00
Adj. R^2	0.01	0.07	0.42	0.43
Trust joint p	.	0.00	.	0.03
Age joint p	.	.	0.00	0.00
Race joint p	.	.	0.00	0.00

Cross-sectional OLS. Heteroskedasticity-robust standard errors. Dependent variable is the IHS of average total income (deflated, winsorized) over waves 2002–2022. Columns *(1)–(2)*: trust only (linear and quadratic). Columns *(3)–(4)*: adds full controls. Age-bin coefficients omitted from display.

Appendix B

Supplementary Appendix to Chapter 3

B.1 Proofs

B.1.1 Proposition 1

Proof. Let

$$W^A(\{p_s\}; U) \equiv \min_{\lambda \in C} \int_S \Big(\mathbb{E}_{p_s}[U]\Big)\, d\lambda, \qquad C \subseteq \Delta(S),\ C \neq \emptyset,$$

and fix any U with $U' > 0$ and $U'' \leq 0$.

Suppose that, for every state $s \in S$,

$$\int_0^z \Big[P_s(y) - P_s^*(y)\Big] dy \leq 0 \quad \text{for all } z \in [0, \bar{y}],$$

where

$$P_s(y) = \int_0^y p_s(t)\, dt, \qquad P_s^*(y) = \int_0^y p_s^*(t)\, dt.$$

For each fixed state s, this is exactly the single-state second-order stochastic dominance condition. Hence,

$$\mathbb{E}_{p_s}[U] \geq \mathbb{E}_{p_s^*}[U] \quad \text{for every } s \in S.$$

Let $\lambda \in C$ be arbitrary. Since λ is a probability measure on S,

$$\int_S \mathbb{E}_{p_s}[U]\, d\lambda \geq \int_S \mathbb{E}_{p_s^*}[U]\, d\lambda.$$

Because this inequality holds for every $\lambda \in C$, taking minima over C gives

$$\min_{\lambda \in C} \int_S \mathbb{E}_{p_s}[U]\, d\lambda \geq \min_{\lambda \in C} \int_S \mathbb{E}_{p_s^*}[U]\, d\lambda.$$

Therefore,

$$W^A(\{p_s\}; U) \geq W^A(\{p_s^*\}; U).$$

Since U was arbitrary in the class $\{U : U' > 0,\ U'' \leq 0\}$, the claim follows.

$\square$

Remark 1. *The argument above establishes a sufficient condition. In general, the converse fails under a maxmin aggregator: the minimizing prior may assign zero weight to states where state-by-state dominance fails. An equivalence requires stronger assumptions (for example, comparisons for every λ in a sufficiently rich set of priors that identifies each state).*

B.2 Single-state benchmark

Proposition 2. *For a single-state problem with distributions f and f^* on $[0, \bar{y}]$, define*

$$F(y) = \int_0^y f(t)\, dt, \qquad F^*(y) = \int_0^y f^*(t)\, dt.$$

Then

$$\mathbb{E}_f[U] \geq \mathbb{E}_{f^*}[U] \quad \textit{for all } U \textit{ with } U' > 0,\ U'' \leq 0$$

if and only if

$$\int_0^z \Big[F(y) - F^*(y)\Big] dy \leq 0 \quad \textit{for all } z \in [0, \bar{y}].$$

Moreover, strict preference for all such U requires strict inequality for some $z \in (0, \bar{y})$.

B.3 The Analogy between Income and Wealth Inequality Aversion

To understand the problem, I begin with describing the model which proposes measures of income inequality using the choice under objective uncertainty literature.

Analytical framework

Denote the set of outcomes $Y = [0, \bar{y}]$. Then, the choice set is given by:

$$L = \Bigg\{ p : 2^{[0,\bar{y}]} \to \mathbb{R} \ \mid \ p(\cdot) \text{ is an income frequency distribution} \Bigg\}.$$

We may define a binary relation over both sets Y, L :

$$\succsim_y \subseteq [0, \bar{y}] \times [0, \bar{y}] \subseteq \mathbb{R} \times \mathbb{R}$$

$$\succsim \subseteq L \times L.$$

Notice that $\succsim_y$ may be represented by a real-valued utility function. This is the social welfare $U(y)$ which is a key object of analysis in this paper.

Axiomatization

V 1 (Weak Order)**.** *$\succsim$ is complete and transitive.*

V 2 (Continuity)**.** *For every $p(\cdot)$, $p^*(\cdot)$, $p'(\cdot) \in L$, if $p(\cdot) \succ p^*(\cdot) \succ p'(\cdot)$, there exists $\alpha, \beta \in (0,1)$ such that*

$$\alpha p + (1-\alpha)p' \succ p^* \succ \beta p + (1-\beta)p'.$$

V 3 (Independence)**.** *For every $p, p^*, p' \in L$ and every $\alpha \in (0,1)$ such that*

$$p \succsim p^* \iff \alpha p + (1-\alpha)p' \succsim \alpha p^* + (1-\alpha)p'.$$

Expected utility representation of the ranking over income distributions

The ranking over income distributions will be represented using the vNM-expected utility representation. That is,

$$p(y) \sim \int_0^{\bar{y}} U(y)p(y)dy \equiv W^O.$$

Formally, note the slight modification of the vNM-EU theorem in this setting of ranking income distributions:

Theorem 2. $\succsim \subseteq L \times L$ *satisfies* $(V1)$ ***weak order****,* $(V2)$ ***continuity****,* $(V3)$ ***independence*** *if and only if there exists* $U : Y \to \mathbb{R}$ *such that, for every* $p(y), p^*(y) \in L$*,*

$$p(y) \succsim p^*(y) \iff \int_0^{\bar{y}} U(y)p(y)dy \geq \int_0^{\bar{y}} U(y)p^*(y)dy.$$

3.6.1 Ratio measure D

Recall the total social welfare under the two choice models: choice under objective uncertainty and choice under ambiguity.

$$W^O \equiv \int_0^{\bar{y}} U(y)\, p(y)\, dy$$

$$W^A \equiv \min_{\lambda \in C} \int_S \mathbb{E}_{p_s}[U]\, d\lambda.$$

Then the ratio measures are

$$D^O = \frac{W^O}{U(\mu)}, \qquad D^A = \frac{W^A}{U(\mu)}.$$

3.6.2 Equally distributed equivalent measure

Define the EDE wealth levels by

$$U\Big(w^O_{EDE}\Big) = W^O = \int_0^{\bar{y}} U(y)\, p(y)\, dy, \qquad U\Big(w^A_{EDE}\Big) = W^A = \min_{\lambda \in C} \int_S \mathbb{E}_{p_s}[U]\, d\lambda.$$

Hence,

$$w^O_{EDE} = U^{-1}(W^O), \qquad w^A_{EDE} = U^{-1}(W^A).$$

The corresponding inequality measures are

$$I^O = 1 - \frac{w^O_{EDE}}{\mu}, \qquad I^A = 1 - \frac{w^A_{EDE}}{\mu}.$$

3.6.3 Closed form under homothetic preferences

For

$$U(y) = \begin{cases} \alpha + \beta \dfrac{y^{1-\epsilon}}{1-\epsilon}, & \epsilon \neq 1, \\ \alpha + \beta \ln y, & \epsilon = 1, \end{cases} \qquad \beta > 0,$$

we obtain the objective and ambiguity formulas by substituting the corresponding welfare aggregator.

If $\epsilon \neq 1$:

$$w^O_{EDE} = \left[\frac{1-\epsilon}{\beta}\Big(W^O - \alpha\Big)\right]^{\frac{1}{1-\epsilon}}, \qquad w^A_{EDE} = \left[\frac{1-\epsilon}{\beta}\Big(W^A - \alpha\Big)\right]^{\frac{1}{1-\epsilon}},$$

$$I^O = 1 - \frac{1}{\mu}\left[\frac{1-\epsilon}{\beta}\Big(W^O - \alpha\Big)\right]^{\frac{1}{1-\epsilon}}, \qquad I^A = 1 - \frac{1}{\mu}\left[\frac{1-\epsilon}{\beta}\Big(W^A - \alpha\Big)\right]^{\frac{1}{1-\epsilon}}.$$

If $\epsilon = 1$:

$$w^O_{EDE} = \exp\left(\frac{W^O - \alpha}{\beta}\right), \qquad w^A_{EDE} = \exp\left(\frac{W^A - \alpha}{\beta}\right),$$

$$I^O = 1 - \frac{1}{\mu}\exp\left(\frac{W^O - \alpha}{\beta}\right), \qquad I^A = 1 - \frac{1}{\mu}\exp\left(\frac{W^A - \alpha}{\beta}\right).$$

So the extension from objective uncertainty to ambiguity is algebraically the replacement

$$W^O \longrightarrow W^A,$$

with the EDE inversion and inequality-index algebra unchanged.

B.4 Median Wealth Gap for the Restricted Sample

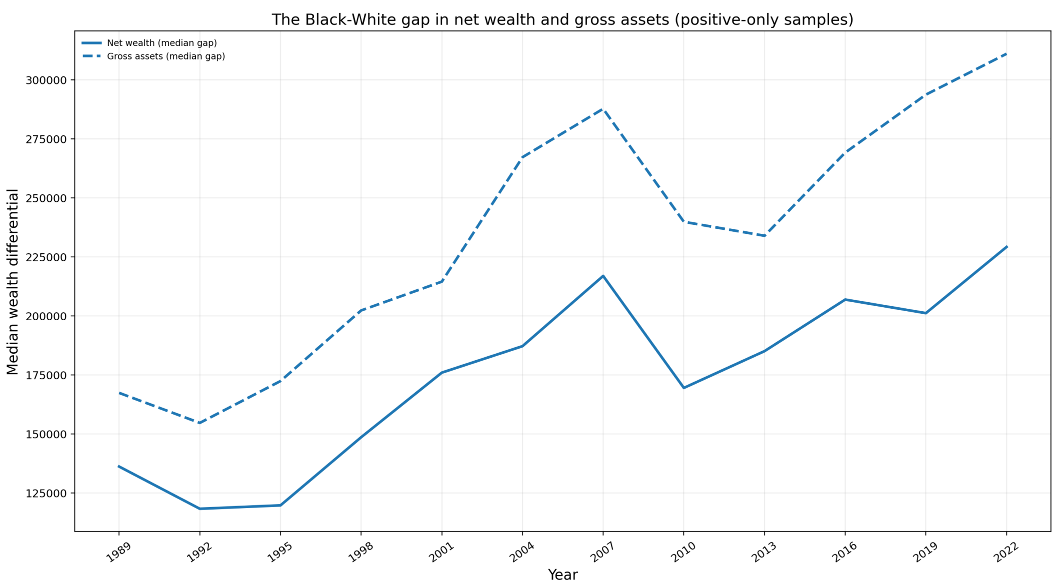


Figure B.1: Black-White median wealth differential in net wealth and gross assets, positive-only samples (SCF 1989–2022).

Even after imposing positive-only restrictions, Figure B.1 shows that both median-gap series still display visible time variation and an upward tendency over the sample. This provides a useful diagnostic benchmark for interpreting the unusually stable, high restricted-sample model-based inequality indices in the main text.

B.5 Diagnostics for Handling Non-positive Wealth

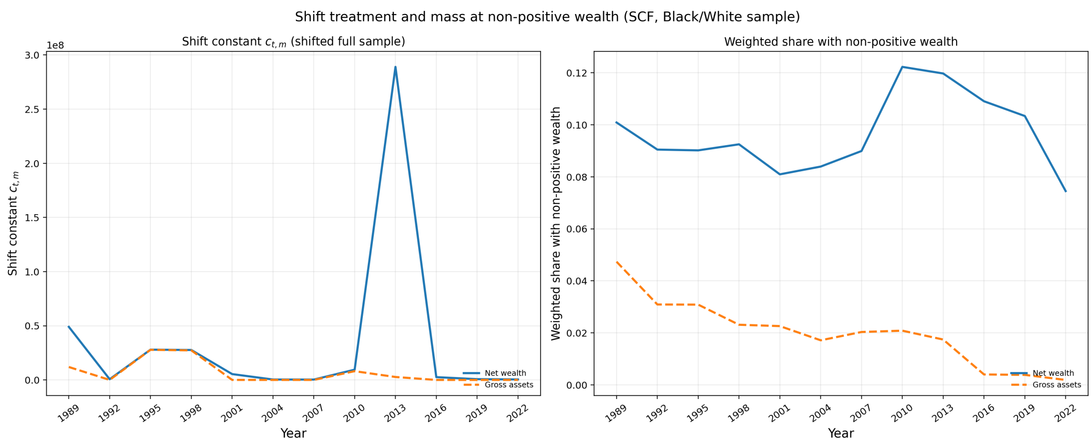


Figure B.2: Left: year-specific shift $c_{t,m}$ applied in the shifted full-sample treatment. Right: weighted fraction of households with non-positive raw wealth for each measure (before shifting).

Together with the median-gap figures, these panels are meant as a simple check on how the sample is affected when households with non-positive wealth are kept in the shifted full-sample treatment (by adding a year-specific constant so utility is well-defined) rather than dropped as in the positive-only restriction. The left panel shows how large that shift had to be each year; the biggest values line up with 2010–2016, which is the period when balance-sheet stress and indebtedness were most salient in the aggregate SCF. The same window shows up in Figure 3.1 as an episode where mean gross assets are comparatively soft in levels while mean net wealth is also compressed—the configuration you expect when liabilities are weighing heavily on net positions. The right panel shows that the weighted share of observations at or below zero in raw wealth is largest in those years as well, for both net wealth and gross assets, so the mechanical "mass at zero or below" lines up with when the shift has to do the most work.

B.6 The Linear Case ($\epsilon = 0$)

When $\epsilon = 0$, the CRRA class collapses to linear utility, $U(y) = \alpha + \beta y$ with $\beta > 0$. Under this case, the objective index satisfies

$$I^O = 0$$

by construction, since the equally distributed equivalent equals the mean. This is the expected no-inequality-aversion benchmark.

In contrast, the ambiguity index remains sensitive to cross-state asymmetry because maxmin aggregation selects the lower state-level expected utility. In the SCF computations, this yields positive I^A values even at $\epsilon = 0$, with notably larger values in the restricted positive-only samples than in most shifted full-sample years.

The $\epsilon = 0$ output is therefore used as a calibration benchmark for the main text: it isolates the linear, no-inequality-aversion case while preserving the ambiguity-vs-objective comparison.